Robotics Research Lab
Department of Computer Science
University of Kaiserslautern-Landau

Master Thesis

Learning-Based Online Adaptation for Control of a Hydraulic Manipulator

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Hereby I declare that I have self-dependently composed the Master Thesis at hand. The sources and additives used have been marked in the text and are exhaustively given in the bibliography.

Friday 4$^{\text{th}}$ September, 2026 – Kaiserslautern

(Kevin Kudakwashe Murera)

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Abstract

Hydraulic manipulators are difficult to control precisely: the actuator response is nonlinear, depends on load and supply pressure, and varies between machines. Faithful simulation of those effects is possible but far too slow for reinforcement learning, while commanding joint positions directly discards the behaviour that makes the machine difficult.

This thesis investigates how hydraulic actuator dynamics learned from a high-fidelity Simscape model can be reduced, transferred into an efficient robotic simulation and used for reinforcement-learning control of a verge mowing arm. Two actuator representations are identified from the same data: a modular inverse and forward chain, and a physics-structured reduced-order model that keeps pressure, force, velocity and extension propagation explicit while learning valve realisation and unmodelled loading. The inverse and forward pair transfers from Simscape frozen, without deployment-specific retraining; the reduced-order model receives a bounded calibration of its correction head against the deployment plant. A model can also score well on supervised prediction while ignoring the valve command a controller must use, so excitation design and command ablation are established as admission tests for a learned actuator.

Both representations are integrated into the deployment simulation and used for policy learning. A damped least-squares controller supplies privileged demonstrations during training and is then removed, leaving the distilled policy as the sole deployed controller, emitting desired cylinder motion into learned actuator dynamics. Both stacks support closed-loop terrain following, and a matched comparison over held-out meshes, trajectory families and reserved plant configurations shows that the choice of representation changes tracking, command behaviour, dynamic response and robustness under plant variation. The transferred models and trained policies already absorb much of the tested variation unaided.

Online adaptation is examined last, as two mechanisms rather than one. The explicit mechanism estimates a plant latent from recent interaction history and corrects the inverse command with it. It is informative but undeliverable: latent-conditioned adapters beat a zero-context control on the adapter objective on all five seeds, yet no closed-loop metric changes, because the inverse command is saturated for most of the scored window and the correction is clipped before it reaches the plant. The implicit mechanism conditions a bounded residual on a causal history window and applies it to the valve command, where authority remains. It improves tracking on four of five training seeds under the pre-registered rule, while a capacity-matched residual reading zeroed history reproduces most of that gain. Effective control of this machine therefore depends not only on model accuracy and policy robustness, but on where the correction acts and on whether live history adds information beyond residual capacity.

Contents

1 Introduction **1**

1.1 The Role of Mechanised Verge Maintenance 1
1.2 Advances in Learning-Based Control for Hydraulic Manipulators 1
1.3 Challenges of Precise Tracking with Hydraulic Mowing Arms 2
1.4 Contribution of This Thesis 3
1.5 Thesis Structure 7

2 Fundamentals and Related Work **11**

2.1 Manipulator Actuation and Control Fundamentals 11
2.1.1 Application Context and Implement Carriers 11
2.1.2 The Electro-Hydraulic Actuation Chain 12
2.1.3 High-Fidelity Simulation as a Data Source 12
2.2 Kinematics and Task-Space Tracking 13
2.2.1 Differential Kinematics 13
2.2.2 From Joint Space to Cylinder Space 13
2.3 Model-Based and Conventional Control Approaches 14
2.3.1 Fixed-Gain Cascades 14
2.3.2 Nonlinear Cancellation 14
2.3.3 Online Parameter Estimation 14
2.3.4 What Is Retained 15
2.4 Data-Driven System Identification and Learned Dynamics 16
2.4.1 Learned Forward and Inverse Dynamics 16
2.4.2 Closed-Loop Identifiability 16
2.5 Reduced-Order Modelling 16
2.6 Domain Randomisation and Latent Adaptation 17
2.6.1 Robustness Through Randomised Training 17
2.6.2 From Robustness to Adaptation 18
2.7 Reinforcement Learning and Structured Priors 19
2.7.1 Policy Optimisation 19
2.7.2 Structured Priors for Policy Learning 19
2.7.3 Reinforcement Learning on Hydraulic Machines 20
2.8 Research Gap and Thesis Positioning 21

3 System Fundamentals and Problem Formulation 24
3.1 Manipulator Platform . . . 24
3.2 Kinematic Structure and Coordinate Mappings . . . 25
3.2.1 Joint Space and Cylinder Space . . . 25
3.2.2 Task Space and the Calibrated Cutter Jacobian . . . 27
3.3 Actuation and Measured State . . . 27
3.3.1 Sensing and Observability . . . 28
3.4 High-Fidelity Simscape Model . . . 28
3.4.1 The Command ROM as Deployment Actuator . . . 28
3.4.2 Mechanical Subsystem . . . 30
3.4.3 Hydraulic Subsystem . . . 30
3.4.4 Validated Scope of the Model . . . 31
3.5 IsaacLab Environment . . . 33
3.5.1 Simulation and Control Rates . . . 33
3.5.2 Environment Configuration Details . . . 33
3.6 The Verge-Following Task . . . 33
3.6.1 Task Definition . . . 33
3.6.2 Task Feasibility and Plant Capability . . . 33
3.7 Control Architecture . . . 34
3.8 Problem Formulation . . . 35
3.9 System-Level Evaluation Criteria . . . 35
3.9.1 The Eight System-Level Evaluation Gates . . . 36
3.9.2 Cylinder and Model Checks . . . 37

4 Data Generation and Preparation 39
4.1 Dataset Objectives and Design Requirements . . . 39
4.2 Randomised Parameter Space of the Simscape Campaign . . . 41
4.2.1 Sampling Design . . . 41
4.2.2 Hydraulic Parameters . . . 41
4.2.3 The Compressibility Axis . . . 42
4.2.4 Mechanical Parameters . . . 42
4.3 Closed-Loop Data Collection . . . 43
4.3.1 The Controller as a Data-Collection Instrument . . . 43
4.4 Open-Loop Excitation Experiments . . . 43
4.4.1 Limitations of the Closed-Loop Batch . . . 43
4.4.2 Design of the Excitation Signal . . . 44
4.4.3 Plant Draws and Acceptance . . . 45
4.5 Combining Closed-Loop and Open-Loop Datasets . . . 45
4.6 Closed-Loop Calibration Data for the Command ROM . . . 45
4.7 The IsaacLab Deployment Collections . . . 46
4.7.1 Provenance of the Deployment Data . . . 47
4.7.2 The Physical Factor Vector . . . 47
4.7.3 Domains and Factor Design . . . 47
4.7.4 Collection Procedure: The Task-Controller Runs . . . 48
4.7.5 The Open-Loop Context-Identification Collection . . . 50

4.7.6 The Frozen Dataset . . . 50
4.7.7 Learning Timebase and Hydraulic State . . . 50
4.8 Feature and Target Construction . . . 51
4.8.1 The Actuator Models . . . 51
4.8.2 Features and Targets for the Adaptation Module . . . 51
4.8.3 Features and Targets for the Prediction Models . . . 52
4.8.4 On-Policy AF Adaptation Rollouts . . . 52
4.9 Preprocessing and Data Splits . . . 53
4.10 Dataset Diagnostics . . . 53
4.11 The System-Dynamics Dataset . . . 54
4.11.1 Dataset Separation Across Plants . . . 54

5 Learned Models and Online Adaptation 57
5.1 Modelling Overview and Common Notation . . . 57
5.2 Inverse Dynamics Model . . . 59
5.2.1 Paired Control Fine-Tuning . . . 60
5.2.2 Transfer of the Frozen Inverse Model . . . 60
5.3 Forward Dynamics Model . . . 61
5.3.1 Assessment of the Supervised Score . . . 62
5.4 Reduced-Order Model . . . 63
5.4.1 Closed-Loop Calibration Against the Deployment Plant . . . 65
5.5 Offline Plant-Context Identifiability Study . . . 66
5.5.1 Information Content of the Estimated Latent . . . 67
5.6 Latent-Delivery Test Through the Inverse Model . . . 67
5.6.1 Model-Level Adaptation Benefit . . . 68
5.6.2 Closed-Loop Evidence and Its Bound . . . 69
5.6.3 Propagation of the Latent Advantage . . . 69
5.7 Training and Model-Selection Protocol . . . 69

6 Control Framework and Simulation Integration 72
6.1 The DLS Teacher and Its Two Roles . . . 72
6.1.1 Scope and Role of the Controller . . . 72
6.1.2 Privileged Information . . . 72
6.1.3 From Task Error to Cylinder Command . . . 73
6.1.4 Execution Through the Student Interface . . . 74
6.1.5 Teacher Performance . . . 74
6.2 Policy Formulation . . . 75
6.2.1 Policy Definition . . . 75
6.2.2 The Observation Vector . . . 75
6.2.3 The Action and Its Bounds . . . 77
6.2.4 The Reward Function . . . 78
6.2.5 Optimisation . . . 80
6.3 Training Protocol and the Distillation Finding . . . 80
6.3.1 Distillation by DAgger . . . 80
6.3.2 Objective Mismatch in Unconstrained Fine-Tuning . . . 81

6.4 Reference-Anchored PPO . . . 81
6.4.1 Critic Warm-Up . . . 82
6.4.2 The Reference Anchor . . . 82
6.4.3 Trust Regions and Rollback . . . 82
6.4.4 The Block Ratchet . . . 83
6.4.5 Theoretical Status . . . 83
6.5 R: Reference Actuator Baseline . . . 85
6.6 IF: Non-Adaptive Learned Baseline . . . 85
6.7 AF: The Adaptive Stack . . . 86
6.7.1 Omission of the Inverse Model . . . 87
6.7.2 Sufficiency of the Forward Model . . . 90
6.7.3 The Residual Controller . . . 91
6.7.4 Freezing the Base Policy . . . 91
6.7.5 Training of the Adaptation Encoder . . . 92
6.7.6 Consistency With the Plant Dynamics . . . 92
6.7.7 Scope of the Claims . . . 92
6.8 IsaacLab Integration and Timing . . . 93
6.9 Constraints and Fair-Comparison Rules . . . 93
6.9.1 Limits and Safeguards . . . 93
6.9.2 Matching Rules and Comparison Controls . . . 94

7 Experimental Evaluation: Representation, Robustness, and the Value of Adaptation 97
7.1 Experimental Protocol . . . 97
7.1.1 Test Conditions and Comparisons . . . 97
7.1.2 Comparative Scope and Scoring . . . 97
7.1.3 Scoring Horizon . . . 98
7.2 Closed-Loop Task Performance . . . 98
7.3 Actuator Representation and Dynamic Response . . . 101
7.3.1 Plant Delay and Step Response . . . 101
7.3.2 Control Effort . . . 101
7.4 Generalisation and Robustness . . . 103
7.4.1 Delivered Stress . . . 103
7.4.2 Actuator Authority . . . 103
7.4.3 What Robustness Leaves for Adaptation . . . 105
7.5 Evaluation of Online Adaptation Mechanisms . . . 105
7.5.1 The Explicit Mechanism and Its Delivery Limit . . . 105
7.5.2 Experiment and Measurement . . . 105
7.5.3 History-Conditioned Residual Across Training Seeds . . . 106
7.5.4 Robustness to the Checkpoint Rule . . . 107
7.5.5 Capacity Versus Live History . . . 107
7.5.6 Dependence on Cutter-Mass Dose . . . 108
7.6 What Determines Effective Online Adaptation . . . 111

8 Conclusion and Future Work **113**
8.1 Summary of Findings . . . 113
8.2 Design Implications . . . 115
8.3 Contributions . . . 117
8.4 Limitations . . . 118
8.5 Future Work . . . 119

A Dataset Ranges and Audit **121**
A.1 Randomised Plant Ranges . . . 121
A.2 Open-Loop Export Audit . . . 122
A.3 Frozen Deployment Collections . . . 123

B Implementation and Reproducibility Record **124**
B.1 Timing Contract . . . 124
B.2 Frozen Artifacts . . . 125
B.3 Learned-Actuator Interfaces . . . 125
B.4 Policy Observation Layout . . . 126
B.5 Internal Model Pressure State, Not a Physical Pressure . . . 126
B.6 Derivation of the Reference-Divergence Gate Bound . . . 128
B.7 Reproduction Entry Points . . . 129

Bibliography **131**

1. Introduction

1.1 The Role of Mechanised Verge Maintenance

Road verges are cut because they are infrastructure. Vegetation left to grow closes sight lines at junctions and bends, obscures signs and barriers, blocks the drainage channels that keep water off the carriageway, and encroaches on the running surface itself. A national road network carries tens of thousands of kilometres of verge, and each kilometre is cut several times in a growing season, so the work is continuous, repetitive and performed almost entirely by machine.

It is also performed from a moving vehicle. Stopping to cut each section would require closing the lane, so a carrier drives the road at a few kilometres per hour while a boom-mounted mowing head works the verge alongside it. The head must hold a cut height of a few centimetres over ground that rises and falls continuously, follow a grass line that curves with the road, and do so without striking the surface hard enough to damage the tool or lift the machine.

The demands on the manipulator follow from that. The machine studied here, shown in Figure 1.1, carries a heavy rotating tool at the end of a long boom, works in intermittent contact with the ground, and must place that tool accurately while its own base is in motion. Hydraulic actuation is the standard answer at this scale: it supplies the force density and the tolerance to shock loading that electric drives do not, and it does so with components rugged enough for continuous outdoor operation. Precision, however, is not what hydraulics are good at, and that tension is the subject of this thesis.

1.2 Advances in Learning-Based Control for Hydraulic Manipulators

Two developments have changed what is achievable on machines of this kind.

The first is that policy optimisation in simulation has become practical for contact-rich manipulation. GPU-parallel physics engines run thousands of environments simultaneously, which makes reinforcement learning feasible on tasks that would take years of real-machine

experience, and randomising the simulator across a family of plants produces policies that transfer without being tuned to one machine. The second is that the actuator itself can be learned. Rather than deriving valve and cylinder behaviour from first principles and fitting its parameters, a network trained on recorded motion can represent the command-to-motion relationship directly, including the nonlinearities that analytical models approximate away.

Combining the two is attractive and not straightforward. A learned actuator model must be cheap enough to evaluate inside the training loop, which rules out the high-fidelity models used for offline analysis, and it must remain valid across the family of machines the policy will meet. This has motivated a line of work in which a compact code describing the current plant is inferred from recent interaction history and supplied to the controller, so that a single policy can serve a family rather than a single machine. Such methods report substantial gains in legged locomotion and in manipulation, and they are the direct antecedent of the architecture examined here.

Figure 1.1: MULAG FME 500 front mower on a Unimog during roadside verge maintenance. Source: [MULAG Fahrzeugwerk 26].

1.3 Challenges of Precise Tracking with Hydraulic Mowing Arms

A valve command does not produce a proportional cylinder motion. The relationship is nonlinear, depends on load and supply pressure, and is subject to dead time between the command and any measurable response. It also differs between machines: seal condition, fluid temperature, hose compliance and the attachment fitted all change it, so two machines of the same model respond differently and one machine responds differently across a season. A controller that inverts only the geometry therefore leaves the largest source of tracking error unaddressed.

That difficulty admits two responses, and they are routinely conflated. The first is robustness: train across a randomised family so that a single fixed model is adequate everywhere. It costs nothing at run time and gives up the difference between the family and the individual. The second is adaptation: infer which member of the family is present from the interaction history and condition the model or the policy on that estimate. It promises to recover that difference, at the cost of an estimator that must work from observable signals alone, since a deployed machine has no access to its own plant parameters.

Whether adaptation repays that cost depends on three things. Plant identity must be present in the signals produced under the available excitation. The robust controller must leave a control-relevant deficit to recover. The correction path must retain enough authority to change the realised command. This thesis measures these conditions on a hydraulic mowing arm and then evaluates whether live history adds value beyond matched residual capacity.

1.4 Contribution of This Thesis

The difficulty described in Section 1.3 is a modelling problem before it is a control problem. A hydraulic actuator can be simulated accurately, but the models that do so are too slow to sit inside a reinforcement-learning loop requiring millions of environment steps, while the usual alternative of commanding joint positions directly removes the load and pressure dependence that makes the machine difficult to control. What is needed is a representation cheap enough to train against and faithful enough that the resulting policy controls something close to the real actuator.

Learned actuator models and latent-conditioned adaptation are both established techniques. What remains unclear is how such models should be identified so that they retain the command dependence a controller needs, whether they transfer between simulators without retraining, and where a correction retains command authority once the models are used in control. Representation and transfer are established first because the residual is applied to an identified plant rather than replacing one. Online adaptation is evaluated last so that any remaining gain can be interpreted against the transferred model and the robustness of the fixed controller. The order follows the dependencies among the claims. The research statement is as follows:

Comparison notation.

The three policy–actuator stacks carry both a short label and a full name, used interchangeably throughout: *R* is PPO-ROM, *IF* is PPO-ID-FD, and *AF* is the Adaptive Forward stack. R uses the command-conditioned reduced-order model; IF uses the frozen inverse/forward pair; and AF consists of the frozen direct-valve base actor, a jointly trained causal history encoder and bounded residual head, and the frozen forward model. The offline AAM of Chapter 5 is a diagnostic plant-context estimator and is not loaded in the AF command path. The labels name complete stacks, not only their actuator models.

To determine how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation improves on the robustness those learned models already provide.

The statement decomposes into six research questions, which are answered in the order given. The first three concern how the actuator is represented, the fourth concerns control through that representation, and the last two concern what remains for adaptation once the first four are settled. The six questions are:

1. How can data from a high-fidelity Simscape model be used to learn computationally tractable representations of hydraulic actuation while preserving the command dependence a controller requires?
2. To what extent do actuator models identified from Simscape transfer to an IsaacLab manipulator without deployment-specific retraining?
3. Can a physics-structured reduced-order hydraulic model supply dynamics that are accurate and stable enough for GPU-parallel policy training?
4. Can a policy trained from a privileged classical teacher become the sole deployed controller while acting through learned actuator dynamics?
5. How does the choice of actuator representation affect tracking, command behaviour, latency and robustness under plant variation?
6. Where plant variation survives robust training, which online adaptation mechanisms exploit it, and how do the representation of the plant context and the location of the correction determine whether their benefit reaches closed-loop tracking?

RQ6 compares two adaptation mechanisms, and they are kept apart throughout because the thesis studies both and deploys only one. The *explicit* mechanism estimates a low-dimensional plant latent decodable against the true factors, which is what the AAM of Section 5.5 produces, and corrects the inverse command with it. It is the mechanism the identifiability evidence concerns, and the one whose benefit stops at the command interface. The *implicit* mechanism, which is the deployed one, conditions a bounded residual on a causal state–action history window and trains its encoder end to end for control (Section 6.7.3). Its internal representation is not constrained to identify physical parameters and is not tested for that, so it is described throughout as history-conditioned adaptation with an implicit control-relevant context, not as latent identification. Neither label is extended to the third arm used below. A residual reading a zeroed history window receives no information about the current plant, so it is a capacity control rather than an adaptation mechanism, and the frozen stacks are robust controllers rather than adaptive ones.

The first question is answered in Chapters 4 and 5, the second and third in Chapter 5, the fourth in Chapter 6, and the last two in Chapter 7.

RQ6 is the question named in the title and is deliberately placed last, since it is only worth asking once the preceding questions have established that the learned models transfer, that a policy can control through them, and how much plant variation the closed loop already absorbs unaided. Its answer separates two levels. Online plant context is usable at the model level, and whether that benefit reaches the closed loop depends on the interface the correction acts through: plant variation must remain control-relevant after robust training, be observable from the available history, and reach a correction path with sufficient authority. Any apparent gain must also be separated from the effect of adding residual capacity. The work rests on the following contributions:

Transferable Hydraulic Actuator Learning: A pipeline that identifies hydraulic actuation from high-fidelity Simscape data and transfers it into IsaacLab through two representations, a modular inverse and forward chain and a physics-structured reduced-order model. The inverse model is trained on Simscape data and evaluated on deployment runs drawn entirely outside that training distribution, where it reproduces most of the valve-command variance with an accuracy that is nearly flat across plant domains. It requires no deployment-specific retraining and remains frozen in every downstream stack. The reduced-order model is transferred differently and the difference is stated wherever it matters: it receives a small bounded calibration against the deployment plant, which is why it serves as a reference condition rather than as an independent competitor to the inverse and forward pair.

Control-Relevant Identification and Validation: A demonstration that predictive accuracy can conceal the loss of exactly the property a controller needs, together with the procedure that detects it. The first forward model reached a near-perfect supervised score while learning to ignore the valve command and reconstruct motion from the state history instead. The validation loss did not show this; ablating the command did. Redesigning the excitation and retraining recovered genuine command sensitivity, and the pairing of excitation design with input ablation is established here as a required test for learned hydraulic models intended for control rather than for prediction.

Reduced-Order Actuator Model: A command-conditioned grey-box actuator in which pressure, force, velocity and extension propagation remain explicit while learning covers valve realisation, unmodelled loading and bounded residual effects. It is accurate enough to serve as the deployment plant, stable in recursive rollout, and cheap enough to evaluate inside GPU-parallel policy training. Zeroing or shuffling its command collapses its extension and velocity accuracy, which establishes that the command authority is real rather than inferred from context.

Policy-Only Control Through Learned Hydraulics: A damped least-squares controller supplies privileged demonstrations during training and is then removed, so that the distilled policy emits desired cylinder motion directly into a learned actuator representation and constitutes the entire deployed controller. Both representations support closed-loop terrain following in this configuration, although two learned models placed in series could have compounded their errors.

A Decision Procedure for Online Adaptation: A matched protocol that first asks whether plant identity is present in the identification data, then measures what a privileged plant encoding does in the closed loop, and only then trains a deployable history estimator. Either of the first two measurements can show that a residual will have nothing to act on.

In this work the identification data do contain plant structure, and the privileged encoding is informative. The procedure therefore does not stop the adaptation path; it determines where the correction must be applied.

Where Online Adaptation Is Effective: A comparison of two online adaptation mechanisms on the same machine, separating the level at which each one acts. The explicit mechanism is informative and undeliverable. Its latent improves the adapter objective on all five seeds, and the closed-loop effect is absent because the correction is added to a command already resting against its own limit for most of the scored window. The binding constraint on latent-conditioned adaptation here is therefore delivery rather than estimation. Removing the inverse model allows an end-to-end history-conditioned residual to act directly on the valve command. The deployed residual improves closed-loop tracking relative to its frozen base on four of five training runs, so the median run improves under both checkpoint rules. Matched ablations locate most of that improvement in the added residual capacity. Live history separates from an equal-capacity zero-history control on one run of five, and a plant-dependent lateral term appears in the cutter-mass dose study, although that design couples starting mass with the later change. The supported method-level claim is therefore a bounded residual on the valve path. The supported adaptation-specific claim is narrower: live history helps where residual capacity alone does not, and where load variation remains visible in the scored channel.

The causal plant estimator, the bounded adaptation path and the eight-gate evaluation harness are described where they are used, as technical enablers rather than separate claims.

Figure 1.2 condenses the logic that connects these contributions. The sequence is a cycle rather than a one-way pipeline: evidence from robustness, identifiability and adaptation feeds back into the next actuator representation and the transfer assumptions made for it.

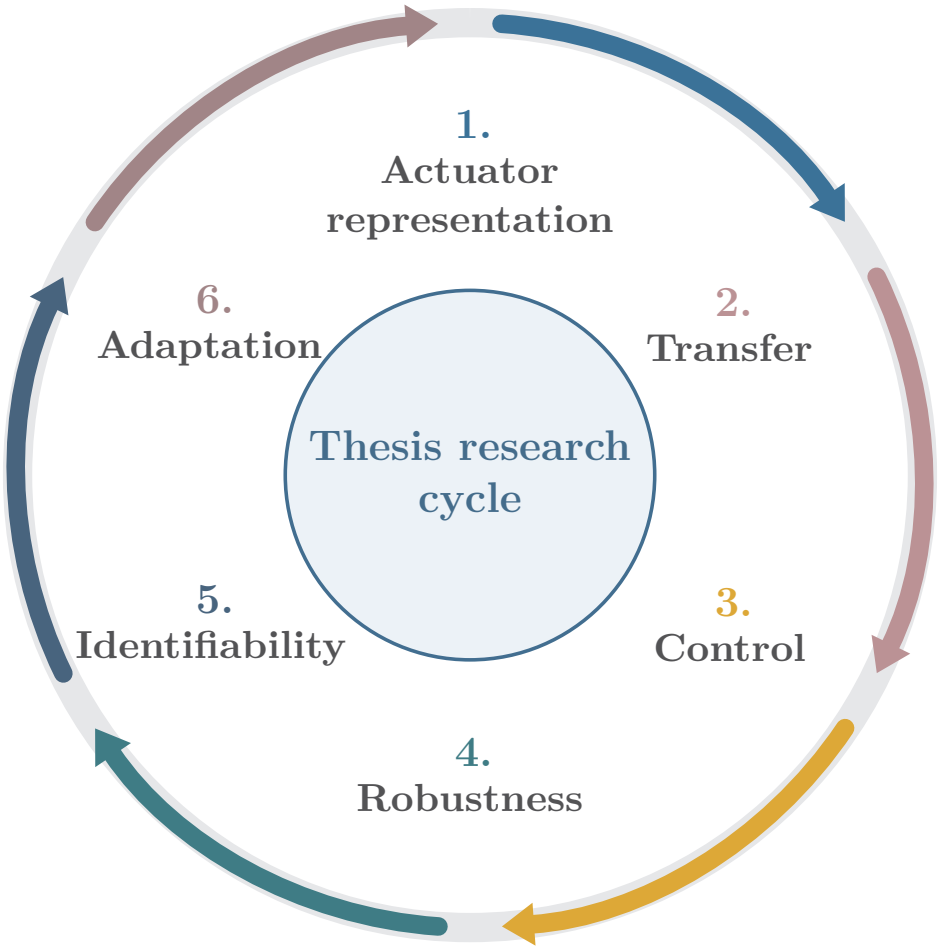

Figure 1.2: Research cycle connecting actuator representation, transfer, closed-loop control, robustness, identifiability, and adaptation. Evidence from the final stages informs the next actuator representation and its transfer assumptions.

1.5 Thesis Structure

This thesis investigates how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, grounding its investigation in the identifiability of the actuator models and in the robustness of the controllers built on them. The document is organised as follows:

Chapter 2 establishes the background against which the remainder is read. It introduces the machine class and its electro-hydraulic actuation, recalls the kinematic tools underlying the classical controller, and reviews how conventional model-based control handles hydraulic uncertainty. It then treats the learning methods this work builds on, namely data-driven dynamics models and the pitfalls of identifying them from closed-loop data, reduced-order and grey-box modelling, domain randomisation and latent adaptation, and policy optimisation with structured priors. The chapter closes by stating the gap that remains and the position of the three control stacks examined here.

Chapter 3 analyses the platform, using the MULAG FME 500 mounted on a Unimog carrier as the case study. It describes the rigid bodies and the serial chain they form, the

coordinate mappings between task space and cylinder space, and the closed kinematic loops introduced by the cylinder mounts. It specifies what the controller can observe, separating measured quantities from quantities produced by a model, and presents the high-fidelity Simscape model from which the actuator data derives. The chapter also states the task, the plant capability limits and the system-level evaluation gates against which every later result is scored.

Chapter 4 presents the data generation strategy and answers the first research question. It describes the excitation design, the randomised factor set defining the family of plants, and the procedures that keep training, validation and reserved test configurations separate. It then reports the identifiability audit applied to the resulting collection, including the measured finding that excitation gathered under a task controller does not separate plants, and the open-loop collection that repairs this.

Chapter 5 develops the learned models and answers the first three research questions. It presents the inverse model that maps a desired cylinder motion to a valve command, the forward model that predicts the response, and the evidence that supervised accuracy alone does not establish that either model retains command authority. It then presents the command-conditioned reduced-order model that serves as the deployment plant inside GPU-parallel training. The chapter closes with the material required for the final research question, namely the privileged latent and its measured limits, the causal estimator that infers a deployable substitute from observable history, and the bounded adapter through which that estimate first reached the actuator path. That adapter produces no closed-loop effect, and the chapter ends by establishing why: the limit is the delivery of the correction rather than the quality of the estimate, which is the finding the following chapter acts on.

Chapter 6 describes the control framework and its integration with the simulator, and answers the fourth research question. It presents the damped least-squares teacher used for demonstration and data collection, the shared observation interface and the action interface the two non-adaptive stacks hold in common, the distillation of the teacher into a student policy, and the reference-anchored fine-tuning that makes further policy optimisation admissible without discarding the distilled behaviour. It then assembles the three deployed control stacks, including the removal of the inverse model from the adaptive command path that the preceding chapter motivates, and states which comparison each stack can carry.

Chapter 7 reports the experimental evaluation and answers the last two research questions. It states the protocol and what each stack can carry, then compares the matched pair on closed-loop tracking and control effort, measures dead time through the actuation chain and the response of each stack to a commanded change, and audits how much of the randomised factor draw reaches each plant. It closes on online adaptation, measured within the adaptive stack against its own frozen base and a capacity-matched control: the contribution of live history, the component of the gain that varies with cutter-mass dose, the replication across five training seeds, and the pressure channel each stack exposes.

Chapter 8 concludes the thesis. It summarises what the work establishes and what it does not, states the design implications that follow from the measured results, records the limitations of the evidence, and sets out the future work, which centres on transferring the learned actuator models and the distilled policy to the physical machine and fine-tuning them on data collected from it.

Appendix A lists the dataset manifests and the factor ranges that define the family of plants used in Chapters 4 and 7. **Appendix B** records the timing and input contracts, frozen artifact identifiers and reproduction entry points used by the model, control and evaluation chapters.

2. Fundamentals and Related Work

Chapter 1 argued that precise task-space tracking with a hydraulic mowing arm is limited less by kinematics than by the actuator. The path from a valve command to actual cylinder motion is nonlinear and load dependent, and it differs from one machine, season, or attachment to the next. This chapter collects the background needed to judge that argument and to understand the design decisions made later in the thesis. Section 2.1 introduces the machine class and its electro-hydraulic actuation. Section 2.2 recalls the kinematic tools behind the classical controller, and Section 2.3 reviews how conventional model-based control deals with hydraulic uncertainty. The remaining sections cover the learning methods this work builds on. Section 2.4 treats data-driven dynamics models and the pitfalls of identifying them from closed-loop data. Section 2.5 covers reduced-order and grey-box modelling, Section 2.6 domain randomisation and latent adaptation, and Section 2.7 policy optimisation with structured priors. Section 2.8 then states the gap that remains and how the three control stacks of this thesis (R, IF, and AF) address it. As defined in Chapter 1, the first two abbreviate PPO–ROM and PPO–ID–FD; AF denotes the deployed Adaptive Forward stack rather than the offline AAM. Only material that is needed for the later chapters is included. Background and results are kept apart: where a later measurement is the reason a line of work was left behind, it is quoted forward and marked as such, and no argument in this chapter rests on it. The concrete system definition and problem formulation follow in Chapter 3.

2.1 Manipulator Actuation and Control Fundamentals

2.1.1 Application Context and Implement Carriers

The MULAG FME 500 mounts a multi-link hydraulic boom and a 1.2 m flail head on a Unimog carrier [MULAG Fahrzeugwerk 26]. It reaches about 6.5 m laterally, below road level and over obstacles, while an operator adjusts cutter height, tilt and lateral position as the vehicle moves. Automating this continuous tracking task is the practical motivation of the thesis.

Hydraulic cylinders suit the required force, shock tolerance and existing vehicle power supply, but their command gain changes strongly with load and configuration. This trade-off between force density and precision is central to hydraulic manipulator control [Mattila 17].

2.1.2 The Electro-Hydraulic Actuation Chain

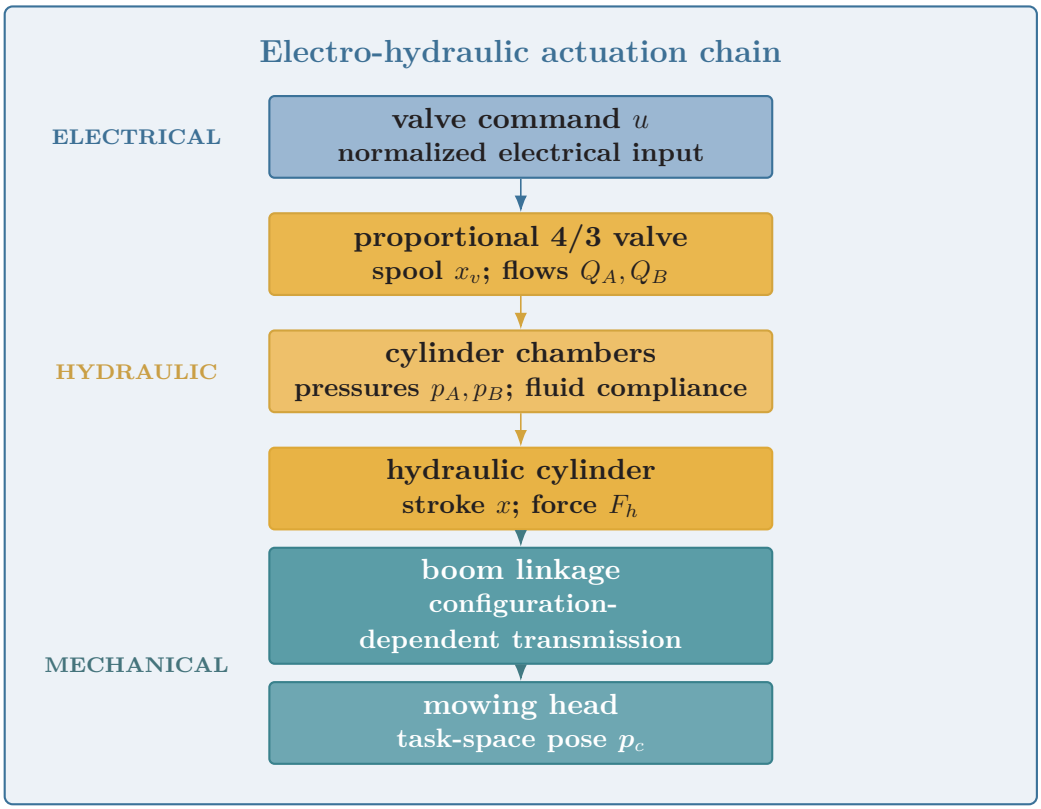

Figure 2.1: The electro-hydraulic actuation chain from valve command to mowing-head motion. The block colors distinguish the electrical, hydraulic and mechanical stages; each subtitle identifies the principal quantity passed to the next stage.

In Figure 2.1, an electrical command moves a proportional valve spool and meters flow to the cylinder chambers. The orifice law [Merritt 67] makes flow depend on spool position and the square root of the pressure margin. Valve authority is therefore load-dependent and directional; overlap creates a dead zone near null and the spool saturates at large commands [Jelali 03].

Compressibility, changing chamber volume, relief limits, leakage and seal friction further separate flow from motion. Several of these effects are non-smooth, while bulk modulus, leakage and friction drift with temperature and wear. The mechanical linkage then converts cylinder translation through configuration-dependent lever arms. Chapter 5 consequently models the hydraulic realization in cylinder space, before kinematic effects are mixed into cutter error.

2.1.3 High-Fidelity Simulation as a Data Source

The component-level Simscape plant [The MathWorks, Inc. 24a] resolves the stiff hydraulic dynamics but runs one machine at small integration steps. Policy optimisation instead

benefits from thousands of GPU-parallel plants [Makoviychuk 21]. The detailed model is therefore used offline to generate identification data; compact actuator models carry its behaviour into parallel training, as summarised in Figure 2.2.

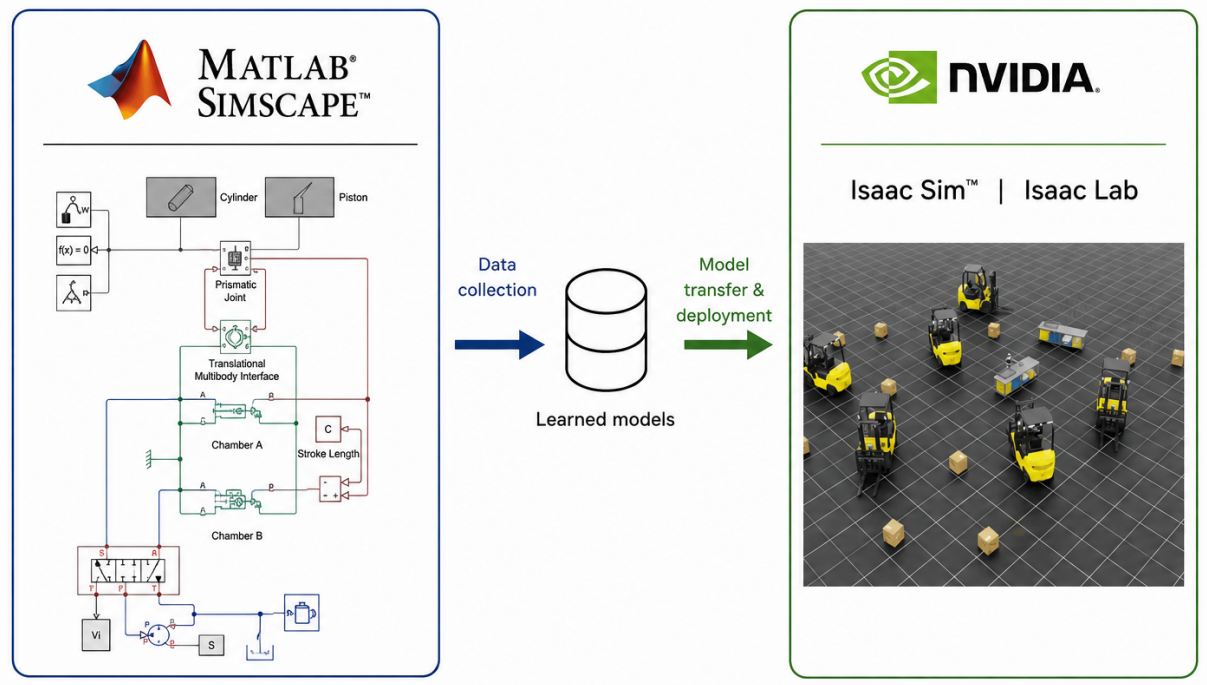

Figure 2.2: Simscape supplies high-fidelity hydraulic data; compact actuator models transfer that behaviour to GPU-parallel training in IsaacLab. Product images: MathWorks and NVIDIA.

2.2 Kinematics and Task-Space Tracking

2.2.1 Differential Kinematics

The cutter path is specified in vehicle coordinates, so the manipulator Jacobian maps joint rates to the required tool velocity [Siciliano 09]. Near singular configurations the ordinary pseudo-inverse produces excessive commands. The damped least-squares inverse of Wampler [Wampler 86] and Nakamura and Hanafusa [Nakamura 86] trades some local accuracy for bounded commands and is retained as the classical controller.

2.2.2 From Joint Space to Cylinder Space

The actuated coordinates are cylinder lengths, and their nonlinear cylinder-to-joint map has a configuration-dependent lever arm. A passive head joint further limits independently controllable directions. The calibrated cutter Jacobian in Chapter 3 composes these mappings. It still assumes that commanded cylinder motion is realized; the learned actuator models address that remaining gap.

2.3 Model-Based and Conventional Control Approaches

Three families of conventional control are relevant here: fixed-gain cascades, nonlinear cancellation, and online parameter estimation. Each of them works on hydraulic manipulators. Each also carries a maintenance or excitation cost, and those costs are the reason this thesis learns the actuator instead.

2.3.1 Fixed-Gain Cascades

Industrial implements commonly map joystick input directly to valve current. The standard improvement is a cascade in which an outer position loop sets a velocity or pressure reference and a fast inner loop tracks it [Jelali 03]. The construction is cheap, needs no plant model beyond a few tuned gains, and is what a commercial machine realistically ships with.

The cost is that the gains are tuned around one operating point. The orifice law makes command-to-velocity gain scale with the square root of the pressure margin, so it changes with load and with direction of travel, and the linkage lever arm changes with configuration. A gain set tuned at the stiff end of that range responds weakly at the compliant end; tuned at the compliant end it oscillates at the stiff end. Scheduling the gains over configuration and load recovers some of the range, but the schedule is machine and attachment specific and has to be remeasured whenever either changes.

2.3.2 Nonlinear Cancellation

Nonlinear model-based control removes the dependence on an operating point by cancelling the known dynamics rather than tuning around them [Slotine 91]. For a hydraulic actuator the cancellation has to span the orifice nonlinearity, chamber compressibility and the rigid-body load, which places the valve command several integrations away from the position error and makes backstepping the usual construction [Jelali 03]. The survey of Mattila et al. [Mattila 17] reports this line, together with virtual decomposition control, as the state of the art for heavy-duty hydraulic arms, and it does deliver accurate tracking on instrumented machines.

Cancellation is only as good as the model it inverts, and the terms it depends on are the terms that drift. Bulk modulus, leakage and seal friction change with temperature and wear over a working day. Some of the structure is non-smooth and cannot be cancelled at all: valve overlap produces a dead zone that has to be inverted explicitly, as Yang et al. [Yang 26] do alongside state constraints. The learned actuator models of Chapter 5 carry an explicit dead-zone term for the same reason. Maintaining this identification per machine, season and attachment is what makes the family expensive on a commercial implement [Mattila 17], not any weakness in the control law.

2.3.3 Online Parameter Estimation

Adaptive control estimates the uncertain parameters during operation instead of fixing them in advance [Åström 95]. Yao's adaptive robust control (ARC) pairs a projection-bounded parameter estimator with a robust feedback term, so that a guaranteed transient tracking

bound does not wait for the estimate to converge, and it is the established result for single-rod hydraulic actuators [Yao 00]. On the face of it this is the closest classical answer to the problem of this thesis, because it targets exactly the drifting parameters just described, and because it is the one classical family that ships a proof rather than an empirical trend, in direct contrast to the residual studied here (Section 6.7.7). ARC buys that guarantee by assuming a known parametric structure for the plant: the orifice, chamber and friction terms must be derived correctly in advance, with only their coefficients left uncertain. That is exactly the assumption Chapter 5 finds hard to hold on this machine, where the reduced-order model's own orifice and friction parameters settle at their optimisation clamps rather than at physically plausible values (Section 5.4).

Two properties of the task limit it. Parameter convergence requires persistent excitation, and mowing supplies smooth, low-bandwidth commands that do not provide it; the redundancy between command and state that Section 2.4.2 describes for learned models constrains a recursive estimator in the same way. The valve also saturates for much of the scored window on this machine (Section 5.6), where the regressor stops reflecting the applied input and adaptation has to be suspended or explicitly compensated [Sofrony 24]. Both limits belong to the task rather than to the estimator, so they carry over to the learned adaptation of this thesis. They are measured there (Section 5.6.2) rather than assumed away.

The scope of that argument should be stated plainly. No classical adaptive baseline is run in this thesis. The experiments of Chapter 7 compare actuator representations inside one reinforcement-learning stack, so nothing here shows that adaptive robust control tracks worse than the learned route on this machine. What is claimed is narrower: the two conditions the classical estimator needs, informative excitation and an unsaturated regressor, are not supplied by the mowing task, and the maintenance burden the family carries per machine, season and attachment is the burden the thesis is trying to remove.

The experiment that would settle the comparison is concrete rather than hypothetical. It needs a per-cylinder differential-pressure loop, since ARC's estimator is driven by the same pressure signal that Chapter 5 shows this simulation only produces as an uncalibrated model state (Appendix B.5); an open-loop identification campaign on the physical implement to fit the orifice and friction structure ARC assumes rather than the clamped values the reduced-order model settles on; a projection bound and adaptation gain tuned against that identified structure; and a second control implementation run on the same excitation schedule and evaluation gates as Chapter 7, so the two families are compared under one protocol rather than one against the literature's reported numbers for the other. None of that instrumentation exists on the target machine yet (Chapter 8), which is why the comparison is left open rather than run.

2.3.4 What Is Retained

The classical methods are not replaced wholesale. Kinematics is reliable and is kept: the DLS controller of Section 2.2 acts as teacher and as the data-collection controller. What is learned is the part the model-based families struggle to maintain: the mapping from cylinder command to realized motion, and the policy that issues that command (Section 2.7).

2.4 Data-Driven System Identification and Learned Dynamics

2.4.1 Learned Forward and Inverse Dynamics

Robot model learning distinguishes forward models, which predict the next state from state and command, from inverse models, which predict the command for a desired transition [Nguyen-Tuong 11]. Section 5.1 fixes both interfaces. One-step fit is insufficient when a forward model is recursively rolled out, because small errors compound [Nagabandi 18]. The models here are therefore trained with multi-step objectives. Probabilistic ensembles represent growing rollout uncertainty [Chua 18]; recurrent architectures instead retain unobserved pressure and valve history [Hochreiter 97, Cho 14], as used by the reduced-order model.

2.4.2 Closed-Loop Identifiability

Informative identification requires inputs that excite the plant independently of feedback [Ljung 99, Isermann 11]. In closed-loop data, command and state history are correlated; a flexible predictor can score well by ignoring the command, and validation from the same distribution will not reveal the shortcut [Schoukens 19].

That failure occurred here: a closed-loop forward model reached one-step $R^2 \approx 0.999$ yet showed almost no command dependence under ablation. Chapter 4 therefore uses open-loop valve excitation, and every learned model is checked with zeroed and shuffled commands. Prediction score alone is not treated as evidence of control authority.

2.5 Reduced-Order Modelling

Reduced-order models retain dominant behaviour at lower cost [Benner 15]. Here the full Simscape plant is reduced to per-cylinder extension, velocity and pressure difference. The construction is grey-box: known structure is embedded so that data fits what remains unknown [Karniadakis 21]. As in the structured dynamics network of Lutter et al. [Lutter 19], explicit physical layers improve interpretability. The hydraulic ROM retains the orifice, pressure and integration structure; learned terms represent valve realization, configuration-dependent load and bounded residuals.

Hwangbo et al. [Hwangbo 19] combined analytic rigid-body mechanics with a learned actuator response, a pattern later scaled in GPU-parallel simulation [Rudin 22]. Yamamori et al. [Yamamori 26] instead shaped the physical actuator to match the simulation. A commercial valve and pump cannot be redesigned in that way, so this thesis learns the hydraulic realization while PhysX retains the rigid-body mechanics (Figure 2.3).

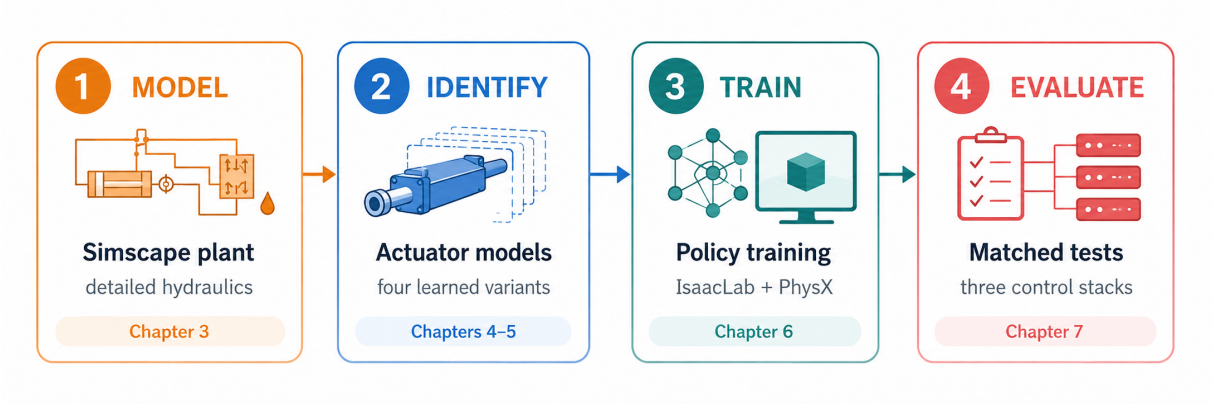

Figure 2.3: Workflow from the hydraulic plant model to matched evaluation. The separation between rigid-body physics and learned actuation follows the actuator-network approach in [Hwangbo 19]. This figure was generated using AI.

2.6 Domain Randomisation and Latent Adaptation

2.6.1 Robustness Through Randomised Training

A model trained on one simulated plant inherits its parameters as assumptions. Domain randomisation instead trains across a family, first for visual appearance [Tobin 17] and then for dynamics such as mass, friction, gain and latency [Peng 18]. The hydraulic family in Chapter 4 varies bulk modulus, valve gain, leakage, friction and payload, producing robustness to physically plausible plant variation.

2.6.2 From Robustness to Adaptation

Figure 2.4 contrasts a policy trained for one robust response across the plant family with an adaptive stack that conditions its response on the current plant.

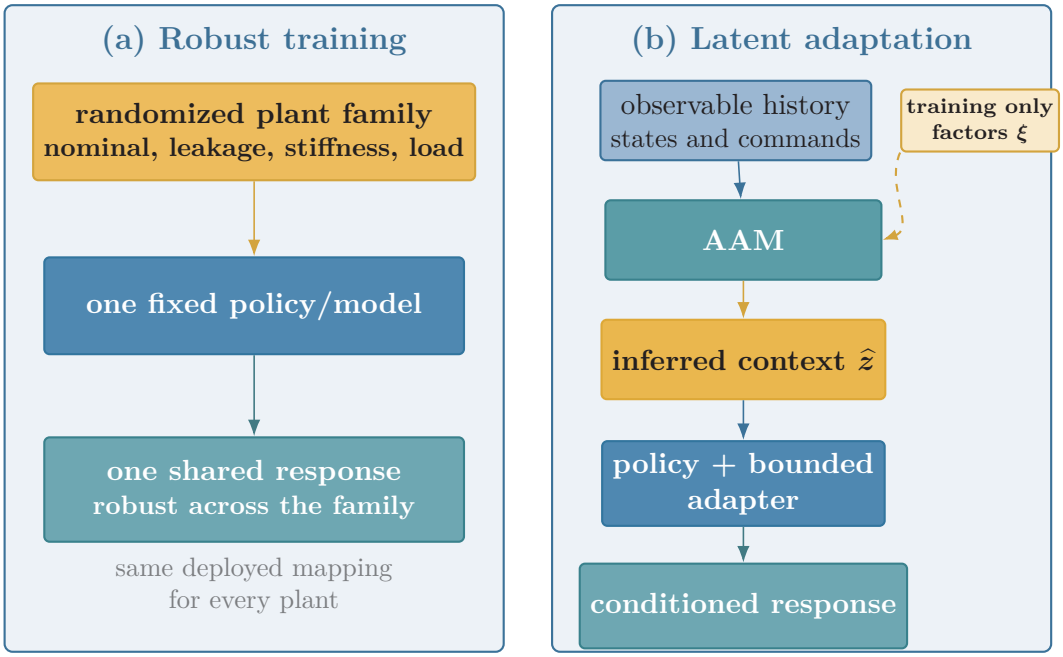

Figure 2.4: Canonical latent-adaptation architecture contrasted with robust training. The architecture motivates the offline identifiability study; the final AF implementation instead learns its context representation end to end from the control objective.

Teacher–student methods encode known simulation factors into a privileged latent and train an estimator to recover it from observable history. Lee et al. [Lee 20] used this construction for hidden terrain properties; Kumar et al. [Kumar 21] developed rapid motor adaptation. Both assume that the factor directions the estimator retains are the directions the controller needs. Liu et al. [Liu 25] report the failure when it does not hold, where a low-dimensional adapter drops factors that act through separate channels, and add training phases to recover them. Chapter 5 finds the same mismatch here in a sharper form, and the finding is quoted forward because it is why the deployed stack departs from this line: what the estimate carries is mainly seal condition, which acts through pressure, and pressure is the one channel the controller does not integrate (Section 5.5.1). Further adapter phases would not close that gap, because the loss is at delivery rather than at estimation.

Unlike classical online parameter estimation [Åström 95], the learned latent need not have a direct physical interpretation. It can be given one. Hu et al. [Hu 26] estimate named physical quantities on an excavator arm with Bayesian reinforcement learning, which buys interpretation at the price of the explicit plant model that Section 2.3 identified as the maintenance burden. The latent used here is anonymous and is therefore audited for content rather than read as parameters. A latent can also condition a dynamics model instead of a policy, as in the contextual neural ODE of Yu et al. [Yu 26]. That is the nearest relative of the route taken here, with one difference that decides the outcome: in

AF the conditioning enters the command path rather than the predictor, because a better prediction helps only if something downstream acts on it.

The offline AAM developed in Chapter 5 tests whether one second of observable history contains recoverable plant information. It is not the encoder deployed in AF, which trains its own history encoder end to end with the residual under the task objective rather than against an offline target (Section 6.7.5 details the training scheme). The two routes therefore answer different questions: the former tests identifiability; the latter tests control-relevant adaptation. The experiments also keep robust randomised training distinct from both routes.

Randomised training produces a single controller for a family of plants. Identification of that family is what makes a member-specific correction possible. In canonical latent adaptation, adaptation is the online use of that identification: the controller infers which plant is present from observable history and conditions the command on that estimate. The two responses are therefore sequential, not alternatives.

2.7 Reinforcement Learning and Structured Priors

2.7.1 Policy Optimisation

Reinforcement learning optimises a policy from observations to actions for expected cumulative reward [Sutton 18]. This thesis uses PPO [Schulman 17] with generalized advantage estimation [Schulman 16]; Section 6.2.5 gives the exact objective. GPU-parallel simulation supplies the required throughput [Makoviychuk 21, Rudin 22], which is why compact actuator surrogates replace the full hydraulic plant during training.

2.7.2 Structured Priors for Policy Learning

Learning from scratch would discard reliable kinematics. Residual RL instead adds a bounded learned correction to a controller at runtime [Johannink 19, Silver 18]. Policy distillation moves the prior to training: a privileged DLS teacher supplies demonstrations, then the student acts alone [Ross 11]. This thesis takes the second route. The teacher is distilled and PPO updates are anchored to the frozen student (Section 6.4), and the deployed actor remains the sole controller: no analytic controller runs underneath it at deployment, so the runtime correction of residual RL is not the construction used here.

One clarification belongs here, because the vocabulary collides. The adaptive stack of Chapter 6 does add a bounded term to its own frozen actor, but that term is not a residual in the sense of this subsection. There is no classical controller beneath it to correct, and its purpose is not to supply a prior that accelerates learning. It is an online adaptation mechanism internal to the reinforcement-learning policy, whose only function is to carry the dependence on inferred plant context that a frozen actor cannot express. It therefore belongs to the latent-adaptation line of Section 2.6.2 and is compared against that literature, not against residual policy learning.

The idea parallels real-world fine-tuning that retains a simulation-trained value function [Yin 25], but here the retained object is the reference policy. Anchoring was introduced because unconstrained PPO improved reward while degrading the physical tracking criterion (Section 6.3).

2.7.3 Reinforcement Learning on Hydraulic Machines

Deep RL has reached full-size hydraulic machines. What distinguishes those stacks from the one built here is not scale but the level at which the policy commands the machine, because that level decides how much of the hydraulic chain of Section 2.1.2 the policy has to represent at all.

Egli and Hutter [Egli 20] trained an excavator-arm policy in simulation and transferred it to a full-size walking excavator, then generalised the construction across excavation tasks [Egli 22]. The policy acts on joint-level velocity references, and a low-level tracking layer on the machine converts them into valve currents. Spinelli et al. [Spinelli 24] keep that division on a commercial material handler and add an underactuated tool, so the policy has to exploit passive tool dynamics rather than command them, while the actuated joints are still driven through the same kind of tracking layer. In both cases the orifice nonlinearity, the dead zone, the directional asymmetry and the load dependence live below the learned policy, in a layer that was identified and calibrated for that machine before the policy existed.

That division is sound where the tracking layer already exists. It is not available here for two reasons. The FME 500 is a series road-maintenance implement with no calibrated joint-velocity layer to inherit, and building one is precisely the per-machine identification that Section 2.3 identified as the maintenance cost of the model-based families. More importantly, a velocity interface hides the quantity this thesis sets out to measure. If the layer beneath the policy already owns the command-to-motion mapping, the question of whether a learned actuator retains command authority under saturation and dead zone cannot be posed at all; it has been answered by construction, once, by hand. The stacks here therefore command the actuator directly, either as a cylinder increment realized by a learned model or, in AF, as the valve command itself.

Maitan et al. [Maitan 26] take the opposite position and train on the physical cylinder, bounding exploration with a contraction-metric filter. That work does command the actuator at the low level, and training on hardware removes the transfer question outright. The result is force control on a single actuator on a test bench, and it does not obviously scale to a multi-link boom on a moving carrier, where the exploration the policy needs is the exploration the filter has to prevent.

These studies settle scale and feasibility. In each of them the actuator representation is a fixed design choice inside the stack rather than a variable, so none separates the transfer result from the representation that produced it. That separation is criterion V below.

None of these stacks varies the plant within a deployment. They are trained for robustness across one machine and its payload or soil conditions, and a plant that changes during a run is outside the construction. What remains open is the combination: a randomised plant family, online actuator adaptation, a privileged kinematic teacher, and a matched comparison of actuator representations under one protocol.

A policy can also compensate for a poor actuator model. Chapter 7 therefore reports rollout error, command magnitude and smoothness, and cylinder-level residuals alongside task tracking. Equal task error alone is not evidence of equal model quality.

2.8 Research Gap and Thesis Positioning

The literature leaves a specific combination unresolved. Hydraulic control provides structure but requires maintained plant models; learned dynamics replace derivation with data but depend on identifiable excitation; actuator surrogates and latent adaptation are established mainly for electric legged robots; and hydraulic RL studies do not isolate actuator representation under a matched protocol. Table 2.1 summarises this position.

The five criteria are: (I) learned hydraulic actuation, (II) data audited for causal input identifiability, (III) online adaptation to plant variation at deployment, (IV) a classical-control prior transferred through policy distillation, and (V) a matched actuator comparison.

What the table does and does not assert is worth fixing before it is read. The rows are lines of work, not individual papers, and the criteria are the ones this thesis needs rather than a neutral taxonomy of the field. Every criterion is individually established somewhere, several of them for decades, and single works cover more of a row than a coarse mark can show. The latent adaptation row in particular samples a large and active literature; it is read here for the assumption those constructions share, that the factor directions an estimator retains are the directions the controller needs, and not for coverage. The claim is the combination on one machine under one protocol, and Column III shows how narrow even that claim becomes once it is measured rather than designed for.

Table 2.1: Positioning against the reviewed lines of work. Parenthesized marks denote partial or implicit coverage. Column III is parenthesized for this thesis because the deployed route is history-conditioned residual control rather than the identified-latent adaptation of [Lee 20, Kumar 21], and because the plant-dependent part of its benefit is resolved in the lateral channel only (Section 7.5).

Line of work	I	II	III	IV	V
Model-based hydraulic control [Mattila 17, Jelali 03]	—	(✓)	(✓)	✓	—
Actuator networks for legged robots [Hwangbo 19, Rudin 22, Yamamori 26]	—	✓	—	—	—
RL for hydraulic machines [Egli 20, Egli 22, Spinelli 24, Maitan 26]	✓	(✓)	—	—	—
Latent online adaptation [Lee 20, Kumar 21, Yu 26, Liu 25, Hu 26]	—	—	✓	—	—
Residual RL and policy distillation [Johannink 19, Silver 18, Ross 11, Yin 25]	—	—	—	✓	—
This thesis (R / IF / AF)	✓	✓	(✓)	✓	✓

Column III carries a parenthesis, and the reason is specific. Causal recovery of the latent was established (Section 5.6.3). The closed-loop benefit of that latent was not, and on the inverse-model path it could not be, for the saturation reason developed in Section 6.7.1. What AF deploys is therefore not identified-latent adaptation but a bounded residual on the valve command, conditioned on one second of observable history. That residual improves task tracking against its own frozen base on four of five training seeds under the pre-registered criterion, so the median seed passes, and the direction survives the

alternative checkpoint rule. Capacity-matched controls then place most of that gain on the added residual capacity rather than on the live history: at equal capacity, history separates from zeros on one seed of five. A cutter-mass dose experiment resolves a lateral component that varies with the plant condition, although starting mass and the later change are coupled (Section 7.5). The correction is demonstrated; its attribution to online adaptation is not.

The contribution remains the complete experimental chain: identifiable cylinder data, physics-structured models fast enough for parallel training, online adaptation and a common evaluation protocol. Adaptation is reported as a conditional mechanism, with positive and negative training seeds kept in the same analysis. The three tested stacks are:

R. A command-conditioned ROM realizes the policy's desired cylinder motion. It is the reference and the only actuator calibrated to the deployment plant (Section 6.9).

IF. The same policy interface drives the frozen inverse model followed by the strict forward model, forming the robust learned baseline.

AF. The policy emits the valve command directly and the inverse model leaves the path. A causal history encoder conditions a bounded residual on that command, which the same forward model realizes ahead of PhysX.

All stacks share the task, terrain, trajectories, deployment factors and evaluation gates. R and IF additionally share the action contract and form the matched pair; AF does not, because its policy chooses a valve command rather than a cylinder increment, and it is therefore compared with its own frozen base rather than ranked against the others. The ROM is calibrated to the deployment plant, so causal architectural claims rely on the matched pair and on within-stack paired ablations rather than on a three-way contrast (Section 6.9.2). Chapter 3 now grounds this comparison in the concrete machine.

The remaining gap concerns the interaction between these established techniques: whether hydraulic actuators identified for control retain command authority after transfer, and under which plant conditions a history-conditioned correction on the identified valve path adds value beyond randomisation and matched residual capacity.

3. System Fundamentals and Problem Formulation

Chapter 2 established the control and modelling concepts used in this thesis. This chapter converts that background into a fixed system definition for the remaining work. It specifies the physical mechanism, coordinate mappings, available measurements, actuator models, task geometry, and evaluation interfaces before the controllers are compared. Keeping these definitions together prevents later chapters from changing the plant, observation set, or success criteria when interpreting results.

The physical machine and its two simulation representations are distinguished explicitly. Simscape is the high-fidelity hydraulic source used for actuator identification. Here, high fidelity is relative to the reduced actuator representations used for policy training. The Simscape model is a component-level hydraulic source model; this thesis does not claim that its complete hydraulic state, particularly absolute pressure, has been validated against measurements from the physical machine. IsaacLab/PhysX is the parallel environment used for policy training, adaptation-data collection, and closed-loop evaluation. Both use the cylinder ordering, kinematics, action contract, and cutter-tip task variables defined here, but they are not assumed to be identical plants.

Sections 3.1–3.3 define the mechanism, coordinate spaces, and signals. Sections 3.4 and 3.5 describe the two simulation models. The remaining sections define the verge-following task, the R, IF, and AF control stacks, the control objective, and the evaluation gates used in the experiments.

3.1 Manipulator Platform

The MULAG FME 500 is modeled as a serial-link mechanism with five revolute joints. Four hydraulic cylinders actuate the slew, two boom, and head-pivot joints. The *Messerkopf* rotation has no independent actuator and is locked at the mowing configuration; it is therefore not a controlled degree of freedom. The cylinder order used throughout the thesis is slew (*Drehzapfen*), boom I (*Ausleger I*), boom II (*Ausleger II*), and head pivot (*Messerkopf-Schwenk*).

Table 3.1 gives the cylinder stroke ranges used in the training data and the fixed limit of the *Messerkopf* joint. Task-space errors are measured at the cutter tip, which lies 0.79 m below the *Messerkopf* body origin.

Table 3.1: Cylinder stroke ranges in the training data (Chapter 4) and the fixed *Messerkopf* joint limit used for the mowing task.

i	Joint	Function	Limit
1	*Drehzapfen*	slew	0.018–0.493 m
2	*Ausleger I*	boom I	0.275–0.791 m
3	*Ausleger II*	boom II	0.309–0.857 m
4	*Messerkopf-Schwenk*	head pivot	0.269–0.689 m
5	*Messerkopf*	head rotation (locked)	$q_{5,\min} = q_{5,\max} = q_{5,\text{lock}}$

Figure 3.1 identifies the links, joints, and four actuator channels used throughout the model.

Figure 3.2 separates the rigid bodies that form the assembled mechanism. The catalog makes the L0–L5 body assignment explicit and shows the representative barrel–rod geometry used for cylinders C1–C4.

Manufacturer CAD supplies the link geometry, joint locations, masses, centers of mass, and inertia tensors through Phobos [von Szadkowski 20]. The same CAD-derived bodies and joint frames form the mechanical basis of the Simscape and IsaacLab models, keeping dimensions, axes, masses, and cylinder lever arms consistent while the hydraulic implementations remain model-specific.

3.2 Kinematic Structure and Coordinate Mappings

3.2.1 Joint Space and Cylinder Space

The configuration is $\boldsymbol{q} \in \mathbb{R}^5$; the actuated coordinates are the four cylinder extensions $\boldsymbol{x} \in \mathbb{R}^4$. Cylinder i joins CAD attachment points $\boldsymbol{p}_A$ and $\boldsymbol{p}_B$ on adjacent links. Its length and extension are

$$\ell_i(\boldsymbol{q}) = \Big\|\boldsymbol{p}_B(\boldsymbol{q}) - \boldsymbol{p}_A(\boldsymbol{q})\Big\|, \qquad x_i = \ell_i(\boldsymbol{q}) - \ell_{0,i}, \tag{3.1}$$

with $\ell_{0,i}$ the retracted reference length. Differentiating (3.1) gives the scalar mapping between joint motion and extension rate,

$$\dot{x}_i = J_{\text{cyl},i}(\boldsymbol{q})\,\dot{\boldsymbol{q}}, \qquad J_{\text{cyl},i} = \boldsymbol{e}_i^{\top}\Big(\boldsymbol{J}_B - \boldsymbol{J}_A\Big), \tag{3.2}$$

where $\boldsymbol{e}_i$ is the cylinder-axis unit vector and $\boldsymbol{J}_A$ and $\boldsymbol{J}_B$ are attachment-point Jacobians. Virtual work gives $\boldsymbol{\tau}_{\text{act}} = \boldsymbol{J}_{\text{cyl}}^{\top}\boldsymbol{F}_h$. The mapping varies with configuration and stroke.

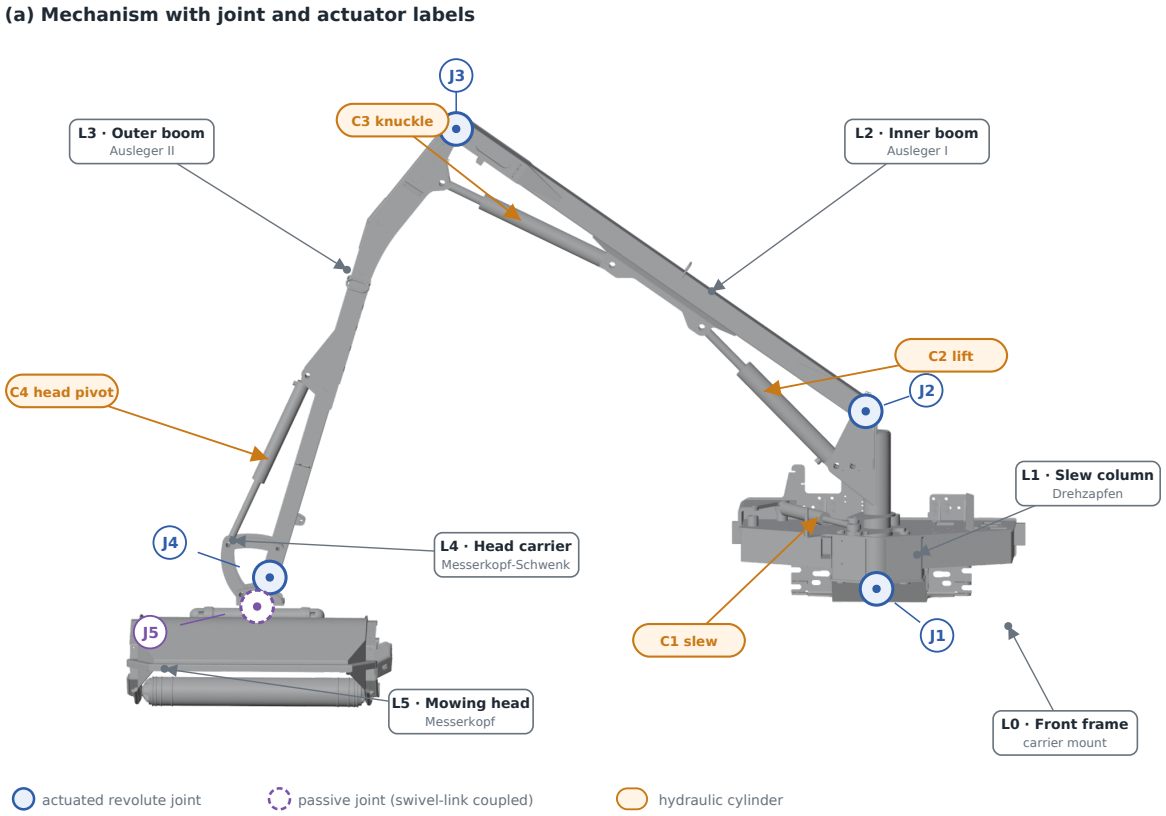

Figure 3.1: Annotated CAD view of the MULAG FME 500. Links L0–L5, joints J1–J5, and the four hydraulic cylinders C1–C4 are identified; J5 denotes the *Messerkopf* rotation, which is locked in the mowing configuration used in this thesis.

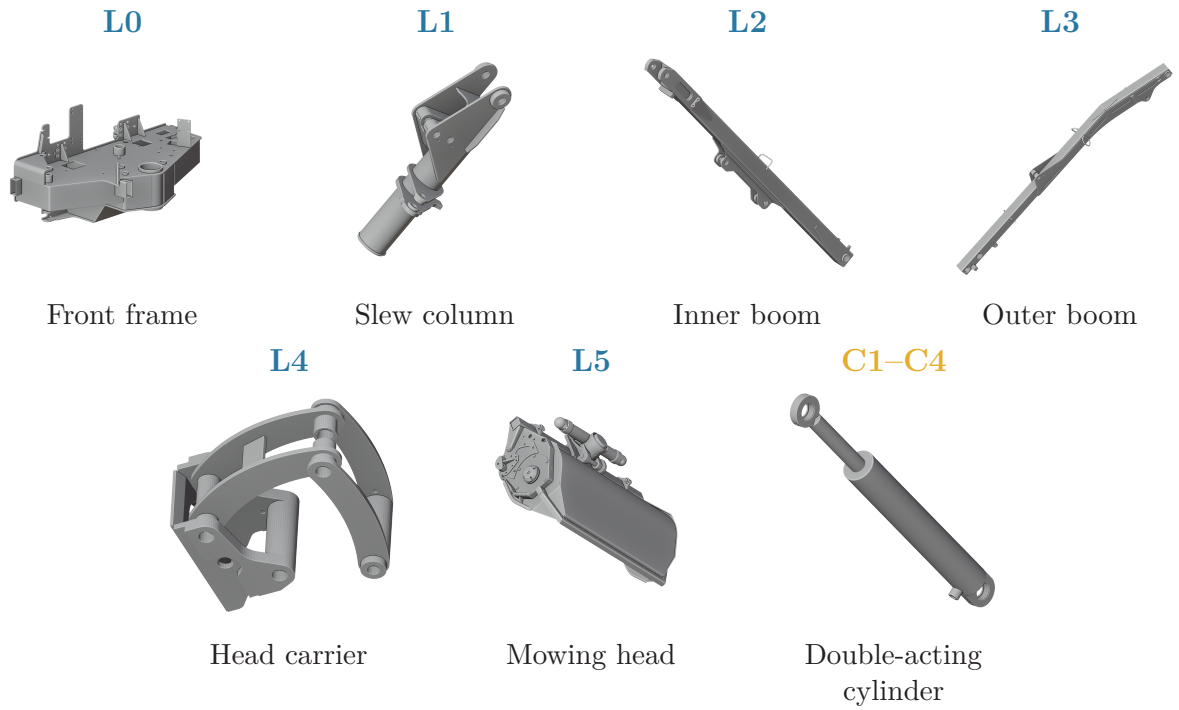

Figure 3.2: Individual CAD-derived bodies of the MULAG FME 500 shown at a common orthographic scale. L0–L5 are the front frame, slew column, inner boom, outer boom, head carrier, and mowing head. The C1–C4 panel shows the representative double-acting cylinder barrel–rod pair.

3.2.2 Task Space and the Calibrated Cutter Jacobian

The cutter tip is expressed in the truck chassis frame (`base_link`); using the moving slew-base frame would corrupt the tracking error. The controlled point includes the $-0.79\,\mathrm{m}$ tool-tip offset. Vehicle motion supplies the along-track direction.

The calibrated Jacobian $\boldsymbol{J}_c \in \mathbb{R}^{3\times4}$ maps cylinder extension rates to cutter-tip velocity. It was measured at the head-down mowing pose by stepping each cylinder target and recording the settled cutter displacement. Its singular values are 8.24, 5.69, and 2.79, giving rank 3. The damped least-squares controller in Chapter 6 uses this same Jacobian.

3.3 Actuation and Measured State

Table 3.2 lists the signals and their availability.

In IsaacLab, valve commands and differential pressures are learned-actuator states, not direct measurements. R produces them internally; IF obtains them from the inverse/forward pair, and AF from the frozen forward model alone. The physical factor vector $\mathbf{e}$ and privileged latent $\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$ are unavailable to every deployed controller. The estimated latent $\mathbf{z}_{\mathrm{AAM}}$ is used only in the offline identifiability and delivery studies of Chapter 5; it is not an input to final AF. Final AF instead receives causal observable history through its own end-to-end trained encoder.

Table 3.2: Signal inventory of the modeled system. "S" marks signals present in the Simscape data exports, "I" signals available live in the IsaacLab environment, "M" quantities that exist there only through the learned models of Chapter 5, and "P" *privileged* quantities that exist in simulation but must never enter a deployable policy or estimator input.

Signal	Dimension	Availability	Meaning
$\boldsymbol{u}$	$[-1,1]^4$	S, M	normalised proportional-valve commands
$\boldsymbol{x}$, $\dot{\boldsymbol{x}}$	$\mathbb{R}^4$	S, I	cylinder extensions and velocities
$\Delta\boldsymbol{p}$	$\mathbb{R}^4$	S, M	chamber pressure differences
$\boldsymbol{F}$	$\mathbb{R}^4$	S	logged rod forces
$\boldsymbol{q}$, $\dot{\boldsymbol{q}}$	$\mathbb{R}^5$	S, I	joint positions and velocities
$\boldsymbol{p}_c$	$\mathbb{R}^3$	S, I	cutter-tip position, truck frame
$\boldsymbol{h}$	$\mathbb{R}^9$	I	forward terrain-height profile ahead of the cutter
$\boldsymbol{c} = [\Delta\boldsymbol{x}^*, \Delta\dot{\boldsymbol{x}}^*]$	$\mathbb{R}^8$	I	desired cylinder command
$\mathbf{e}$	$\mathbb{R}^6$	P	physical randomisation factors of the active plant
$\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$	$\mathbb{R}^{d_z}$	P	privileged dynamics latent, $d_z = 2$
$\mathbf{z}_{\mathrm{AAM}}$	$\mathbb{R}^{d_z}$	M	offline diagnostic estimate of $\mathbf{z}_{\mathrm{SD}}$; not deployed in AF

3.3.1 Sensing and Observability

The deployable observation contains articulation state, terrain rays, contact signals, the desired cylinder command, and actuator-model pressure state. Plant parameters are not observed; AF infers their effect from causal history.

Figure 3.3 separates deployable measurements from privileged plant information. Articulation state is read from the physics engine without encoder noise or quantisation.

3.4 High-Fidelity Simscape Model

The actuator models use a multi-domain Simscape model of the FME 500 developed in the preceding master project. Table 3.3 distinguishes the data source of each frozen component.

Simscape supplies the frozen ID and strict FD checkpoints. The ROM receives an additional bounded IsaacLab calibration (Section 5.4.1). The adaptation models are trained from IsaacLab data collected through the ROM and PhysX.

3.4.1 The Command ROM as Deployment Actuator

The command ROM plus PhysX is the deployment plant. Simscape is too slow for policy training and bare PhysX has no hydraulics. The ROM is therefore both the plant model and the actuator of R:

- R uses the deployment plant itself and is the reference condition.
- IF replaces it with an independently identified inverse/forward pair.
- AF drops the inverse model, has the policy emit the valve command directly, and adds online adaptation on the forward model.

The R–IF gap measures the transfer cost of the independently identified pair; it is not a general ranking of the two model classes.

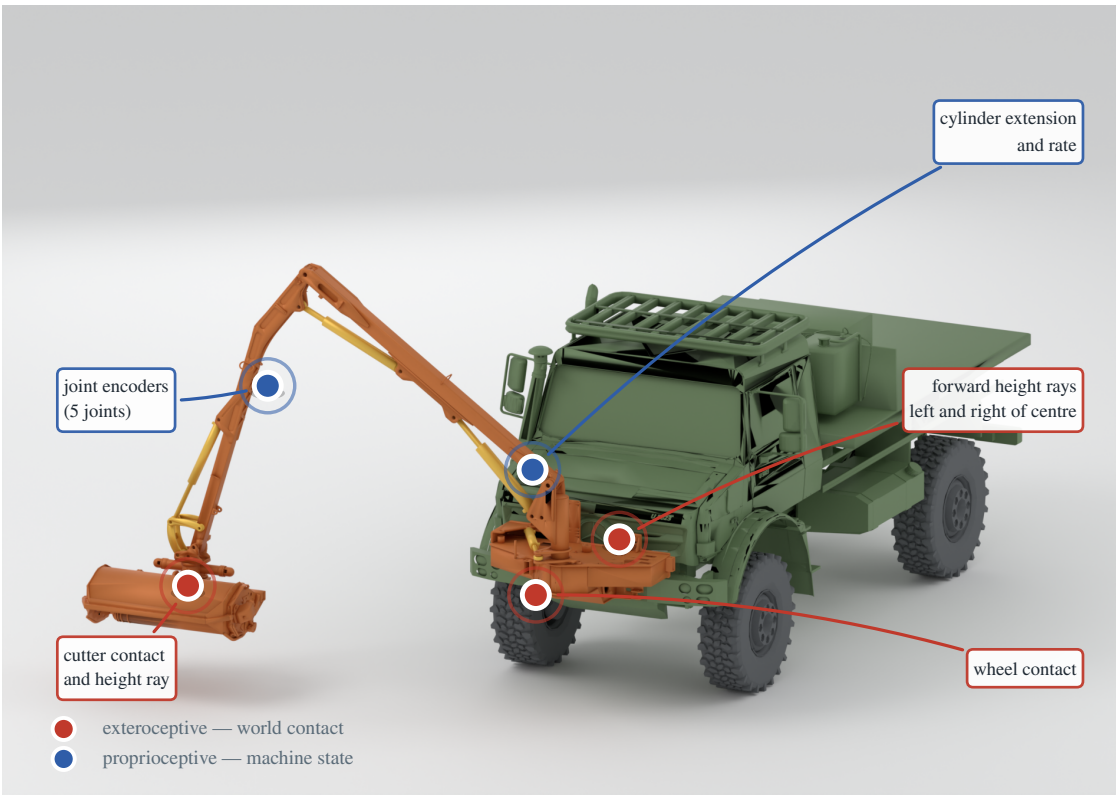

Figure 3.3: Sensor and observability map for the deployed controller. Blue callouts identify proprioceptive inputs: joint encoders and cylinder extension and rate. Red callouts identify exteroceptive inputs: forward terrain-height rays and cutter and wheel contact.

Table 3.3: Provenance of each frozen actuator component. The command ROM is the only one that has been calibrated against the deployment plant; the inverse/forward pair is Simscape-identified throughout. This asymmetry is a property of the comparison and is carried into the discussion of Chapter 7, not glossed over.

Component	Identified on	Calibrated on
Command ROM	Simscape closed-loop export (grey-box, full model)	IsaacLab/PhysX closed-loop rollouts, via a bounded command-gated correction head only
Control ID	Simscape closed-loop export	—
Strict FD	Simscape closed-loop export, retrained on the strict open-loop excitation export	—
SD_0, $\mu(\mathbf{e})$, SD_z, AAM	IsaacLab/PhysX adaptation collection (from scratch)	—

3.4.2 Mechanical Subsystem

The mechanism uses the CAD-derived links, joints, and cylinder attachment frames of Section 3.1. It is simulated in Simscape Multibody [The MathWorks, Inc. 24b]. The joint-space dynamics take the standard manipulator form

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{\tau}_{\text{fric}}(\dot{\boldsymbol{q}}) = \boldsymbol{\tau}_{\text{act}} - \boldsymbol{J}^{\top}(\boldsymbol{q})\,\boldsymbol{F}_{\text{ext}}, \tag{3.3}$$

with inertia $\boldsymbol{M}$ from the imported CAD. Friction combines viscous, Coulomb, and Stribeck terms. The cutting load scales with feed velocity and vegetation density. Each cylinder uses a spherical–prismatic–cylindrical loop and hard stops enforce the limits in Table 3.1.

3.4.3 Hydraulic Subsystem

Figure 3.4 shows four 4/3 valve-driven asymmetric cylinders fed by a variable-speed, fixed-displacement pump ($Q_p = D_p\omega$), accumulator, and pressure-relief valve.

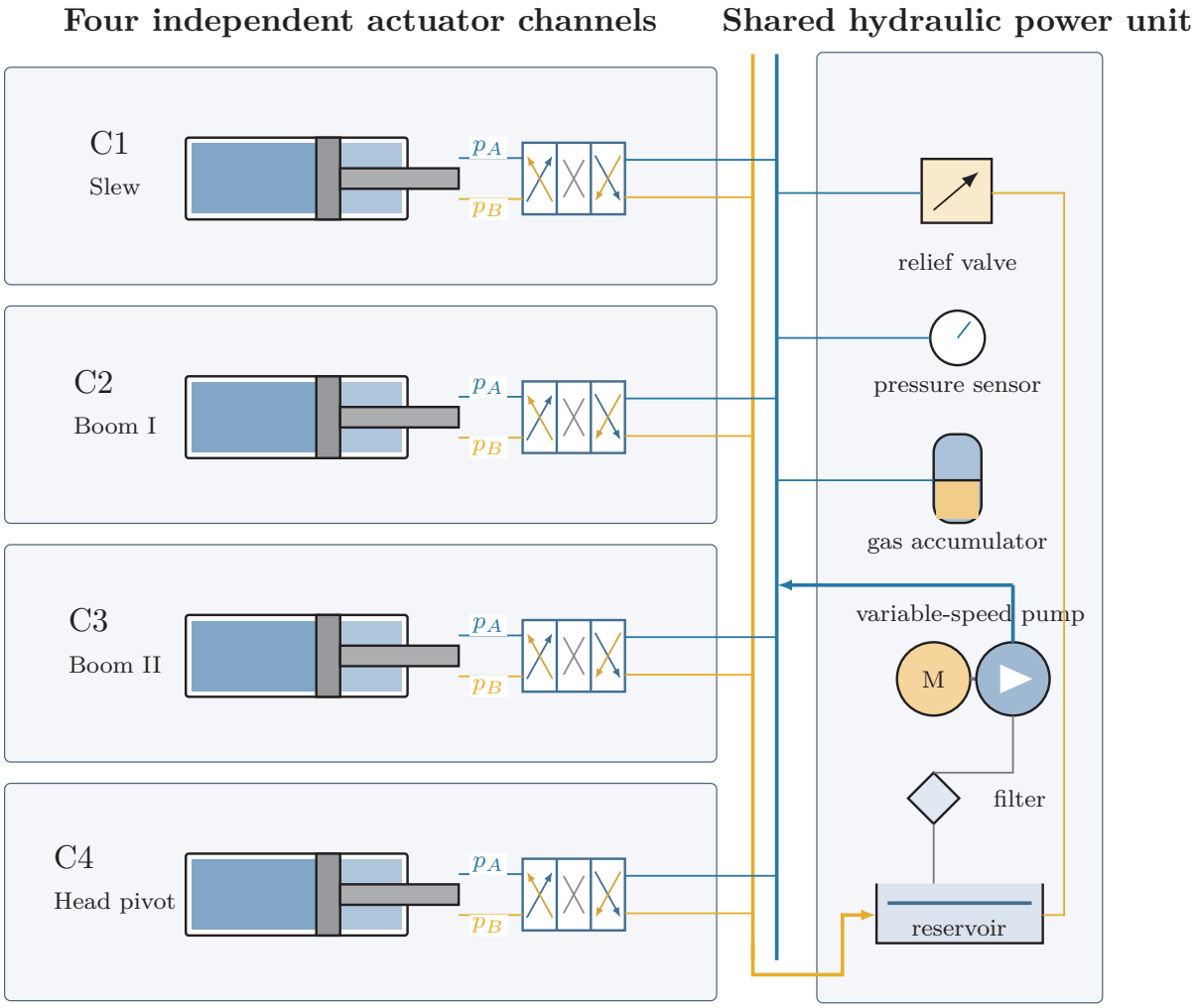

Figure 3.4: Hydraulic system schematic. Four proportional-valve cylinder channels share a variable-speed pump, reservoir, accumulator, sensor, and pressure-relief valve. Blue and orange rails denote supply and return.

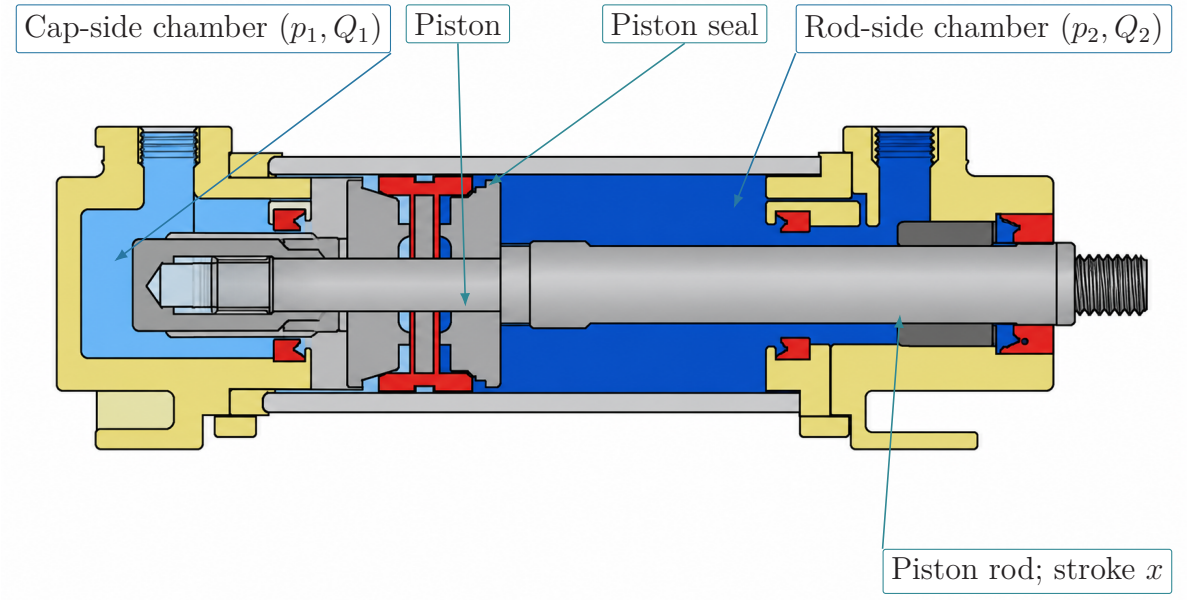

Figure 3.5: Annotated cutaway of an asymmetric double-acting hydraulic cylinder. The cap-side and rod-side chamber pressures generate the piston force and rod stroke. The labels are the states of (3.4), and the four cylinder channels of Figure 3.4 are of this type.

Figure 3.5 identifies the cylinder chambers, piston, seal, and rod used in the actuator model.

The 4/3 valve in Figure 3.4 meters flow to the two chambers of each asymmetric double-acting cylinder, cap side (p_1, Q_1) and rod side (p_2, Q_2). The chamber pressures follow

$$\dot{p}_1 = \frac{E_{\text{eff}}}{V_1(x)}\big(Q_1 - A_1\dot{x} - C_{\text{leak}}(p_1 - p_2)\big), \qquad \dot{p}_2 = \frac{E_{\text{eff}}}{V_2(x)}\big(-Q_2 + A_2\dot{x} + C_{\text{leak}}(p_1 - p_2)\big), \tag{3.4}$$

where $V_{1,2}(x)$ include dead volumes, $A_1 \neq A_2$, and C_{leak} is internal leakage. Entrained air is represented by

$$E_{\text{eff}}(p) = \left(\frac{1}{E_f} + \frac{\alpha}{p}\right)^{-1}, \tag{3.5}$$

Valve flow follows the orifice law of Section 2.1.2; spool dynamics are $\dot{x}_v = (u - x_v)/\tau_v$. The piston force is $F_h = p_1 A_1 - p_2 A_2 - F_{\text{fric}}(\dot{x})$. The coupled model is stiff and uses a variable-step implicit solver, which prevents RL-time simulation.

3.4.4 Validated Scope of the Model

Two checks fix what this model may be used for. The first is a parameter sensitivity study carried out in the preceding master project. Under a fixed integral-backstepping controller at 0.01 s, sweeping $\beta \in \{0.5, 1.0, 1.5, 2.0\}$ GPa moves slew-axis delay from about 0.85 s to 0.65 s, the system delay index from 0.45 s to 0.35 s, and task-space tracking error by about 11 %, with the improvement saturating beyond 1.5 GPa. The model responds to bulk modulus in the direction and with the saturation that the compressibility term of (3.5) predicts.

The second check asks whether the exported behaviour is what the configured plant should produce. It runs on the strict open-loop excitation export that identifies the forward model (Chapter 4): four runs, 25 600 rows on a 100 Hz grid, 93.7 % of rows valid. The reference is the collector's own declared parameter set, so the check tests the assembled model rather than the parameter file. Table 3.4 reports it.

Table 3.4: Behavioural checks on the strict open-loop excitation export against the parameters the collector configures. Values are pooled over the four runs; ranges span the four cylinders. The last two rows are the limits of the validated scope, not passes.

Property	Measured	Expected
Motion onset after command onset	within two log samples, $\leq$ 20 ms	the nominal 8 ms spool constant plus transport
Rise to 63 % of peak velocity	30 ms–80 ms	spool constant plus chamber fill
Velocity sign follows command	100 % above 0.05 m/s, $\geq$ 95.8 % above 0.01 m/s	no inversion once the cylinder moves
Speed against rising load at $\lvert u\rvert > 0.5$	falls by two orders of magnitude across $\lvert\Delta p\rvert$ terciles	orifice law: flow follows the pressure margin
Stroke visited	13 %–34 % of the configured stroke	bounded by the assembled linkage, not by the valve
$\lvert\Delta p\rvert$ ceiling	116 MPa–122 MPa, repeatable per channel	relief valve set at 27.6 MPa

Timing, direction and load dependence hold. Motion starts within two log samples of the command, velocity reaches most of its plateau inside 80 ms, the velocity sign never inverts against the command once the cylinder is actually moving, and speed collapses as the pressure the cylinder works against rises. That last property is the orifice law acting through the pressure margin, and it is the reason a fixed command gain does not describe this plant. These are the behaviours the learned actuator models have to reproduce, and the export has them.

Two properties are outside the validated scope. Stroke coverage is narrow: the excitation visits 13 % to 34 % of the configured cylinder stroke because the assembled linkage cannot reach the rest, and the head pivot is the narrowest channel. Chapter 4 carries that forward, and it is the channel the strict forward model predicts least well (Section 5.3).

Absolute pressure is not validated at all. Differential pressure does clip at a ceiling that repeats per channel, which is the signature of a relief valve, but the ceiling is 116 MPa to 122 MPa against a configured setting of 27.6 MPa, and about half of the valid rows sit above twice that setting. The pressure signal has the right shape and the wrong level, and the level is inherited from this source rather than introduced by the learned models. No result in this thesis is stated in absolute pressure units: every learned model consumes Δp through a fitted normaliser, and Appendix B.5 reads the deployed pressure state for shape only, for this reason.

Beyond these checks the model represents one machine and omits thermal dynamics. The logged rod-force channel is not carried into the learned actuator models: force is generated internally by the PhysX plant at deployment, and the logged channel also fails

the piston-area consistency audit (Section 5.4). Chapter 4 introduces variation through parameter randomisation.

3.5 IsaacLab Environment

Policy training, adaptation-data collection, and evaluation run in IsaacLab [Mittal 23] on PhysX. The USD articulation contains the Unimog carrier, five arm links, and four prismatic cylinder joints. Learned actuator models supply the hydraulic behaviour between the controller and PhysX cylinder targets.

3.5.1 Simulation and Control Rates

Four clocks run concurrently. PhysX integrates at 120 Hz and the policy acts at 60 Hz under a decimation of two. The learned actuator models and the AAM update every three policy steps, on the 20 Hz grid at which the identification data were sampled, and their prediction is held across the intervening PhysX updates. Data collection uses that same 50 ms grid. "The control frequency" is therefore not a well-defined quantity for this system and no scalar rate is reported for it. Table B.1 summarises these rates and distinguishes the offline AAM timing from the deployed AF history path.

3.5.2 Environment Configuration Details

The arm root is spawned at $z = 1.26\,\mathrm{m}$; ground-height spawning buries the vehicle. Cylinder targets are integrated in reference space so gravity does not erase commanded increments. Vehicle motion is scripted because the authored tyres and suspension do not provide reliable PhysX traction; a planar fixture holds lateral position and heading.

3.6 The Verge-Following Task

3.6.1 Task Definition

The task is moving verge following. The truck drives a 26 m strip at a speed sampled from 2 to 7 km/h; development uses 4 km/h. Episodes allow 2808 steps (46.8 s), while scoring stops when the strip is completed. The arm must keep the cutter at the target height, on the verge centre line, in contact with the ground, and within its travel limits. Terrain height is randomised per environment (Figure 3.6).

Eleven reference-path families span straight, periodic, spline, sweep, and lane-change motion. Results use the worst path family and terrain mesh rather than a pooled score. A regression test verifies that all eleven families are distinct.

3.6.2 Task Feasibility and Plant Capability

The action contract limits cylinders 1–3 to 4 mm per policy step, which bounds lateral cutter speed at

$$v_{\mathrm{lat}}^{\max} = \frac{4\,\mathrm{mm}}{1/60\,\mathrm{s}} = 0.240\,\mathrm{m/s}. \tag{3.6}$$

Figure 3.6: IsaacLab realization of the moving verge-following task. The scripted truck travels on the level road while the four-cylinder arm places the cutter above the procedurally generated verge and follows the white reference path. Randomised grass, bushes, soil patches, twigs, and logs form a non-colliding visual layer; the underlying verge mesh supplies the terrain disturbance and height-sensing geometry used by the controller.

This uses the 1/60 s policy interval. Early path settings exceeded this limit; the worst case (`boom_sweep`) required 3.7× the available rate. Path and terrain parameters are now derived from (3.6), and a preflight check rejects infeasible tasks.

Equation (3.6) is a property of the cylinder-increment contract, so it describes the action chains of R and IF directly. AF bounds its command in valve space instead, and the same arithmetic does not apply to it. The paths and terrain derived from the ceiling are nonetheless shared by all three stacks, so what the task *demands* is identical in every case; what differs is only the space in which the request is bounded.

3.7 Control Architecture

The vehicle advances at the scripted speed and the reference generator supplies the verge centre line and cut height. The policy maps normalised observations to four bounded actions and has no runtime baseline controller. In R and IF those actions are desired cylinder increments and the action contract bounds them as such; R uses the command ROM and IF the frozen inverse/forward pair to realise them. AF instead emits the valve command directly and realises it through the frozen forward model, adding a bounded residual conditioned on causal history. All paths set the same PhysX cylinder target. The DLS controller is used only for teaching and identification.

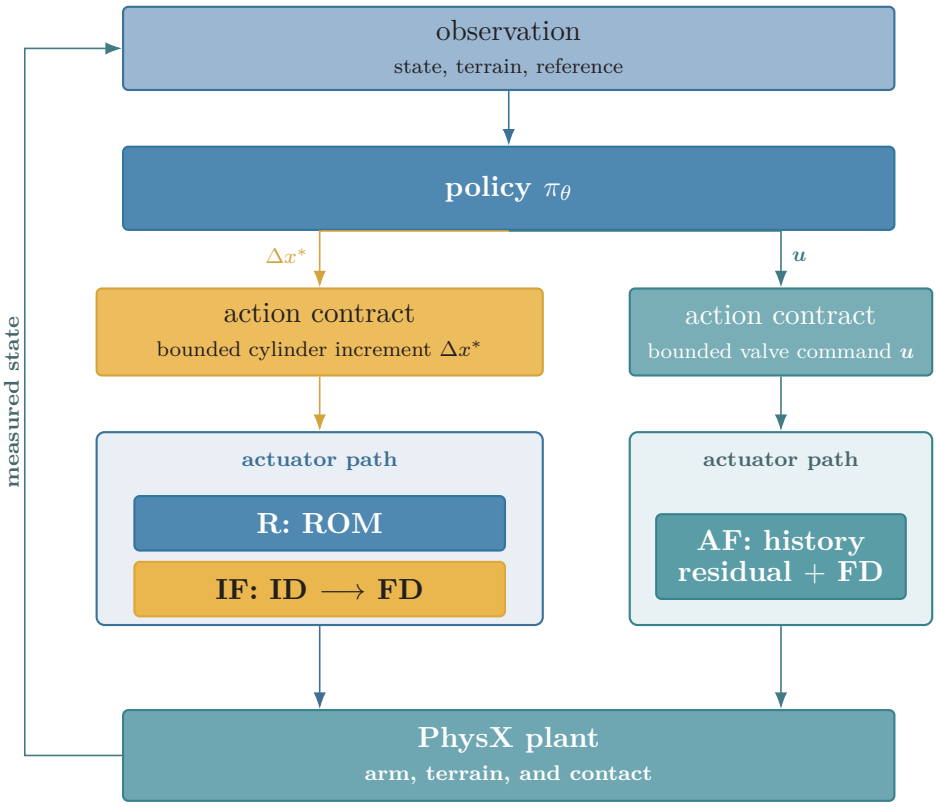

Figure 3.7: Deployed control stacks. R and IF share one action contract, a bounded cylinder increment, and differ only in how they realise it. AF does not share it: the policy emits the valve command itself and a bounded residual conditioned on causal history acts on that command before the frozen forward model. All three set the same PhysX cylinder target, and measured state closes the loop. The differing contract is why AF is compared within itself rather than against the other two (Section 7.1.2).

3.8 Problem Formulation

The reference is the verge centre line $y_{\text{ref}}(t)$ and cut height h_{ref}, expressed in the truck frame. With cutter-tip position $\boldsymbol{p}_c(t)$ and terrain height $g(\cdot)$, the scored errors are

$$e_{\text{lat}}(t) = p_{c,y}(t) - y_{\text{ref}}(t), \qquad e_{\text{cut}}(t) = \Big(p_{c,z}(t) - g(\boldsymbol{p}_c(t))\Big) - h_{\text{ref}}. \tag{3.7}$$

The objective is to minimize the mean and tail of both errors across path families and terrain while maintaining contact, head alignment, and smooth motion. Learned models are evaluated on one-step and recursive cylinder predictions. The stacks are tested across the randomised plant family. R and IF are compared with each other on held-out factors, and AF is compared with its own frozen base under a plant change, for the reasons given in Section 6.9.2.

Constraints are cylinder limits and headroom (Table 3.1), the action contract, checkpoint output limits, and the head-down condition.

3.9 System-Level Evaluation Criteria

Controllers and models are evaluated separately. Controller scores can hide actuator-model errors through policy compensation.

The system-level criteria cover three aspects of operation. Cut-height and lateral errors measure task accuracy. Contact, head alignment, impact rate, and cutter acceleration describe tool–terrain interaction and motion quality. Cylinder headroom and safety terminations check whether the task is completed without driving the mechanism into its limits. The metrics are evaluated over the scored part of each episode and are kept in physical units where possible. This makes a failed gate traceable to a specific limitation rather than to a single combined reward value.

3.9.1 The Eight System-Level Evaluation Gates

Table 3.5 defines the eight physical gates used throughout this thesis. Their role is to supply a criterion that is *not* the reward. A policy is optimised against the scalar return of Section 6.2.5, but it is selected and promoted against this vector of physical quantities, and Section 6.3.2 shows that the two point in measurably different directions. The gates are therefore an instrument for checkpoint selection and for the block-ratchet promotion of Section 6.4, and that is how every result in this thesis uses them. They are used to rank and to promote candidates, not to certify a stack as accepted, and no claim of certified acceptance is made anywhere in this work.

All controller stacks are evaluated with the same path, terrain, and randomisation conditions. A candidate is preferred when it satisfies more of the gates at a wider margin; strong mean tracking does not compensate for loss of contact, inadequate cylinder headroom, excessive impacts, or safety terminations. Worst-case aggregation prevents easy terrain or path cases from masking a local failure. Mean absolute errors quantify sustained tracking performance, while the lower and upper percentiles capture short alignment, clearance, or acceleration events that a mean can hide. Contact and termination measures are reported as fractions so that episodes of different completed length remain comparable.

Table 3.5: Eight system-level evaluation gates. Each gate uses at least three terrain meshes and the worst mesh; tracking-critical gates also use the worst path family. The last column states where each number comes from, because a threshold read off an earlier baseline does not carry the same weight as one fixed by the task or by the machine.

Gate	Metric	Threshold	Aggregation	Origin
Cut height	mean $\lvert e_{\mathrm{cut}} \rvert$	$\leq 0.040\,\mathrm{m}$	worst mesh	mowing spec
Lateral tracking	mean $\lvert e_{\mathrm{lat}} \rvert$	$\leq 0.300\,\mathrm{m}$	worst mesh, worst family	earlier baseline
Ground contact	contact fraction	≥ 0.700	worst mesh, worst family	mowing spec
Head alignment	p01 head-down alignment	≥ 0.900	worst mesh	tool geometry
Hard impacts	hard-impact step fraction	≤ 0.350	worst mesh	wear tolerance
Tool acceleration	p95 cutter acceleration	$\leq 120\,\mathrm{m/s^2}$	worst mesh	measured floor $\times 1.2$
Cylinder headroom	p01 travel margin	$\geq 0.005\,\mathrm{m}$	worst mesh	machine limit
Safety terminations	termination rate	≤ 0.080	worst mesh	engineering tolerance

The origins are not equivalent, and the difference decides how much each gate can be asked to do. Cut height and ground contact come from what verge mowing requires of the tool. Head alignment follows from the cutting head's geometry, which must stay down to cut at all. Cylinder headroom is a machine limit: the arm must not drive into its own travel stops. Tool acceleration and safety terminations are engineering tolerances, the first set at roughly 1.2 times the acceleration floor measured on a working stack, the second at the rate of raw envelope violations judged acceptable per control step.

The lateral gate is the exception and is treated as one throughout. Its 0.300 m was anchored on an earlier and narrower trajectory library, and once that library was widened the number stopped describing the task and started describing whichever family the widening made hardest. It still ranks candidates usefully, because every candidate meets the same number, but it is not a task requirement and no claim in this thesis rests on passing it. Section 7.2 shows both stacks failing it on a single family each, and a different family in each case, which is what a stale threshold looks like.

3.9.2 Cylinder and Model Checks

Cylinder checks use fresh ± 10 mm commands, coordinated 80 mm motions, return error, and saturation. Model checks use one-step and rollout accuracy, input ablation, and Jacobian conditioning (Chapter 5). Chapter 7 reports these with the evaluation contract. The small command checks local response and deadband; the coordinated motion checks coupling between actuator channels. Return error tests reversibility, and saturation checks whether the learned model respects the command and stroke limits. These diagnostics are interpreted alongside the system gates, not as substitutes for closed-loop task performance.

The system is now specified. The next chapter turns to the data the learned models are built from.

4. Data Generation and Preparation

Chapter 3 defined two plant representations: detailed hydraulics in Simscape and fast rigid-body dynamics in IsaacLab. This chapter documents the datasets that connect them, including collection, auditing, feature construction and splitting. Table 3.3 records their provenance.

Simscape campaigns. Closed-loop runs provide realistic operating data for the inverse and reduced-order models. Separate open-loop valve excitation provides the causal command information needed by the strict forward model (Section 4.4).

ROM calibration. After Simscape training, the command ROM was fine-tuned on five closed-loop IsaacLab reference shapes. Only its bounded correction head remained trainable, with Simscape replay retained (Section 4.6). The ID–FD pair was not calibrated in IsaacLab.

IsaacLab deployment data. The system-dynamics surrogate, the privileged encoder and the adaptation module use two 118-run collections recorded in the deployed command ROM–PhysX plant, including dedicated PRBS windows (Section 4.7). No earlier adaptation or system-dynamics model was reused. Figure 4.1 summarises the common preparation workflow.

4.1 Dataset Objectives and Design Requirements

Four requirements governed collection and preparation.

R1: realistic operating distribution. The learned models are deployed inside a tracking controller, so they must be accurate in the states that controller visits, namely coordinated multi-axis motion, gravity-loaded poses and moderate speeds.

R2: causal identifiability of the valve. A forward model is only useful to a policy if it responds to the command. This requires that the valve signal in the training data carry information that the state history does not already carry, which is a property of the *experiment*, not of the network [Ljung 99, Isermann 11].

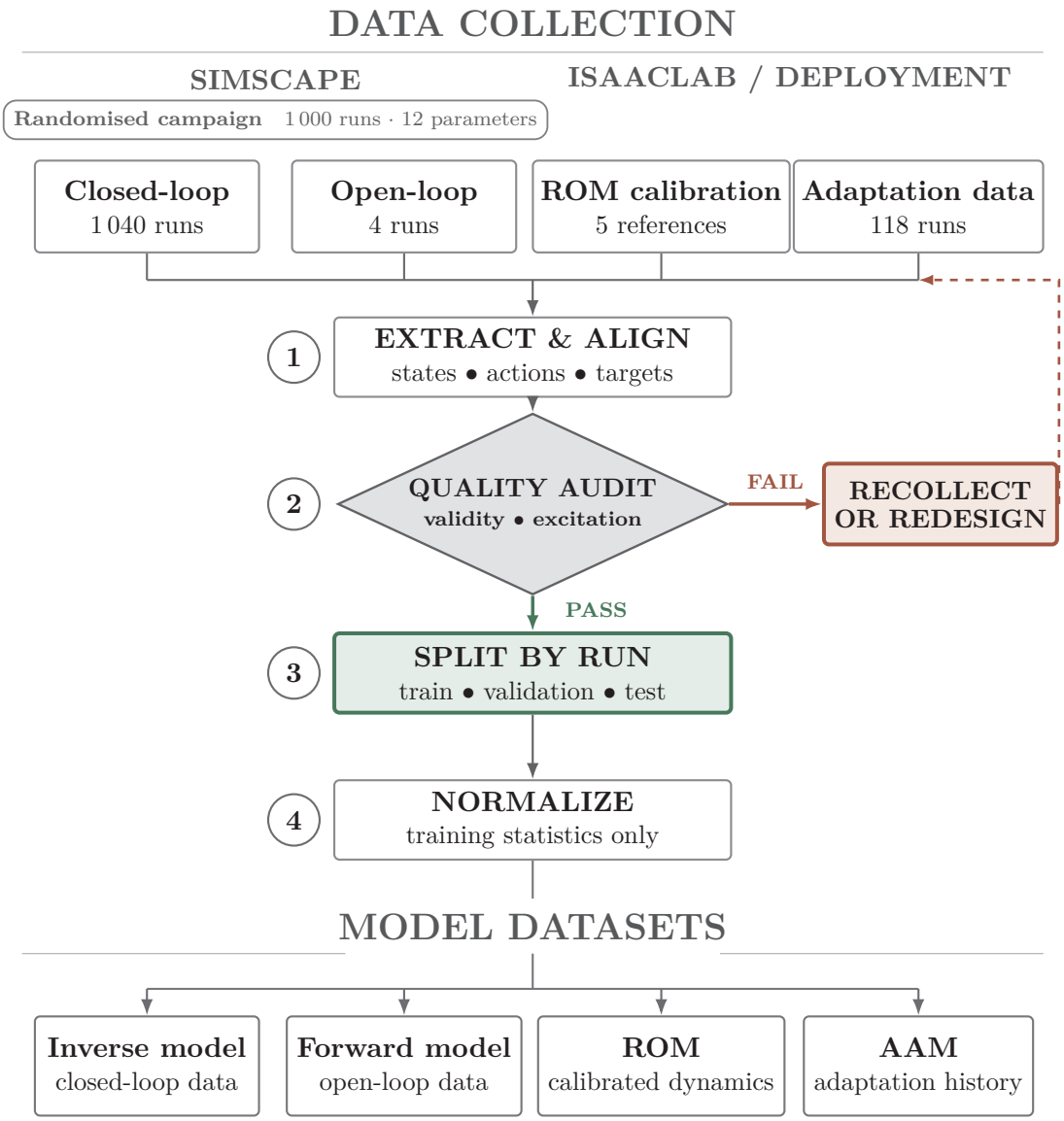

Figure 4.1: Data-preparation workflow. Two Simscape collections and two IsaacLab collections pass through common extraction, audit, run-level splitting and training-statistics normalisation stages. The randomised Simscape campaign is provenance metadata rather than a pipeline stage. A failed audit returns the experiment to collection; accepted material forms the inverse, forward, ROM and AAM datasets.

R3: variation across a plant family. Robust training and latent adaptation both need many plants rather than one. The variation must be physically meaningful, must demonstrably change the measured response, and must be recorded per run, because the privileged plant context of Section 4.11 is computed from those factors.

R4: clean separation. Train, validation and test material must be separable at the level of complete runs, so that no transition window straddles a split boundary and no run contributes to two splits.

R1 and R2 require different experiments. Closed-loop control gives realistic motion but correlates command with state; open-loop excitation breaks that correlation but is less representative of deployment. The two campaigns are therefore retained and combined as described in Section 4.5.

4.2 Randomised Parameter Space of the Simscape Campaign

This section and the next record the Simscape randomisation design and the procedure that executed it. No reported model is trained on this curriculum. It is retained as collection provenance and because its compliance analysis (Section 4.2.3) informed the IsaacLab factor design. The export the actuator models are actually fitted on is a separate closed-loop collection on the nominal plant, described in Section 4.5. Readers concerned only with trained models may continue at Section 4.4.

4.2.1 Sampling Design

The closed-loop campaign draws its plants from a twelve-dimensional parameter space using Latin hypercube sampling [McKay 79], which spreads the draws more evenly over each marginal than independent uniform sampling would at this sample size. The twelve coordinates are fixed once for the whole batch, before any simulation runs, so the parameter table is reproducible and can be audited independently of the data it produced.

The batch is organised as a curriculum rather than a single homogeneous pool. Run 1 is the untouched nominal plant, runs 2–251 randomise only the hydraulics, runs 252–501 only the mechanics, and runs 502–1000 both together (Figure 4.2(a)). This makes it possible to attribute a later modelling failure to the hydraulic or the mechanical axis by retraining on one block at a time.

4.2.2 Hydraulic Parameters

Table A.1 lists the hydraulic axes. They were chosen to correspond to the physical mechanisms that make one machine differ from another in service, in the sense discussed in Section 2.1.2: air works its way into the fluid, spools and seals wear, and relief settings drift.

The seal-wear coordinate is the one place where a single sampled number moves several model parameters together. A worn cylinder does not become leakier without also becoming slower, so $w_{\mathrm{seal},i}$ scales the piston clearance linearly, the valve leakage area with the cube

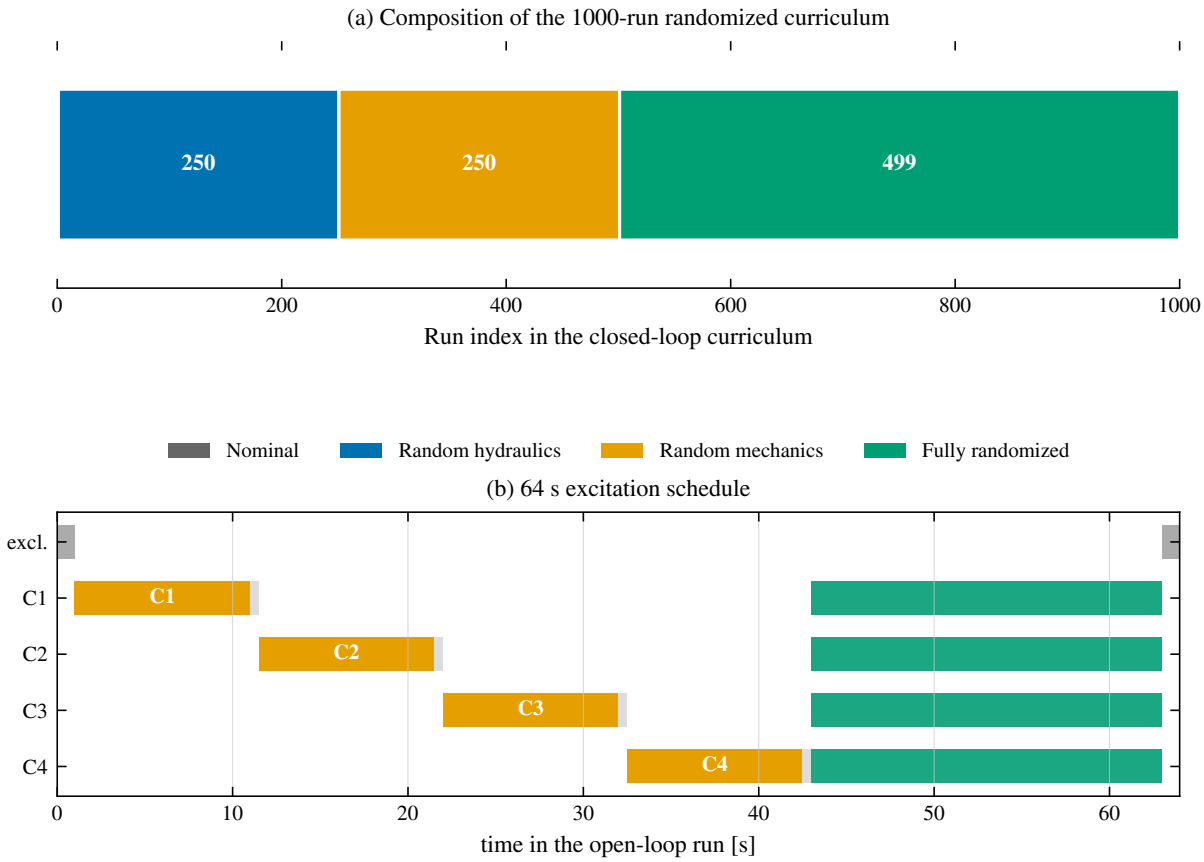

Figure 4.2: Design of the two collection campaigns. (a) Composition of the closed-loop curriculum: one nominal run, then 250 hydraulics-only, 250 mechanics-only and 499 fully randomised draws. (b) Timeline of one open-loop excitation run.

of the same factor, and the spool time constant from 8 ms to about 13 ms at the top of the range. Sampling those three independently would have produced physically incoherent machines, some very leaky and yet perfectly fast.

4.2.3 The Compressibility Axis

The entrained-air and dead-volume coordinates randomise the chamber dead volume, which spans 505 to 579 mL across the batch. This reaches the pressure dynamics through the continuity equation (3.4), which carries the chamber volume: a larger chamber is more compliant and slower, a smaller one stiffer and faster. The compressibility axis therefore enters the batch as capacitance variation.

4.2.4 Mechanical Parameters

The mechanical axes of Table A.2 are applied to all fourteen rigid bodies, namely the six links, the four cylinder barrels and the four pistons. Per body, mass is perturbed by up to $\pm 20\,\%$, the centre of mass by up to $\pm 5\,\%$ of its nominal offset, and the inertia tensor with its products of inertia consistently with the mass change, so that the body remains physically realisable.

Algorithm 4.1: Closed-loop curriculum generation.

input : run budget $N = 1000$; joint limits $\boldsymbol{q}_{\min}, \boldsymbol{q}_{\max}$; nominal parameters $\boldsymbol{\theta}_0$
output : dataset $\mathcal{D} = \{(\text{trajectory}_i, \boldsymbol{\xi}_i)\}_{i=1}^{N}$

$\mathbf{L} \leftarrow \textsc{LatinHypercube}(N, 12)$ `// one design for the whole batch`
for $i \leftarrow 1$ **to** N **do**
  assign i to a curriculum block (nominal / hydraulic / mechanical / full)
  $\boldsymbol{\xi}_i \leftarrow \textsc{DrawParameters}(\mathbf{L}_i, \text{block})$ `//` $\varepsilon_{\text{air}}, \varepsilon_{\text{vol}}, c_d, w_{\text{seal}}, p_{\text{set}}$`; masses, CoM, inertia, stiffness`
end
for $i \leftarrow 1$ **to** N **do**
  inject $\boldsymbol{\xi}_i$ into the Simscape model
  $n_w \leftarrow \textsc{RandInt}(12, 20)$; $\boldsymbol{W} \leftarrow \textsc{LatinHypercube}(n_w, \text{JDOF})$ within $[\boldsymbol{q}_{\min}, \boldsymbol{q}_{\max}]$
  assign each segment a speed class (slow / medium / fast / variable); **no** dwell
  $\boldsymbol{q}_{\text{ref}}(t) \leftarrow \textsc{Interpolate}(\boldsymbol{W})$ over 22 s
  simulate with `ode23t`; log on the 0.05 s grid
  store $(\boldsymbol{u}, \boldsymbol{x}, \dot{\boldsymbol{x}}, \Delta\boldsymbol{p}, \boldsymbol{F}, \boldsymbol{q}, \dot{\boldsymbol{q}}, \boldsymbol{q}_{\text{ref}}, \boldsymbol{\xi}_i)$
  if $i \bmod 100 = 0$ **then** checkpoint $\mathcal{D}$ to disk
end

The joint stiffness and damping terms model compliance in the bearings and linkage rather than in the fluid. They matter here because they set how much of a pressure transient appears as boom oscillation instead of clean motion, which is the coupling that a purely kinematic controller cannot anticipate.

4.3 Closed-Loop Data Collection

Algorithm 4.1 summarises the procedure that produced the curriculum of the preceding section, which no reported result uses. The parameter table is fixed before simulation, and each run receives a new reference path.

4.3.1 The Controller as a Data-Collection Instrument

Every closed-loop run uses the same integral-backstepping controller at 0.005 s. Its gains remain fixed, so response differences are caused by the plant draw rather than controller retuning.

Joint 2 uses PI control with velocity damping and gravity bias because it carries the boom load. A barrier term enforces the URDF limits. Per-axis command limits are [0.95, 0.95, 0.80, 0.80, 0.90]; the first four are also stored in the inverse-dynamics checkpoint (Section 5.2).

4.4 Open-Loop Excitation Experiments

4.4.1 Limitations of the Closed-Loop Batch

The first forward model achieved one-step $R^2 \approx 0.999$ on closed-loop data but was almost insensitive to valve input: zeroing the input changed long-horizon error by only a few

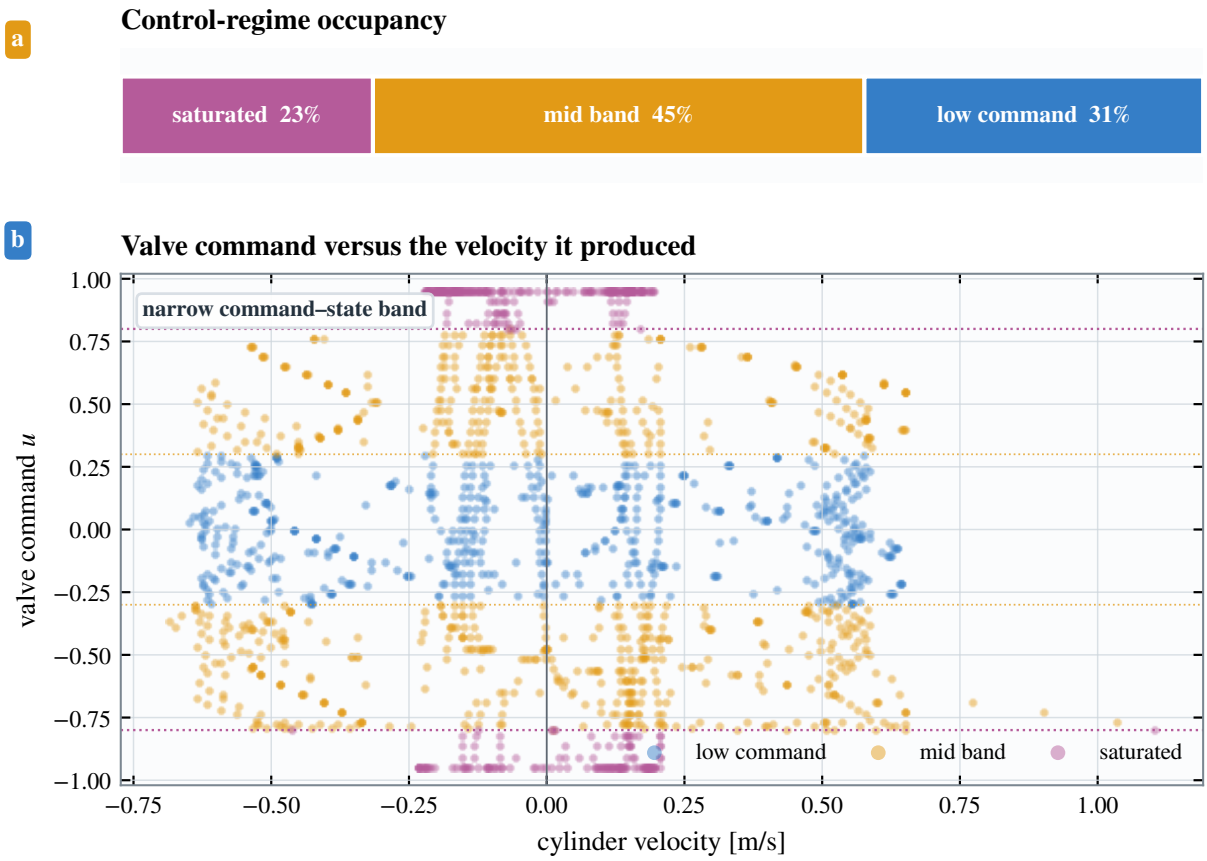

Figure 4.3: The closed-loop batch is diverse in control effort but not identifiable. (a) Fraction of time the valve spends saturated, in the mid band and in low-command creep. (b) The same run in the command–state plane, where the correlation that blocks identification is visible.

percent (Section 5.3.1). In closed loop, command is largely determined by the state history already given to the model, so the data do not separate command causality from state correlation [Ljung 99].

Figure 4.3 confirms that coverage alone is not the issue. The valve is saturated for about one quarter of a pooled run, operates at mid-authority for roughly half, and otherwise remains in low-command creep. Despite this spread, command and cylinder velocity remain concentrated along a narrow band. The data are diverse in control effort but not identifiable, motivating a separate open-loop experiment.

4.4.2 Design of the Excitation Signal

The excitation run lasts 64 s and follows the fixed schedule in Figure 4.2(c). One second of startup is discarded, because those samples describe the initialisation rather than the dynamics. Four sequential slots follow, each giving one cylinder ten seconds of excitation followed by a half-second zero-command gap that makes the segment boundaries unambiguous. A twenty-second phase then drives all four channels at once, and a final second of zero command closes the run. Commands are limited to ± 0.90 of full authority.

An initial rapidly switching signal met the formal persistence condition but kept each piston near one stroke position. The final design instead uses long full-authority plateaus

in the sequential phase and slow, high-authority motion in the simultaneous phase. A slow near-square cycle is combined with distinct medium-frequency channel components, with raised-cosine entry and exit ramps. This provides both input excitation and useful stroke coverage.

4.4.3 Plant Draws and Acceptance

The same excitation is applied to four plant draws from the parameter space of the closed-loop campaign: nominal, randomised hydraulics, randomised mechanics and fully randomised. Holding the input fixed and varying only the plant makes the four runs directly comparable.

Each run is checked for stroke coverage before acceptance, the collector requiring at least 10 % of available stroke on cylinders 1–3 and 8 % on cylinder 4. The retained runs cover 33 % of the stroke on cylinder 1 and 12 % on cylinder 4, the narrowest of the four. Every channel clears the floor, and Section 4.10 confirms the material is informative at these coverages.

The acceptance checks in Section 4.10 verify that every export contains independent state trajectories.

4.5 Combining Closed-Loop and Open-Loop Datasets

The inverse model uses only closed-loop data because its target is the controller command for a measured state. The forward model is retrained on the strict open-loop export while retaining closed-loop replay at weight 0.25. Without replay, four excitation runs do not provide enough pose coverage.

Figure 4.4 shows the difference: closed-loop data covers the reachable range, while open-loop data concentrates at identifiable valve plateaus. The forward model therefore uses both.

For ROM training, the inverse and forward exports are joined by run identifier, time index and sample time. The join is accepted only when their sixteen-dimensional cylinder states match sample for sample. It contains 452 400 transitions at 0.05 s from 1 040 runs, split into 936/52/52 complete runs. Valve command is used only to supervise the ROM's internal valve head; it is not a runtime input (Section 5.4).

This aligned nominal-plant export trains the inverse, forward and ROM models and ships with the project, so every model-level result can be recomputed from the file that produced it. Exact alignment, increment consistency and the identifiability failure were checked directly on that export.

4.6 Closed-Loop Calibration Data for the Command ROM

The ROM drives a PhysX articulation whose compliance and contact response are not present in Simscape. Further Simscape training cannot remove the resulting deployment offset.

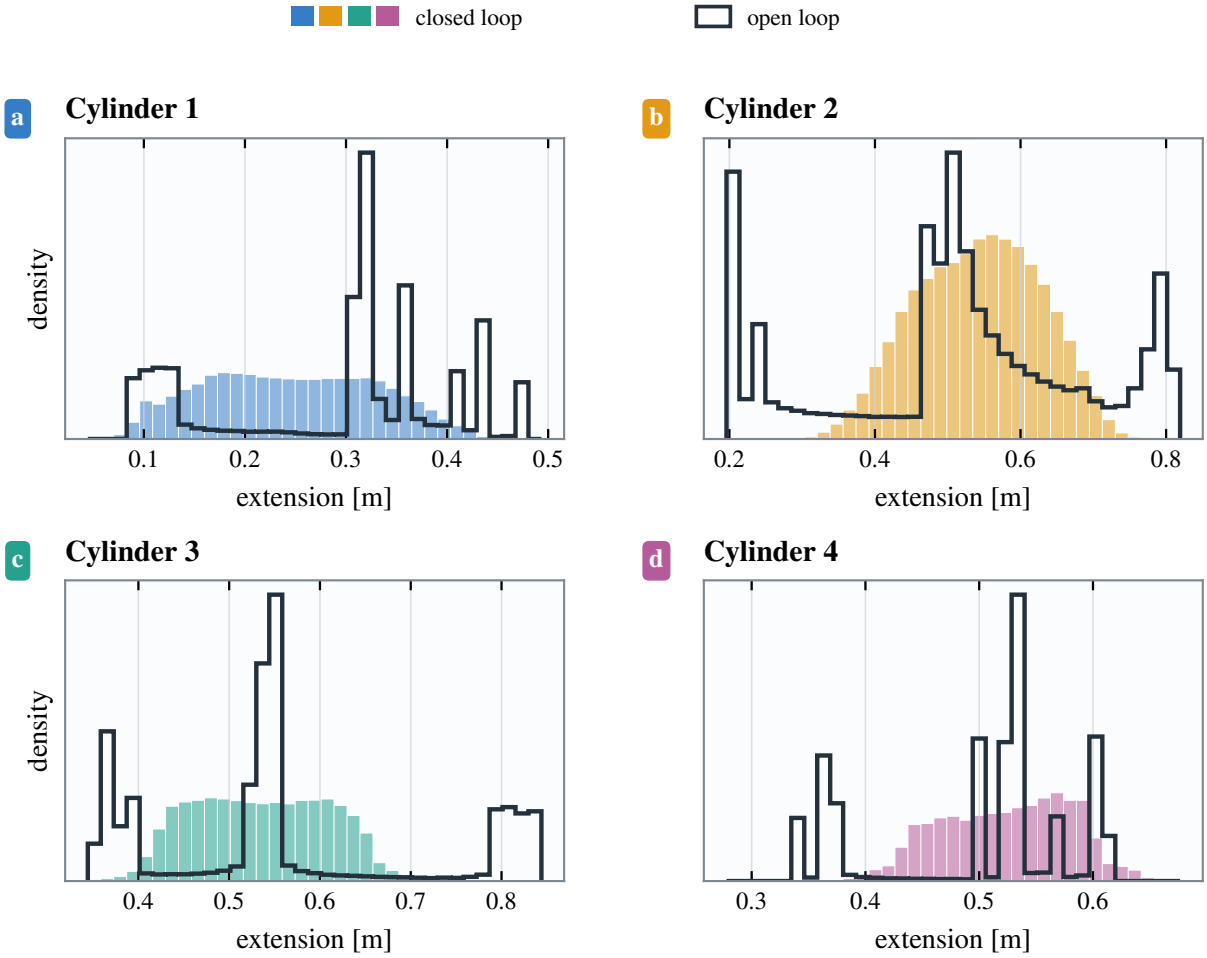

Figure 4.4: Extension coverage of the two campaigns, per cylinder and pooled over runs (closed-loop filled, open-loop outline). The closed-loop data fills the reachable range smoothly; the open-loop data concentrates into the plateaus where the valve is identifiable.

The calibration set therefore comes from closed-loop ROM–PhysX operation on the native 0.05 s grid. Its 4 084 transitions store the six-step runtime history and measured PhysX transition.

Only the command-gated correction head is trainable; all other parameters are frozen and the correction is capped at 3 mm. Original-data replay at weight 0.25 preserves wider pose coverage. The gate makes the correction zero at zero command, retaining stationary hold.

The trajectories predate the moving-verge benchmark and exclude cutter–ground contact. The bounded head limits, but does not remove, this coverage gap. Section 3.4.1 explains the consequence for the three-way comparison.

4.7 The IsaacLab Deployment Collections

Two collections are recorded in the deployment plant. They share their factor design, their domains, their splits and their freezing procedure, and differ in one respect only: how the arm is driven. The first is driven by the task controller and is the collection the inverse

and forward models are scored against in Section 5.2.2; the second is driven by open-loop excitation and is the one the adaptation module is trained on. Why both exist is itself a result, and it is given in Section 4.7.5.

4.7.1 Provenance of the Deployment Data

The models that carry the identifiability argument, namely the system-dynamics surrogate, the privileged factor encoder and the offline history encoder, describe the plant the adaptive controller is closed around. That plant is the command ROM plus PhysX, not the Simscape model. A surrogate identified on Simscape data would be a model of a model, and the latent it defines would describe variation the deployment plant does not necessarily exhibit. Both collections are therefore made *in* the deployment plant, and the frozen ROM, ID and FD checkpoints enter this chain only as fixed components.

4.7.2 The Physical Factor Vector

The randomised quantity is a six-dimensional physical factor vector

$$\mathbf{e} = \left[e_{\text{flow}},\ e_{\text{compliance}},\ e_{\text{seal}},\ e_{\text{pressure}},\ e_{\text{mass}},\ e_{\text{COM}_z}\right], \tag{4.1}$$

with the ranges of Table A.3. Four coordinates are hydraulic and two mechanical, which is what makes the hydraulic-only, mechanical-only and combined domains of Section 4.7.3 separable. Each candidate was admitted only after a one-factor sensitivity screen confirmed that moving it alone produces a measurable change in cylinder extension, velocity, pressure or end-effector motion.

One omission is deliberate. Actuation delay is not a randomised coordinate, because delay here is an *emergent* property of flow, compliance and load rather than a free parameter. It is measured as a response property of each run (Chapter 7) rather than supplied to any model as a known factor: handing an estimator a factor the real machine would never publish is the same mistake as handing it the true latent.

4.7.3 Domains and Factor Design

Runs are organized into five domains (nominal, hydraulic-only, mechanical-only, combined-training, and an *unseen* combined-test domain), and one factor vector is fixed for a complete run, so that no run mixes plants. Factor configurations are drawn with deterministic scrambled Sobol sequences [Sobol' 67] rather than independent uniform sampling, with explicit boundary anchors so that the corners of the space are represented and not merely approached, and with fixed per-domain seeds so the design is reproducible independently of the data it produced. Combined draws that compound into an excessive response gain are rejected at draw time rather than being collected and discarded later.

The resulting design is auditable: 98.3 % of the factor keys are unique, the active-factor matrix is full rank, every factor quartile is populated, and the factor keys of the training, validation, and test splits are disjoint. That last property is the one that matters most for the claims of Chapter 7: a test run does not merely come from a different episode, it comes from a plant the models have never seen.

4.7.4 Collection Procedure: The Task-Controller Runs

Every run is collected through the same interface the policy uses. The DLS controller closes the loop on measured cylinder state and issues its command through the four-dimensional action port into the command ROM and PhysX. No joint-target bypass is used and no PPO policy collects identification data, because a policy trained on the same plant would reintroduce the closed-loop correlation that Section 4.4 spent a campaign removing.

Independent excitation is supplied by dedicated four-channel PRBS windows through that same action port, at small amplitude and a hold of ten control steps, so the excitation is causally independent of the state without leaving the safe envelope. Probe actions are logged separately from controller output. Each run follows a fixed eleven-segment structure (settle, three postures, three PRBS windows, DLS task motion, reach-and-return and stow), giving 11 304 control steps and 188.4 s of simulated time per run.

Figure 4.5 summarises the IsaacLab collection configuration and its frozen run-level outputs.

Figure 4.5: IsaacLab adaptation-data collection. Each run uses one fixed six-factor plant draw and combines DLS task motion with independent PRBS probes. State, action and factor keys are logged, audited and frozen into disjoint train, validation and held-out test splits.

4.7.5 The Open-Loop Context-Identification Collection

A second collection of 118 runs was recorded in the same deployment plant, with the same six-factor design and the same domain and split structure, and it is the offline context-identification collection used to train the AAM. It differs from the collection above in a single respect, and that respect is the point of it: there is no task controller. Each run is driven by open-loop excitation alone, the vehicle is stationary, and no policy is in the loop. The excitation is the same four-channel PRBS used as a probe above, at amplitude 0.02 and held for ten control steps, but it now drives the whole run rather than three windows inside one. Each run is 11 040 control steps, giving 1 302 720 steps across the collection.

The reason for collecting twice is structural. Under a task controller the command is computed from the state, so command and response are close to redundant and a plant descriptor has little left to explain; a model asked to identify the plant from that data is being given a question with no answer in it, and what it reports is a property of the data rather than of the architecture. Removing the controller makes the question well posed, which is why every model-level result in this thesis is fitted on this collection. How rich the excitation must be, and the amplitude beyond which plant identity stops being recoverable, remain outstanding measurements.

What each consumer takes from the recorded feature contract is set out in Section 4.8.2 for the adaptation module and Section 4.8.3 for the prediction models.

4.7.6 The Frozen Dataset

Both collections contain 118 complete runs under the identical domain design: 2 nominal, 40 hydraulic-only, 12 mechanical-only, 48 combined-training and 16 unseen combined-test. Collection ran in four vectorized IsaacLab environments with per-environment factors and logging, and slots that tripped a safety or validity check were discarded and retried individually. All 118 runs passed the final collection gate, which checks finite physics, exact model-grid alignment, command rank, factor-response authority, state coverage and saturation limits.

Each dataset is then *frozen*: the raw runs are checksummed and the run-level split is fixed at 82 training, 20 validation and 16 reserved combined-test runs, with zero factor-key leakage between them. The reserved sixteen do not influence history length, checkpoint, hyperparameter or architecture selection; they are scored only after each model is fixed, and test results are reported with validation results rather than in place of them.

4.7.7 Learning Timebase and Hydraulic State

The learning timebase is the actuator model grid, not the control clock. Samples are exact 50 ms ROM transitions taken at control indices $0, 3, 6, \ldots$: state row t and command row t predict state row $t + 3$. Nothing is interpolated. This is the same alignment rule as Section 3.5.1, and it is enforced rather than assumed, because a model trained across a mismatched interval is wrong by a fixed factor in the quantity it exists to supply.

The hydraulic state used by the system-dynamics surrogates is

$$\boldsymbol{s}_t = \left[\boldsymbol{x}_t^{1:4},\ \dot{\boldsymbol{x}}_t^{1:4},\ \Delta\boldsymbol{p}_t^{1:4}\right] \in \mathbb{R}^{12}, \tag{4.2}$$

with joint position and velocity carried as *context* only, and the prediction target the next 50 ms state delta. Differential pressure is not a measured signal. At runtime it is the actuator model's recursive state, generated by the command ROM or frozen forward model (Section 3.3). The system-dynamics surrogate and AAM may use it because it exists inside the deployed loop, but it inherits actuator-model error. Causal history is therefore retained to represent pressure transients that one sample cannot summarise.

4.8 Feature and Target Construction

4.8.1 The Actuator Models

The three learned actuator models take cylinder-space features, in the ordering and notation fixed in Section 5.1. Their input contracts differ because their questions differ. The inverse model receives the measured cylinder state, the logged rod forces, the desired increments, and the previously applied command, twenty-eight numbers in all. The forward model receives the applied valve command and the measured state, twenty numbers. The reduced-order model receives a six-sample history of state, previously predicted valve, and desired command, twenty-four numbers per step.

Three conventions apply throughout, and they are the same ones the models enforce at inference time.

Increment Targets

Every dynamic target is a within-run one-step increment, $\Delta \boldsymbol{x}_k = \boldsymbol{x}_{k+1} - \boldsymbol{x}_k$, never an absolute state. This removes the large static component of the signals and turns the trivial persistence solution into an explicit baseline rather than a hiding place.

Causal Alignment

Features at step k contain only quantities available at step k. The extraction stage verifies this by recomputing every logged target from the within-run difference of the states and requiring exact agreement, a check reported in Section 4.10.

Time-Base Conversion

The open-loop campaign is logged at 0.01 s but the models run at 0.05 s, so five consecutive raw samples are combined into one transition. The valve input attributed to that transition is the command held at the start of the window, which is the causally correct choice; the window mean is recorded for analysis only. Cylinder extension velocity is not logged directly in the closed-loop archive and is obtained by differentiating extension on the output grid, consistent with the trapezoidal identity that the strict data satisfies to $R^2 = 0.994$–0.9997 (Equation 5.6).

4.8.2 Features and Targets for the Adaptation Module

The adaptation module is trained on the open-loop collection of Section 4.7.5. Its history window carries only quantities the deployed stack produces at runtime: the five joint positions, the five joint velocities, the four cylinder pressure differences and the four

previously applied valve commands, giving the eighteen-value row of Section 5.5. The pressure entry is the learned actuator's own recursive state rather than a measurement (Section 3.3, Table 3.2); it is admitted because the model that generates it runs inside the deployed loop, and the consequence of that choice is examined where the estimator is evaluated. Four classes of signal are excluded by construction, and the exclusion is enforced by an explicit input mask rather than by convention:

- the physical factors $\mathbf{e}$ and the true latent $\mathbf{z}_{\mathrm{SD}}$, because recovering them is the estimator's job;
- any future observation, because the estimator is causal;
- the hidden internal state of the command ROM, because the deployment interface does not expose it;
- the desired task reference, so that the latent describes the plant rather than the trajectory being asked of it.

Windows are contained within a single run, and each carries the stopped-gradient target $\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$ of its own run together with the run identifier, domain, split and valid history bounds. Applied to the 118 frozen runs this yields 444 624 causal history ticks; what is built from them belongs to Chapter 5.

The residual encoder deployed in AF shares this row but uses no collection at all, for the reason given in Section 4.8.4.

4.8.3 Features and Targets for the Prediction Models

The prediction models read the same runs through a third contract: a transition at the 50 ms tick rather than a history window, and the privileged factor vector $\mathbf{e}$, which the adaptation module never sees and which is what allows them to define the target the module is then asked to recover. That contract is set out with the dataset it produces in Section 4.11, because what it is for is only visible alongside the audit it makes possible.

4.8.4 On-Policy AF Adaptation Rollouts

The residual that carries the adaptation result in AF is not trained from the frozen collection above. It has no dataset, no stopped-gradient latent target and no run-level splits, because its encoder never regresses a target: the encoder and the residual head are optimised together under the reinforcement objective, so their training data is the on-policy experience the stack generates as it learns.

What replaces the dataset is a plant design, since on-policy training makes the training distribution a property of the environment rather than of a file. The cutter mass is drawn from two well-separated modes and flips between them *mid-episode*. The flip instant is jittered per environment so it cannot be anticipated on a timer, and the direction is balanced across environments so it cannot be assumed. This design makes recent history potentially informative: the active regime cannot be inferred from a fixed timer, and one fixed compromise need not be optimal in both regimes. It creates an opportunity for

history-conditioned control without assuming that history will be beneficial. Whether live history contributes beyond additional residual capacity is tested explicitly against a capacity-matched zero-history control in Section 7.5.

Two properties of the frozen collection are inherited unchanged. The history row is the same eighteen-value row defined above, with the same four exclusion classes enforced by the same input mask, and the plant is still the deployment plant, so the change of training regime does not substitute a different system.

4.9 Preprocessing and Data Splits

Splitting is done at the level of complete runs, never of individual samples, because a random sample-level split would place transitions from the same trajectory on both sides of the boundary and report a score that mostly measures interpolation within a run. For the command-conditioned dataset the 1 040 runs are divided into 936 training, 52 validation and 52 test runs, giving 407 160, 22 620 and 22 620 transitions, with no run identifier in more than one split. The IsaacLab deployment collection is split by run *and* by factor configuration (82/20/16), the stronger condition: separating runs alone would let two runs of the same plant fall on opposite sides of the boundary and turn a claim about unseen plants into a claim about unseen episodes.

The strict open-loop export needs different treatment because it contains only four runs, and holding one out would remove an entire plant draw from training. Each run is instead divided into twenty temporal blocks, every fifth block is assigned to validation, and a five-sample guard interval is discarded at each boundary so that no 0.05 s transition window spans the split. This preserves comparable target distributions in training and validation.

Normalisation statistics are computed on the training split only and stored inside each checkpoint, so a model always normalises with the statistics it was fitted with; the loader enforces the stored contract rather than recomputing it. Samples are carried with an explicit validity flag rather than silently dropped, 93.7 % of the strict export being marked valid and the remainder being the excluded startup and shutdown seconds and the zero-command gaps. Keeping them with a flag means the time base stays contiguous and the within-run difference checks remain meaningful.

4.10 Dataset Diagnostics

No dataset in this thesis is used before it passes an explicit audit, for the reason given at the end of Section 4.4. The audit of the strict open-loop export is reproduced in Table A.4.

The two rank figures speak directly to requirement R2. The instantaneous command matrix has rank four, so no channel is a linear combination of the others, and the lagged regressor matrix built from the command history has rank eighty, so the excitation is persistently exciting to the order the models need. The condition number of that lagged matrix is about 4.6×10^3, finite and workable rather than comfortable, and it is the price paid for the stroke coverage the long plateaus buy.

The three increment checks return a maximum error of exactly zero, which is expected rather than suspicious: the targets are computed from the logged states by the same within-run difference the check recomputes, so the check verifies that the extraction stage

did what it claims rather than that the simulation is accurate. It catches misalignment, off-by-one indexing and rows reordered or concatenated across run boundaries. Physical fidelity is addressed separately by the validation study of Section 3.4.4.

Two properties of the Simscape material are recorded here because later chapters refer to them. Stroke coverage is 12 % on the head-pivot cylinder, the narrowest of the four channels and also the one the strict forward model predicts least well. The rod-force channel carries zero supervised weight in the reduced-order model, a design choice rather than a data defect: force is generated internally by the PhysX plant at deployment, and the logged channel additionally fails a piston-area consistency check on this export. No result rests on it.

The IsaacLab collection is audited against different failure modes: design adequacy before use (unique factor keys, full active-factor rank, disjoint keys across splits), physical validity per run (finite physics, exact model-grid alignment, full command rank, bounded saturation), and leakage structurally, with privileged channels removed by an explicit input mask rather than by trusting the feature list. Its limitation is scale, 118 runs over six factors being a sparse design, which is why the paired test of Section 4.11.1 is reported with bootstrap intervals and per-run detail rather than as a pooled point estimate. That limitation does not bound the closed-loop adaptation result, which is estimated from paired environments generated on-policy under the plant design of Section 4.8.4.

4.11 The System-Dynamics Dataset

The prediction models read the deployment collection through a third contract, built once and frozen with the rest. It is a derived view rather than a separate collection: the same 118 runs, resampled from the 60 Hz control log to the 20 Hz model tick by taking every third step, giving 3 840 transitions per run and 453 120 in total. The run-level split is inherited unchanged, 82 training, 20 validation and 16 test runs, so a run contributes to exactly one split here and in every other view of this data.

Each sample is one transition. The input is the 12-dimensional hydraulic state, the 10-dimensional joint context and the valve command pair, previous and current, for 30 channels; the target is the hydraulic-state increment to the next tick in the same 12 dimensions. Training draws five consecutive transitions at a time, so the objective is a short rollout rather than a one-step fit. Four per-tick flags record cylinder limit, command saturation, valve saturation and non-finite physics, so a model can be scored on valid ticks alone. Each run carries its own six-factor vector $\mathbf{e}$, which is privileged: it is what allows a target to be defined for the adaptation module of Section 4.8.2, and it is masked out of every deployable input.

4.11.1 Dataset Separation Across Plants

One property of this dataset cannot be checked by an audit of its structure. The checks of Section 4.10 establish that the runs are valid, that the factor design is full rank and that the splits do not leak, but not that the runs *differ by plant* in a way any model could use. If they do not, nothing built on this dataset can recover the difference.

The test is a paired one on the dataset itself. Two prediction models are trained on identical samples, splits and objective; one receives a two-dimensional plant context from

an encoder $\mathbf{z}_{\mathrm{SD}} = \mu(\mathbf{e})$ trained jointly with it, and the matched control receives no such input. Neither is an actuator and neither enters a control stack.

The context helps, and narrowly. Pooled rollout error falls by 7.20×10^5 on the 20 validation runs, with a 95 % interval of $[-1.03 \times 10^6, -4.30 \times 10^5]$, and by 4.41×10^5 on the 16 test runs, interval $[-7.61 \times 10^5, -1.74 \times 10^5]$. Both exclude zero, so the dataset does separate plants. Where it separates them is the more useful result: the per-channel decomposition puts the whole gain in pressure, and the extension and velocity intervals cross zero on both pools.

Table 4.1: Per-channel decomposition of the paired advantage at a 20-tick horizon, pooled over 36 held-out runs. Values are the mean paired difference between the context-conditioned model and the matched control, with a 95 % bootstrap interval. The advantage is confined to pressure.

Channel	mean paired difference	95 % interval	excludes zero
extension	5.2×10^{-5}	$[-3.6 \times 10^{-5},\ 1.4 \times 10^{-4}]$	no
velocity	-2.4×10^{-4}	$[-7.1 \times 10^{-4},\ 2.4 \times 10^{-4}]$	no
pressure	1.75×10^5	$[2.1 \times 10^4,\ 3.3 \times 10^5]$	yes

Two limits travel with this result and both matter later. It compares separately trained models, so it establishes that the context is informative, not that an estimator could recover it or that a controller could use it; Section 5.6.3 substitutes contexts within one model, which is the tighter test. And the informative channel is pressure, which the controller does not integrate. That is the first appearance of the delivery problem the rest of this thesis runs into: this dataset carries plant identity in the one channel that reaches the command path least directly.

With the data described and audited, the next chapter turns to what was learned from it: the inverse and forward models that form the IF actuator path, the command-conditioned reduced-order model that replaces them in R, and the adaptation module that estimates at runtime what the surrogate above was given.

5. Learned Models and Online Adaptation

Chapter 4 established how the datasets were generated and why both closed-loop operation and open-loop valve excitation were required. This chapter develops the learned representations that consume them: the inverse-dynamics (ID) and forward-dynamics (FD) networks of the IF actuator path, the command-conditioned reduced-order model (ROM) that replaces them in R, and an offline Arm Adaptation Module (AAM) study that tests whether plant context can be recovered from causal history. This offline AAM and its ID adapter are diagnostic prototypes. The deployed AF encoder and residual are trained jointly under PPO in Chapter 6.

The chapter continues one evidence chain. Section 4.11 established the first link on the data itself: a compact plant context exists and improves prediction, and the improvement sits in pressure. What follows is whether that context is recoverable causally from observable history, whether the recovered estimate is as informative as a privileged one, and whether it reaches the command. The chain fails at that last link, and the diagnosis specifies the redesign carried out in Chapter 6.

One commitment organises the chapter: no model is accepted on a supervised score, because such scores repeatedly proved uninformative here, a forward model reaching a one-step R^2 near 0.999 while having learned almost nothing about valve authority (Section 5.3). Every model is therefore presented with its contract, training objective and the offline evidence that accepted or refused it. All models operate on cylinder-space signals.

5.1 Modelling Overview and Common Notation

All learned models act on the four hydraulic cylinders of the MULAG arm, indexed $i \in \{1, \dots, 4\}$ in the fixed order slew cylinder (*Drehzapfen*), boom cylinder I (*Ausleger I*), boom cylinder II (*Ausleger II*) and head pivot cylinder (*Messerkopf-Schwenk*). The passive fifth joint is not actuated and enters no model input. At the learned-model sample time $T = 0.05\,\mathrm{s}$ the discrete index k refers to the cylinder extension $\boldsymbol{x}_k$, the extension velocity $\dot{\boldsymbol{x}}_k$, the chamber pressure difference $\Delta\boldsymbol{p}_k$ and the logged rod force $\boldsymbol{F}_k$, each in $\mathbb{R}^4$. The

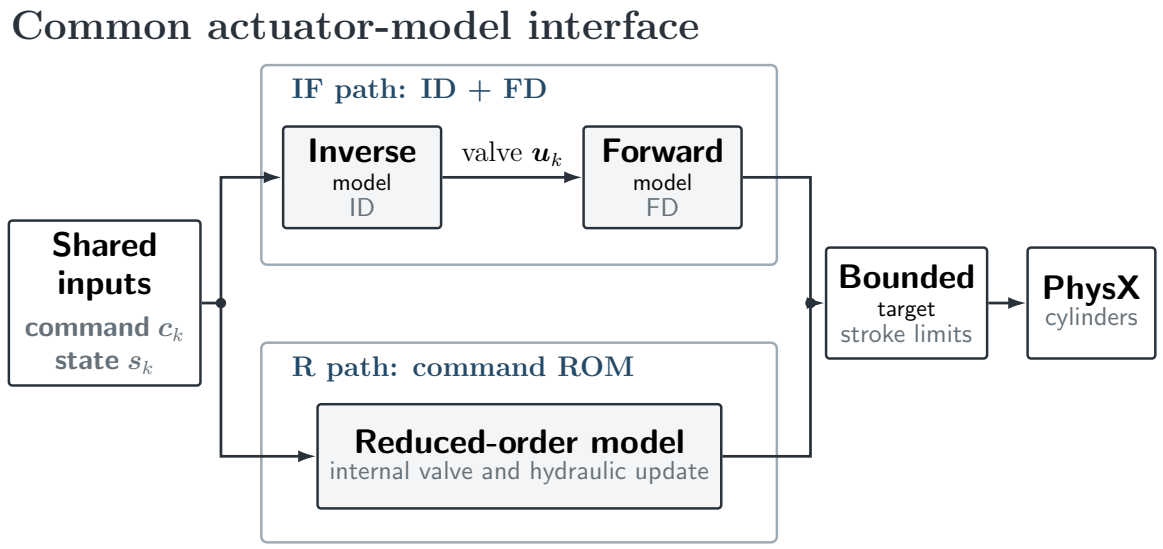

Figure 5.1: Common interface of the two alternative actuator paths. Both receive the desired cylinder command and measured hydraulic state, and both produce the same bounded PhysX cylinder target. The IF path composes the inverse and forward models; the R path uses one command-conditioned reduced-order model.

valve command $\boldsymbol{u}_k \in [-1, 1]^4$ is the normalised spool command of the four proportional valves. The desired cylinder command produced by the task controller is the pair

$$\boldsymbol{c}_k = \left[\Delta \boldsymbol{x}_k^*,\ \Delta \dot{\boldsymbol{x}}_k^*\right] \in \mathbb{R}^8, \tag{5.1}$$

the extension and extension-velocity increments to be realised over the next sample. Predictions carry a hat, and $\Delta(\cdot)_k = (\cdot)_{k+1} - (\cdot)_k$ denotes a within-run one-step increment throughout.

Figure 5.1 compares the actuator paths. In IF, the ID maps the state and desired increment to a valve command, and the FD predicts the resulting increment. In R, one command-conditioned model predicts the valve internally and propagates pressure, velocity and extension. Both paths produce the same bounded PhysX cylinder target in IsaacLab [Mittal 23] under the same policy action contract [Schulman 17].

Three conventions apply throughout. Dynamic targets are within-run increments, with $\Delta\hat{\boldsymbol{x}} = \mathbf{0}$ as the persistence baseline. Each checkpoint uses its own training-set normalisation. Inputs at step k are causal, and targets are measured transitions to step $k + 1$.

Simscape supplies the identification data described in Chapter 4. The exported arrays are audited, frozen and used to train the models in PyTorch [Paszke 19]; Simscape is not part of training or deployment.

The closed-loop Simscape export [The MathWorks, Inc. 24a] contains 452 400 transitions at $T = 0.05\,\mathrm{s}$ from 1 040 runs, split by run into 936/52/52 training, validation and test runs. It supplies the supervised ID, FD and command-ROM data. The FD repair uses four open-loop 64-s runs at 0.01 s with rank-4 valve excitation on four plant draws.

5.2 Inverse Dynamics Model

The ID network maps the current hydraulic state and desired cylinder increment to a valve-command change:

$$\Delta \boldsymbol{u}_k = f_{\mathrm{ID}}\left(\boldsymbol{x}_k,\, \dot{\boldsymbol{x}}_k,\, \Delta \boldsymbol{p}_k,\, \boldsymbol{F}_k,\, \Delta \boldsymbol{x}_k^*,\, \Delta \dot{\boldsymbol{x}}_k^*,\, \boldsymbol{u}_{k-1}\right) \in \mathbb{R}^4, \tag{5.2}$$

a 28-dimensional input. The deployed command is $\boldsymbol{u}_k = \mathrm{clip}(\boldsymbol{u}_{k-1} + \Delta \boldsymbol{u}_k,\, -\boldsymbol{u}_{\max},\, \boldsymbol{u}_{\max})$ with per-axis limits $\boldsymbol{u}_{\max} = [0.95,\, 0.95,\, 0.8,\, 0.8]$ stored in the checkpoint. The change formulation retains the previous valve operating point near the direction-dependent dead zone (Section 2.1.2). Logged achieved transitions serve as desired increments because no independent cylinder-space reference was recorded.

The 512-wide residual MLP [He 16] uses SiLU activations, layer normalisation and dropout [Elfwing 18, Ba 16, Srivastava 14]. Adam minimises a smooth-L_1 loss on normalised targets, with saturated samples up-weighted [Kingma 15, Huber 64].

Figure 5.2 separates the three input groups and shows the four residual blocks explicitly. The network predicts a four-axis valve-command increment, which is accumulated and clipped before use.

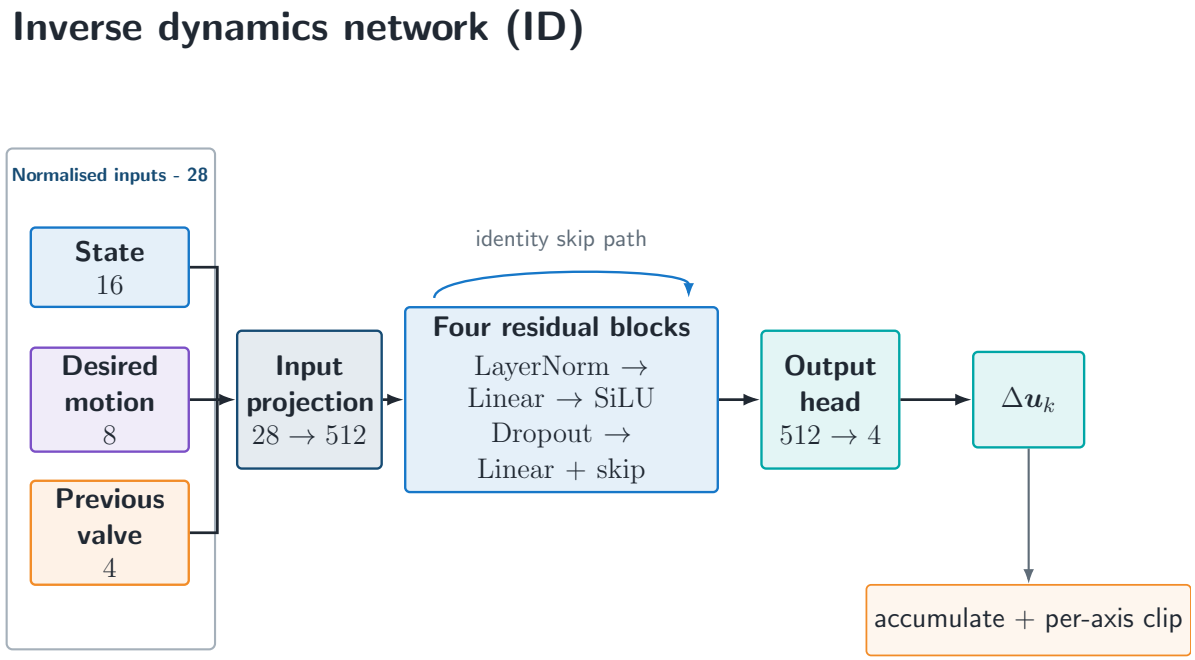

Figure 5.2: Inverse-model architecture. Current hydraulic state (16), desired motion (8) and the previous valve command (4) form the 28 inputs. A 512-wide, four-block residual MLP predicts $\Delta \boldsymbol{u}_k$, which is accumulated and clipped per axis to form the bounded valve command.

Per-valve validation R^2 is 0.9973–0.9996. This shows reproduction of logged commands, not plant authority: in closed-loop data, achieved transitions and logged commands are correlated (Section 2.4.2).

5.2.1 Paired Control Fine-Tuning

The composite is assessed by the *paired differential response*: the FD prediction under the ID command for the requested target minus the prediction under its command for a zero target,

$$\begin{aligned} \boldsymbol{u}_k^{\mathrm{tgt}} &= f_{\mathrm{ID}}(\cdot,\, \boldsymbol{c}_k,\, \boldsymbol{u}_{k-1}), \qquad \boldsymbol{u}_k^{0} = f_{\mathrm{ID}}(\cdot,\, \mathbf{0},\, \boldsymbol{u}_{k-1}), \\ \Delta\hat{\boldsymbol{x}}_k^{\mathrm{pair}} &= f_{\mathrm{FD}}\Big(\boldsymbol{u}_k^{\mathrm{tgt}}, \cdot\Big) - f_{\mathrm{FD}}\Big(\boldsymbol{u}_k^{0}, \cdot\Big). \end{aligned} \tag{5.3}$$

Before fine-tuning, the composite normalised control error was 1.00, equal to the zero-response baseline, and all regression slopes were below 0.004 (Table 5.1). The networks were fine-tuned jointly on $\mathcal{L}_{\mathrm{ctl}} = \|\Delta\hat{\boldsymbol{x}}^{\mathrm{pair}} - \Delta\boldsymbol{x}^{*}\|^2_{\mathrm{norm}} + \lambda_{\mathrm{ID}}\mathcal{L}_{\mathrm{ID}} + \lambda_{\mathrm{FD}}\mathcal{L}_{\mathrm{FD}}$, with the supervised terms retained. Target blocks were shuffled across valid states to break the logged state–target pairing. After 30 epochs, control error fell to 0.109; all slopes exceeded 0.8 and all correlations exceeded 0.92. This frozen checkpoint is the inverse model of the IF actuator path, and of the latent-conditioned adapter route of Section 5.6. It is not part of AF, whose command path contains no inverse model (Section 6.7.1).

Table 5.1: Effect of paired control fine-tuning on the composite differential response (5.3), per cylinder axis on validation data. Before fine-tuning the composite had essentially no control authority despite near-perfect supervised scores of both networks; the composite normalised control error fell from 1.0011 to 0.1086.

	Regression slope		Correlation	
Cylinder axis	before	after	before	after
1 (slew)	0.003	0.818	0.135	0.920
2 (boom I)	0.002	0.928	0.090	0.967
3 (boom II)	0.001	0.849	0.037	0.933
4 (head pivot)	0.004	0.912	0.249	0.958

5.2.2 Transfer of the Frozen Inverse Model

Because the adapter corrects a frozen ID, its transfer was measured on all 118 frozen runs of Section 4.7, comprising 2 nominal, 40 hydraulic-only, 12 mechanical-only and 64 combined runs, of which 16 are held-out combined-test. These runs lie outside the ID's Simscape-only training distribution and are recorded in the plant the adaptive controller is closed around. Two regimes were scored: a teacher-forced one-step transfer supplying the measured cylinder state at every tick, and a closed-loop transfer feeding back the pair's own prediction while supplying the recorded reference. Pressure and force are held at their training means, as the deployed feature contract requires.

Teacher-forced, the pair transfers well and evenly (Table 5.2). One-step extension error is millimetre-scale in every domain and nearly flat across them, from 2.36 mm on the combined runs to 3.36 mm on nominal, and the held-out combined-test group is no worse than the domains used in training, at 2.34 mm. This is the regime in which IF uses the pair, because the policy closes the loop at every tick and never asks it to predict its own future.

Table 5.2: Transfer of the frozen inverse and forward pair across deployment domains, all 118 runs of the deployment collection. The one-step columns are teacher-forced; the saturation column is the fraction of pre-clip valve elements at the limit during closed-loop transfer. The ROM valve recorded on these same runs never saturates.

		one-step Δx error		closed-loop
Domain	runs	RMSE	worst run	valve saturation
combined	64	2.36 mm	5.72 mm	0.888
hydraulic	40	2.43 mm	4.96 mm	0.885
mechanical	12	2.81 mm	4.14 mm	0.919
nominal	2	3.36 mm	4.02 mm	0.929
all	118	—	—	0.891

Closed-loop, it does not transfer at all, and the failure is uniform rather than occasional. Every one of the 118 runs leaves the recorded reference by more than one metre, after about 32 s on the combined and hydraulic domains and sooner on the harder ones. The mechanism is saturation: the pair's pre-clip valve command sits at its limit on 89.1 % of elements over the collection, between 88.5 and 92.9 % by domain, while the ROM valve recorded on those same runs never reaches its bound at all. For scale, the measured plant tracks the recorded cylinder reference at about 0.807 mm on the held-out group, so the closed-loop divergence is a failure of model transfer and not a property of the collection.

Two consequences follow. The pair cannot be promoted as an unadapted autoregressive deployment path, because its own predicted state leaves the support its training data covered, though nothing in that result disturbs the one-step use IF makes of it. What it does disturb is adaptation: a bounded correction added to a command already resting on its stops has nothing left to move (Section 6.7.1).

5.3 Forward Dynamics Model

The FD network predicts the one-step cylinder response to a valve command,

$$\left[\Delta\hat{\boldsymbol{x}}_k,\ \Delta\hat{\dot{\boldsymbol{x}}}_k,\ \Delta\widehat{\Delta\boldsymbol{p}}_k\right] = f_{\mathrm{FD}}\left(\boldsymbol{u}_k,\ \boldsymbol{x}_k,\ \dot{\boldsymbol{x}}_k,\ \Delta\boldsymbol{p}_k,\ \boldsymbol{F}_k\right) \in \mathbb{R}^{12}, \tag{5.4}$$

using the same residual trunk as the ID network. Supervised pretraining on the closed-loop export, with an auxiliary recursive rollout loss [Nagabandi 18], reached grouped one-step R^2 near 0.999 for the extension increments.

Figure 5.3 shows the forward model separately because its input and output contracts differ from the inverse model despite the shared residual trunk.

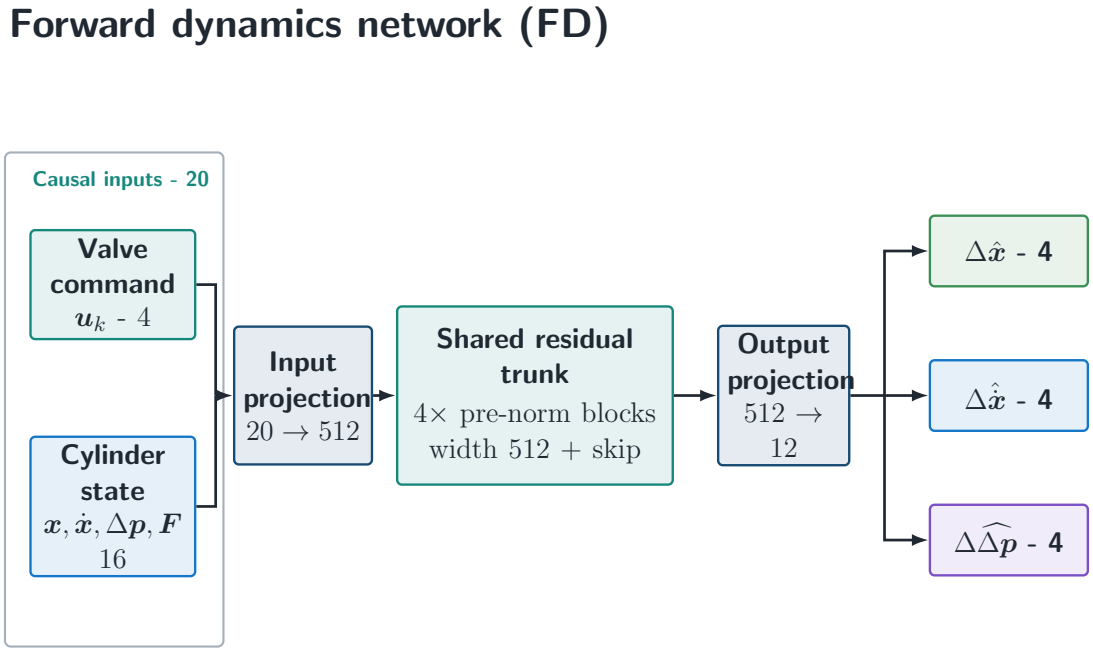

Figure 5.3: Forward-model architecture. The applied valve command and cylinder state condition a one-step hydraulic transition through a 512-wide, four-block residual MLP. Its 12 outputs are extension, velocity and pressure increments. Extension is the integrated controller output; velocity and pressure are diagnostic outputs.

5.3.1 Assessment of the Supervised Score

The supervised score did not establish valve sensitivity. At a 0.5 s horizon, zeroing the valve input changed the rollout-error ratio to only 0.94. The FD had mainly extrapolated the smooth closed-loop kinematics because command and state history were nearly redundant. Closed-loop ablation confirmed that the valve was not identifiable, so repair required new data.

The repair dataset applies rank-4 open-loop valve plateaus to four plant draws. Its trajectories are distinct, valve history predicts pressure with $R^2 \approx 0.80$–0.85, valve-regressor variance-inflation factors are 7–10, and relief saturation occupies 9–22 % of samples.

Starting from the paired-control checkpoint, strict retraining minimised

$$\mathcal{L}_{\mathrm{FD}} = \mathcal{L}^{\mathrm{strict}}_{\mathrm{Huber}} + 0.25\,\mathcal{L}^{\mathrm{MAT}}_{\mathrm{Huber}} + 2.0\,\mathcal{L}^{(10)}_{\mathrm{roll}} + 10.0\,\mathcal{L}_{\mathrm{kin}}, \tag{5.5}$$

combining the one-step loss on the strict data, a replay term on the original closed-loop data, a 0.5 s recursive rollout loss, and a kinematic consistency term enforcing the trapezoidal identity

$$\Delta \boldsymbol{x}_k = T\,\dot{\boldsymbol{x}}_k + \tfrac{1}{2}\,T\,\Delta\dot{\boldsymbol{x}}_k, \tag{5.6}$$

which holds in the strict data with $R^2 = 0.994$–0.9997. Valve authority varies by a factor of 15–75 across pressure quartiles and can change sign, so the ROM represents this dependence by operating point [Merritt 67].

Strict retraining reduces extension rollout RMSE from 75.2 mm to 12.4 mm at 0.5 s and from 157.9 mm to 27.3 mm at 1.0 s. The valve-zero error ratio rises from 0.94 to 1.81, and per-axis one-step extension R^2 improves from $[-5.7, 0.60]$ to $[0.76, 0.98]$. Zero-order hold remains competitive below 0.25 s, and velocity increments remain weak on two axes. Deployment therefore integrates only $\Delta\hat{\boldsymbol{x}}$ and retains velocity as a diagnostic output.

5.4 Reduced-Order Model

The R path uses one command-conditioned model instead of the ID–FD composition. Its structure is grey-box: physical equations propagate pressure, force, velocity and extension. Neural heads estimate valve realisation, load and bounded residuals [Karniadakis 21, Lutter 19]. Valve authority is pressure dependent by construction through the orifice-flow and continuity equations [Merritt 67]. A GRU [Cho 14] reads six samples (0.25 s) of cylinder state, previous predicted valve command and desired command. The measured valve is only an internal-head training target [Hwangbo 19].

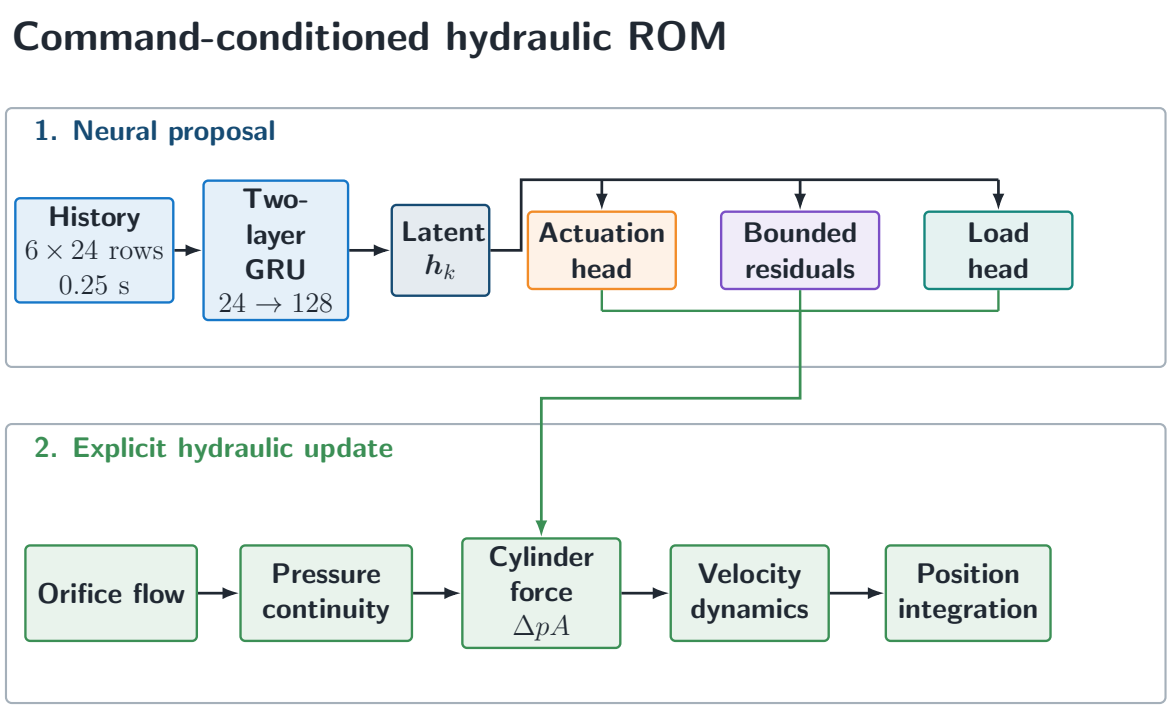

Figure 5.4: Command-conditioned ROM. A recurrent neural proposal encodes a six-step history. Its actuation, bounded-residual and load heads condition the explicit hydraulic update: orifice flow, pressure, force, velocity and position. The explicit chain constrains the predicted transition.

The actuation head uses a learnable smooth dead zone with per-cylinder width and asymmetric gains [Yang 26]. Its output $\hat{a}_i \in [-1, 1]$ drives the orifice-flow and continuity

update

$$q_i = \hat{a}_i\, k_{q,i} \sqrt{\tfrac{1}{2}\big(p_{s,i} - \operatorname{sgn}(\hat{a}_i)\,\Delta p_i\big)_+},$$
$$\Delta\hat{p}_{i,k+1} - \Delta p_{i,k} = \frac{4\beta_i}{V_i(x_i)}\big(q_i - A_i\dot{x}_i - k_{\ell,i}\,\Delta p_i\big)\,T \;+\; r_{p,i}, \tag{5.7}$$

where β_i, $p_{s,i}$, $k_{q,i}$, $k_{\ell,i}$, A_i and $V_i(x_i)$ are the effective bulk modulus, supply pressure, flow gain, leakage, piston area and chamber volume. Pressure then produces force and motion:

$$\hat{F}_i = \Delta\hat{p}_{i,k+1}A_i + r_{F,i}, \qquad \ddot{x}_i = \frac{\hat{F}_i - c_{v,i}\,\dot{x}_i - \ell_i(\boldsymbol{h}_k)}{m_i},$$
$$\dot{\hat{x}}_{i,k+1} = \dot{x}_{i,k} + \operatorname{sat}\big(T\,\ddot{x}_i + r_{v,i}\big), \qquad \hat{x}_{i,k+1} = x_{i,k} + \tfrac{T}{2}\big(\dot{x}_{i,k} + \dot{\hat{x}}_{i,k+1}\big) + r_{x,i}, \tag{5.8}$$

Here $c_{v,i}$ is viscous friction, m_i is effective modal mass and $\ell_i(\boldsymbol{h}_k)$ is the learned configuration-dependent load. Physical parameters are trained in log-space inside fixed clamps. Several settle on a clamp bound and are therefore fitting coefficients rather than measured constants: modal mass and viscous friction both reach their upper clamp of 10^7 (in kilograms and in newton-seconds per metre respectively), the piston area reaches its lower clamp of $1\,\mathrm{cm}^2$ against a $64\,\mathrm{cm}^2$ initialisation, and the supply pressure separates onto both clamps, with two cylinders at 1 MPa and two at 50 MPa. The same values appear in every checkpoint of this family. Pressure and velocity residuals are bounded to 10 % of their step limits, and the position residual to 1 cm, capping the magnitude each head may add per step.

The explicit chain and the neural heads contribute to the transition in very different proportions. With the residual heads inactive the explicit chain alone produces no net travel, its velocity increments falling six orders of magnitude below their nominal values: at the fitted parameters the friction term $c_{v,i}\dot{x}_i$ reaches 1.0 MN at 0.1 m/s, against a pressure force $\Delta\hat{p}_i A_i$ of about 10 kN at 100 MPa. Realised displacement is produced by the residual heads. The pressure equation nonetheless remains active and responsive to its own parameters: with the heads inactive, halving $k_{\ell,i}$ still moves Δp by 63 % of its range. Hydraulic parameters therefore reach the predicted motion indirectly, through the pressure state entering the recurrent encoder, rather than through the force balance. The model has a grey-box structure. Its equations fix the transition form and keep it differentiable in named hydraulic quantities, but they do not identify the machine's physical constants. Put bluntly: at these clamped values the explicit chain is a regulariser on the shape of the residual heads' output, not an identified plant, and the residual heads are what actually moves the cylinder.

Training uses rollout horizons $H \in \{1, 5, 20, 50\}$ and scheduled sampling up to 75 % self-fed [Bengio 15]. Residual penalties retain the physical structure. Logged rod force has zero supervised weight because PhysX generates it at deployment and the export fails the piston-area consistency check.

With the true command, mean one-step R^2 is 0.998 for the internal valve, 0.64 for velocity and 0.97 for extension. Zeroing or shuffling the command reduces velocity R^2 below zero and extension to about 0.5; the extension Jacobian is diagonally dominant with condition number 2.14. At 5 s, terminal extension R^2 is 0.998, 0.998, 0.84 and 0.91 across the cylinders. Pressure-increment R^2 is only 0.06–0.09, so the internal pressure is a dynamic state, not a calibrated pressure estimate (Figure 5.5), consistent with the source Simscape

pressure signal's own wrong-level ceiling (Section 3.4.4) and with the AF residual's own deployed pressure state carrying shape rather than level (Appendix B.5). Any later use of this pressure channel, including the seal-condition decoding of Section 5.5.1, inherits this limit and is read accordingly.

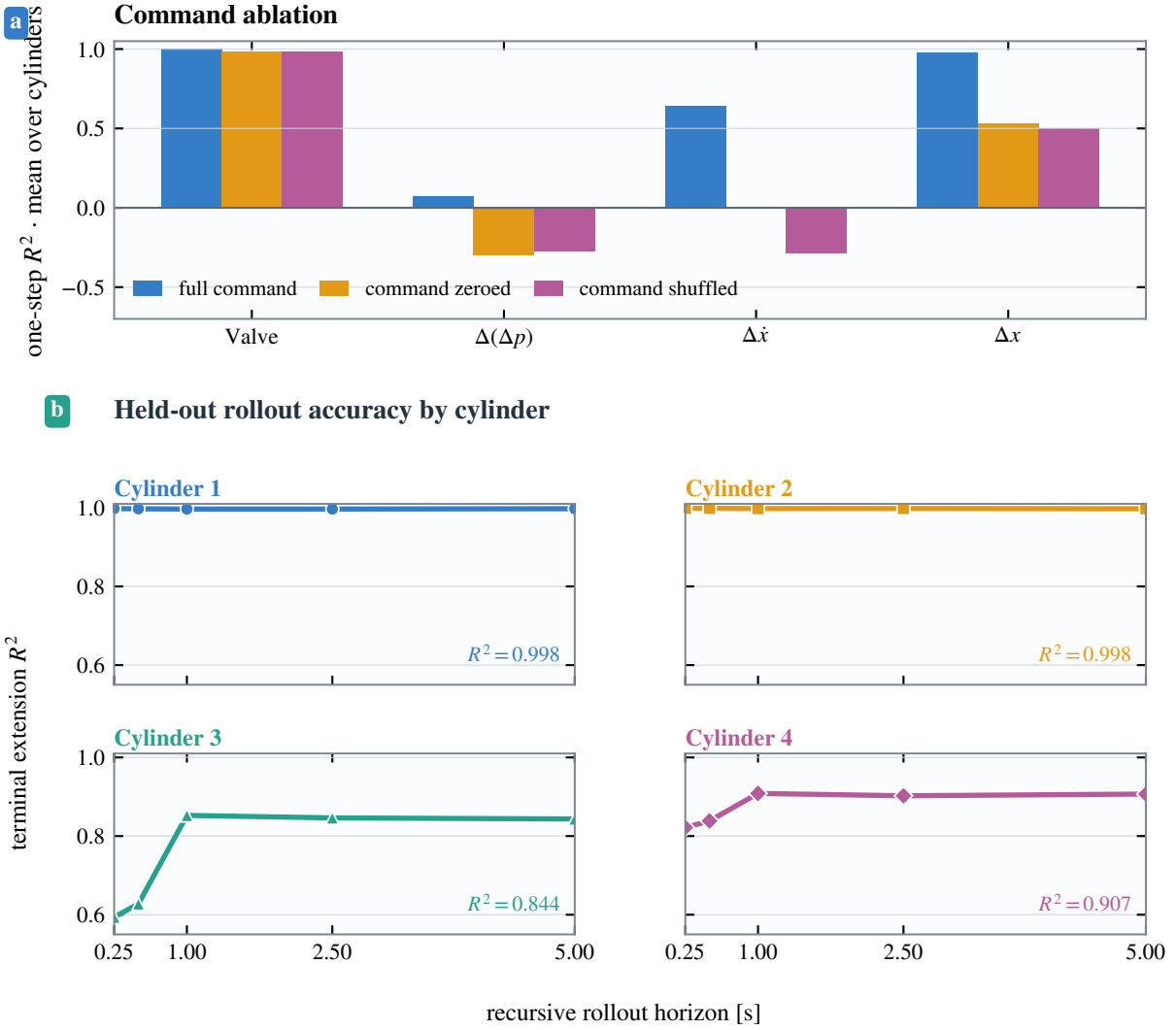

Figure 5.5: Offline validation of the command ROM on held-out runs. (a) One-step R^2 per output group, averaged over the four cylinders, with the full desired command, with the command zeroed, and with the command shuffled across the batch. The collapse of the velocity and extension scores without the true command demonstrates command authority. (b) Terminal extension R^2 of recursive rollouts up to 5 s, shown separately for each cylinder so coincident high-accuracy traces remain visible.

5.4.1 Closed-Loop Calibration Against the Deployment Plant

The Simscape-trained ROM has a systematic extension offset when driving PhysX. The promoted checkpoint adds a bounded, command-gated correction:

$$\hat{x}_{i,k+1} \leftarrow \hat{x}_{i,k+1} + \delta_i(\boldsymbol{h}_k, \boldsymbol{c}_k), \qquad |\delta_i| \le 3\,\mathrm{mm}, \qquad \delta_i\big|_{\boldsymbol{c}_k=\mathbf{0}} = 0, \tag{5.9}$$

It is trained on 4 084 IsaacLab transitions from five reachable reference shapes, with Simscape replay. All hydraulic and integration parameters remain frozen. The gate sets the correction to zero for a commanded hold.

The calibrated command ROM serves as the deployment plant in Section 3.4.1. Unlike the ID–FD path, it has seen PhysX data. It is frozen and has no latent input. The earlier prototype assessed in Sections 5.5 and 5.6 modifies the inverse path. Final AF instead omits the inverse model and corrects the frozen actor's valve command, as described in Section 6.7.

5.5 Offline Plant-Context Identifiability Study

This section describes the encoder trained offline against the plant context of Section 4.11, on the frozen collection of Section 4.7. Its purpose is evidentiary: it establishes that plant identity is recoverable from causal observable history at all, which is a precondition for anything built on top of it. The encoder deployed in AF is trained differently (Section 6.7.5), so the results below measure the information in the history rather than describing the deployed component.

The offline AAM estimates $\mathbf{z}_{\mathrm{AAM}} = \mathrm{AAM}(\boldsymbol{H}_t)$ from deployment history. Each row contains five joint positions, five joint velocities, four cylinder pressure differences and four previous valve commands. A four-layer, width-64 causal convolution processes 20 rows (1.0 s, 360 values).

The pressure values are recursive actuator-model states, not measurements: they come from the command ROM in R and the frozen FD in AF. The factor vector, true latent, reference, future rows and hidden state are excluded. Using the previous valve command keeps each row causal. Validation selected the shortest competitive history from 5, 10, 20 and 40 ticks. Latent RMSE is 0.802 on validation and 0.734 on reserved test.

Table 5.3 collects the trainable size of the four learned models introduced in this chapter, counted directly from each deployed checkpoint's state dictionary rather than estimated from the architecture description. ID and FD share the same 512-wide, four-block residual-MLP family and are consequently within eight parameters of each other despite different input and output widths. The command ROM and the offline AAM are both roughly two orders of magnitude smaller: the ROM's explicit hydraulic chain carries the interpretable structure so its recurrent core need not, and the AAM is deliberately a small probe rather than a capacity-rich estimator, so a positive result could not be attributed to head size alone.

Table 5.3: Trainable parameters of the four learned models, counted from the deployed checkpoint state dictionaries.

Model	Role	Architecture
ID	valve-command increment	512-wide, 4-block residual MLP
FD	extension/velocity/pressure increment	512-wide, 4-block residual MLP
Command ROM	command-conditioned actuator	GRU, hidden 128, 2 layers, + hydraulic ch
Offline AAM	plant-context latent encoder	4-layer causal convolution, width 64

Inference runs at 20 Hz inside the 60 Hz control loop, the prototype ID adapter is bypassed until 20 rows are available, and a causal five-prediction rolling mean adds 250 ms lag. That smoother belongs to this offline encoder and not to AF, whose residual consumes its history window directly, so the latency stated here does not apply to the adaptive result of Chapter 7.

5.5.1 Information Content of the Estimated Latent

Linear decoding from the per-run mean estimate gives $R^2 = 0.956$ for seal condition, 0.448 for compliance, 0.114 for flow, 0.081 for supply pressure, 0.056 for centre-of-mass offset and -0.008 for mass. The latent therefore represents mainly seal condition. It is stable within a run but does not refine over time.

Low-dimensional adapters can lose information when factors act through different channels [Liu 25]. This differs from the rapid-motor- adaptation assumption that only task-relevant factor directions need to be retained [Kumar 21]: here the dominant seal-condition direction acts through pressure, which the controller does not integrate. Estimating named physical quantities would improve interpretation but requires an explicit plant model [Hu 26].

5.6 Latent-Delivery Test Through the Inverse Model

This section evaluates the explicit adaptation mechanism of RQ6, in which a causally estimated plant latent conditions a bounded correction. The earlier prototype applies that correction to the inverse model's output, the direct adaptation point while the inverse model remains in the command path. The closed-loop evidence below shows it has too little authority, and the prototype is retained because that result determines the redesign of Section 6.7.

The frozen ID produces $\boldsymbol{u}_{\text{base}} = \text{ID}(\boldsymbol{s}, \boldsymbol{r})$. The adapter adds a bounded latent-conditioned residual:

$$\boldsymbol{u} = \text{clip}\Big(\boldsymbol{u}_{\text{base}} + g_\psi(\boldsymbol{s}, \boldsymbol{r}, \boldsymbol{u}_{\text{base}}, \mathbf{z}_{\text{AAM}})\Big). \tag{5.10}$$

Training minimises valve-command MSE at the exact ROM tick, with the residual bounded to 0.15, and the ID, FD and policy are frozen. This route uses a separately trained 68-input latent-conditioned policy, so disabling the ID residual restores the baseline actuator path but not the baseline actor. Final AF removes that asymmetry (Section 6.7.3).

Figure 5.6 shows the causal convolutional stack and the bounded adapter that consumes its estimate. It documents the earlier prototype, not the final AF architecture. The final history-residual path is shown separately in Figure 6.6.

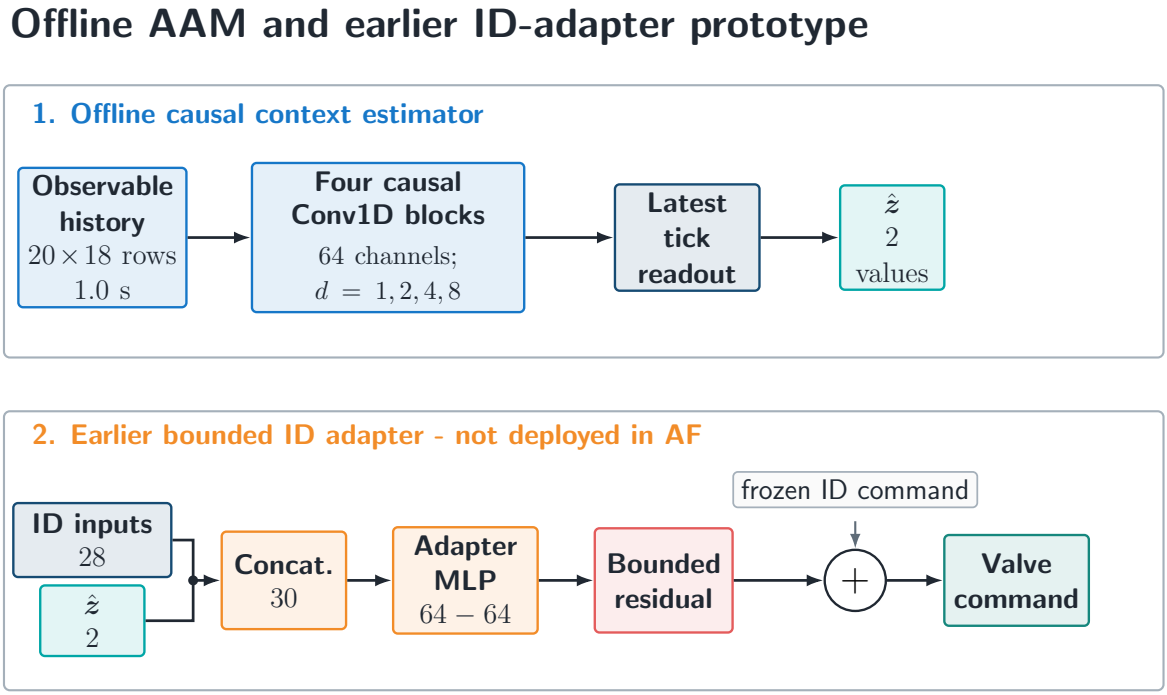

Figure 5.6: Earlier offline AAM and ID-adapter prototype. Four causal convolutional blocks map a 20-step history of 18 observable features to the two-value estimate $\hat{z}$. The bounded adapter combines this estimate with the ID inputs and corrects the frozen ID valve command. This prototype is retained as diagnostic evidence; deployed AF omits the inverse model and trains its history encoder and actor residual jointly under PPO.

5.6.1 Model-Level Adaptation Benefit

The residual cap is 0.15, and the frozen ID already explains about 84 % of valve-command variance (Section 5.2.2). The adapter can therefore make only small corrections. Section 5.6.3 tests latent information without this ceiling.

Three matched adapter arms use a zeroed, estimated or privileged latent. Across five seeds, both latent arms improve validation loss over the zeroed arm in 5/5 cases. Mean paired improvements are 0.0037 for the privileged latent and 0.0118 for the AAM estimate. The effect is consistent but small.

This is the one measurement in which online plant context acts without the closed loop in the way, and it is reproducible across every seed: a causally estimated latent carries information that the model consuming it can use, and the estimate is not worse than privileged context on this objective. The result is stated at the level of the adapter objective. It is not a tracking result, and the sections below measure what happens to it in the loop.

Masking pressure and force does not strengthen the result: the privileged latent helps in only 2/5 seeds and variance increases by about one order of magnitude.

5.6.2 Closed-Loop Evidence and Its Bound

Closed-loop ablations compare the full stack with zeroed, shuffled, frozen and mean latents, plus a sham plant change. None consistently improves lateral error, cut-height error or impacts.

Valve saturation limits the test: cylinder 1 is saturated for about 92 % of the scored window, cylinder 2 for 67–82 %, and cylinder 4 for 50–57 %. Control-effort ratios before and after the plant change are 1.005–1.009. A bounded additive correction has little authority under these conditions [Sofrony 24].

Reducing the lead cap lowers cylinder-1 saturation from 0.91 to 0.14 but triples cut-height error. The task is therefore close to the plant capability limit (Section 3.6.2).

5.6.3 Propagation of the Latent Advantage

The chain gate compares $SD_0(\boldsymbol{s}, \boldsymbol{u})$, $SD_z(\boldsymbol{s}, \boldsymbol{u}, \mathbf{z}_{\mathrm{SD}})$ and $SD_z(\boldsymbol{s}, \boldsymbol{u}, \mathbf{z}_{\mathrm{AAM}})$ under the frozen protocol with paired bootstrap intervals. Neither the true nor estimated latent improves extension or velocity over SD_0 with an interval excluding zero. Within this evaluated surrogate chain, replacing the causal estimate with privileged context does not recover extension or velocity performance. Estimation error is therefore not the dominant observed bottleneck in this test; the measurable bottleneck is delivery through the saturated command path. The latent advantage remains in pressure, as it was for the separately trained pair of Section 4.11.

Every stage of the chain up to the command holds: plant identity is present, a causal estimator recovers it, the estimate is as informative as a privileged one, and it improves a model that consumes it. The benefit stops at delivery. The correction is added to a command already resting against its own clip for most of the scored window, so the authority to act on the information does not exist at the point where the information arrives. A limit of that shape is not addressed by a better estimator or a larger latent, nor by reducing the demand, since lowering the lead cap buys headroom at the cost of tripling cut-height error (Section 5.6.2).

The prototype does not show that the plant estimate is useless. It shows that an estimate obtained from identification does not affect the plant if it is added to a saturated inverse command. The limit is delivery. Chapter 6 therefore removes the inverse model from the adaptive path and applies the residual to the valve command of the identified forward model.

5.7 Training and Model-Selection Protocol

Adaptation data are split 82/20/16 into training, validation and reserved test by complete run and factor configuration; windows do not cross run boundaries. The 16 test runs are not used for model or hyperparameter selection. SD_z and SD_0 are selected on validation rollout, the offline AAM on downstream rollout, and the earlier ID adapter on validation command error. Table 5.4 records the selected checkpoints.

Table 5.4: Provenance of each model in the modelling chain. Frozen components are loaded unchanged and never receive gradients in this chapter.

Model	Status	Selected checkpoint
Control ID	frozen	paired-control, Section 5.2
Strict FD	frozen	strict-excitation, Section 5.3
Command ROM	frozen	PhysX-calibrated, Section 5.4.1
SD_0 (no latent)	scratch	epoch 58
SD_z + encoder	scratch	epoch 48, frozen after the gate
Offline AAM	scratch	epoch 73, 1.0 s causal CNN
ID adapter	scratch	epoch 116

The ID adapter is trained and selected on recorded valve-command error. Each component is evaluated on its stated training contract. The adaptation residual of AF is deliberately absent from Table 5.4, because it has no supervised checkpoint: its encoder and head are optimised jointly under the reinforcement objective, and their provenance is a training run rather than an epoch number.

With the models identified, audited and frozen, the next chapter builds the controllers that consume them. It assembles the reference and non-adaptive stacks from the models above and introduces the final adaptive stack described in Section 5.6.3.

6. Control Framework and Simulation Integration

Chapter 5 left three actuator representations, each with its own contract and offline evidence. This chapter puts a controller in front of them and turns them into three experiments.

Section 6.1 describes the classical controller, because it is where the competence comes from even though it never runs at deployment. Section 6.2 specifies the policy: what it sees, what it emits, what stops it from emitting something dangerous, and what it is paid for. Section 6.3 explains how the first becomes the second, and reports a result about that transfer which changed the training protocol of this thesis. The individual stacks appear only after those three are fixed.

6.1 The DLS Teacher and Its Two Roles

6.1.1 Scope and Role of the Controller

No damped-least-squares layer runs underneath the policy at runtime and nothing is added to the policy's output; if the policy emits zero, the arm does not move. The DLS controller is nevertheless the reason the policies work at all, and it appears twice, both times at training or collection time and never at deployment (Figure 6.1). It is the *privileged teacher* whose behaviour the policies are distilled from (Section 6.3), and it is the closed-loop controller that drove the identification runs of Section 4.7. Calling it a nominal layer would misdescribe the deployed system and would make every reported result a property of a controller pair rather than of the policy.

6.1.2 Privileged Information

The teacher is *privileged* rather than better tuned: it reads two quantities that no deployable controller on this machine could obtain, namely the exact terrain height directly under the cutter, queried from the simulator rather than inferred from a forward-looking scan,

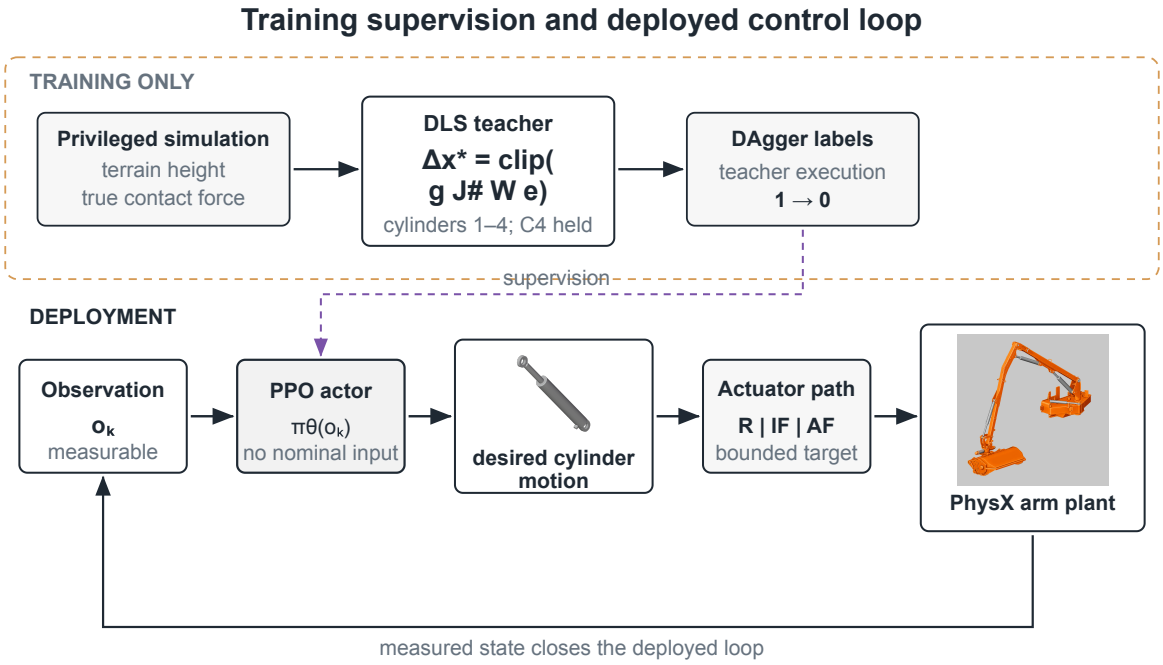

Figure 6.1: Training supervision and deployment. Simulator-only terrain and contact signals drive the DLS teacher, which supplies DAgger labels to the actor. At deployment only the lower feedback loop remains: measured observations pass through the PPO actor, the shared cylinder interface and the selected R, IF or AF actuator path before PhysX updates the arm. No nominal command is added to the policy output.

and the true cutter–ground contact force. Together these turn the hardest part of the task, knowing where the ground is underneath a 1.2 m flail head that is occluding it, into a lookup.

The teacher uses that privilege in a specific way. Rather than commanding a height, it runs a contact regulator: it maintains a vertical offset relative to the ground and adjusts that offset to hold the measured contact force near 250 N within a 100 N deadband, pressing down below the band, backing off proportionally above it, searching downward when contact is lost and retreating immediately on a force spike. The accumulated offset is clamped to $[-0.30, -0.05]$ m so the regulator can neither drive the head into the ground nor lift it clear.

The teacher's advantage is therefore not kinematic. Its inverse is the same DLS solution available to anything; what it has is contact sensing, and that is the quantity the student must infer from a noisy forward scan and its own contact history.

6.1.3 From Task Error to Cylinder Command

The rest of the teacher is conventional and is stated for completeness. A goal point is maintained in the truck frame: laterally it tracks the verge centre line, vertically it follows the measured ground height plus the contact regulator's offset. The task error is the difference between that goal and the measured cutter position, with the lateral component

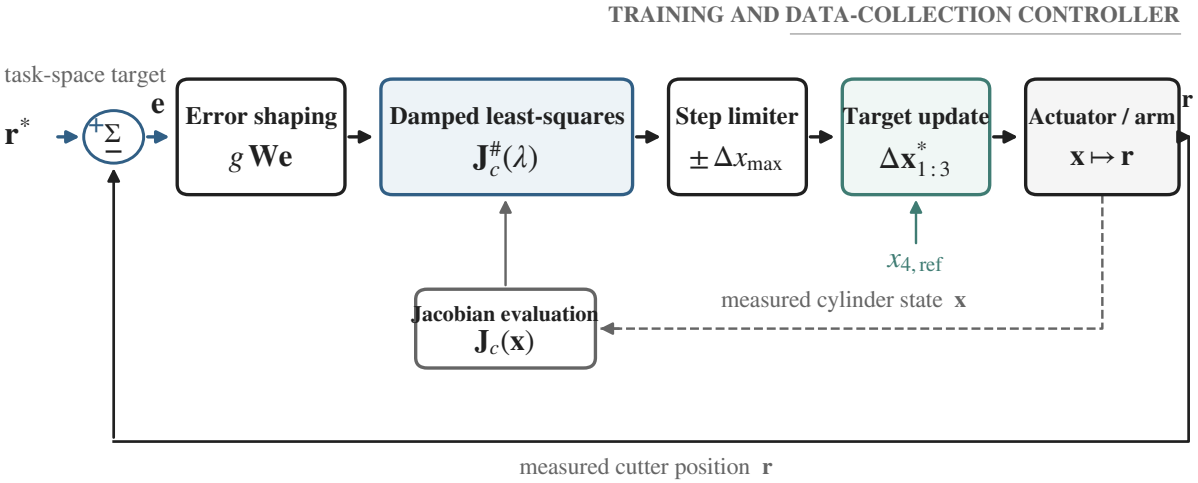

Figure 6.2: Feedback realization of the damped least-squares teacher of (6.1). The weighted task error is scaled by g, mapped to the first three cylinder coordinates by the measured damped inverse $\boldsymbol{J}_c^{\#}(\lambda)$, and clipped to $\pm\Delta x_{\max}$ per tick. Cylinder 4 is held at its head-down reference. Cutter position closes the task-space loop and the measured cylinder state updates $\boldsymbol{J}_c$ at every step. The loop is used only for teaching and data collection, never at deployment.

weighted by a factor of three, because on this machine lateral error is what the acceptance gate measures and vertical error is largely handled by contact.

That error is mapped to cylinder space through the damped least-squares inverse of the measured cutter Jacobian $\boldsymbol{J}_c$ of Section 3.2.2,

$$\begin{aligned} \boldsymbol{J}_c^{\#}(\lambda) &= \boldsymbol{J}_c^{\mathsf{T}}\left(\boldsymbol{J}_c\boldsymbol{J}_c^{\mathsf{T}} + \lambda^2\boldsymbol{I}\right)^{-1}, \\ \Delta\boldsymbol{x}^*_{1:3} &= \operatorname{clip}\Big(g\,\boldsymbol{J}_c^{\#}(\lambda)\,\boldsymbol{W}\boldsymbol{e},\ \pm\Delta x_{\max}\Big), \quad g = 0.30, \quad \Delta x_{\max} = 3\,\mathrm{mm}. \end{aligned} \tag{6.1}$$

with $\boldsymbol{W}$ the lateral weighting and the per-step clamp bounding how far any cylinder may be asked to move in one tick. Only cylinders 1–3 are solved. Cylinder 4 is held at the reference extension that keeps the head down, for the reason given in Section 3.1. The gain g is small on purpose: this is a stiff, heavy mechanism in contact, and a resolved-rate controller that closes quickly on the task error simply excites it.

6.1.4 Execution Through the Student Interface

One implementation detail determines whether distillation converges to something usable. The teacher's output $\Delta\boldsymbol{x}^*$ is not written to PhysX directly: it is converted into the same normalised four-value action the student emits and pushed through the student's entire action chain into the same plant, so every label the student imitates is by construction an action it can produce. A teacher writing targets directly to the plant would generate labels outside the student's reachable set, giving the imitation error a floor no training could remove.

6.1.5 Teacher Performance

Quoting the teacher's performance is necessary because distillation cannot exceed it and the gates of Section 3.9 were anchored to it. Table 6.1 reports the privileged teacher

Table 6.1: The privileged DLS teacher, measured on the stage-6 verge task through the command-ROM plant (128 environments, 480 steps, all eleven path families, seed 17). It is competent, not perfect, and several evaluation thresholds sit deliberately just outside what it achieves.

Quantity	Teacher	Gate
Mean $\lvert e_{\text{cut}} \rvert$	0.0234 m	≤ 0.040 m
p95 $\lvert e_{\text{cut}} \rvert$	0.1524 m	—
Mean $\lvert e_{\text{lat}} \rvert$	0.1221 m	≤ 0.300 m
Ground-contact fraction	0.7807	≥ 0.700
p01 head alignment	0.9638	≥ 0.900
Minimum head alignment	0.8317	—
p95 tool acceleration	64.9 m/s^2	≤ 120 m/s^2

measured on the stage-6 verge task through the ROM plant, over 128 environments and 480 control steps.

The teacher is competent but visibly imperfect: it loses ground contact about a fifth of the time and its cut-height tail is six times its mean. This is why Section 3.9 re-derived several thresholds, since a gate set above these numbers could not be passed by the ground-truth controller itself and would carry no information about the policy.

6.2 Policy Formulation

6.2.1 Policy Definition

The deployed controller is a memoryless Gaussian multilayer perceptron. It maps a 66-value observation to four normalised actions, one desired motion command per hydraulic cylinder, and that is the entire control law. Actor and critic are separate networks with hidden widths 256–128–64 and ELU activations, over empirically normalised observations. This description is of the base policy, and it is shared by all three stacks: the actor width is the same 66 in each, and AF deploys this network frozen and unmodified rather than a widened variant of it. Two things differ for AF and are treated where they arise. Its four actions are interpreted as valve commands rather than as desired cylinder increments (Section 6.7.2), and the deployed controller is this policy plus an additive residual whose own encoder reads a history window carried alongside the observation, leaving the base actor's input unchanged (Section 6.7.3).

The base policy is not itself a residual and has no nominal command to correct. The AF residual is added around this recoverable frozen base. Temporal inference is assigned to a separate encoder (Section 6.7.5), while the truck's forward motion remains scripted.

6.2.2 The Observation Vector

Figure 6.3 lays out the 66 channels by group. The grouping tells the story of what this controller has to reason about: nine terrain-height samples ahead of the cutter, three task-error channels, sixteen channels of cylinder state and remaining travel, sixteen describing the actuator interface, ten of pose and context, and twelve of model error.

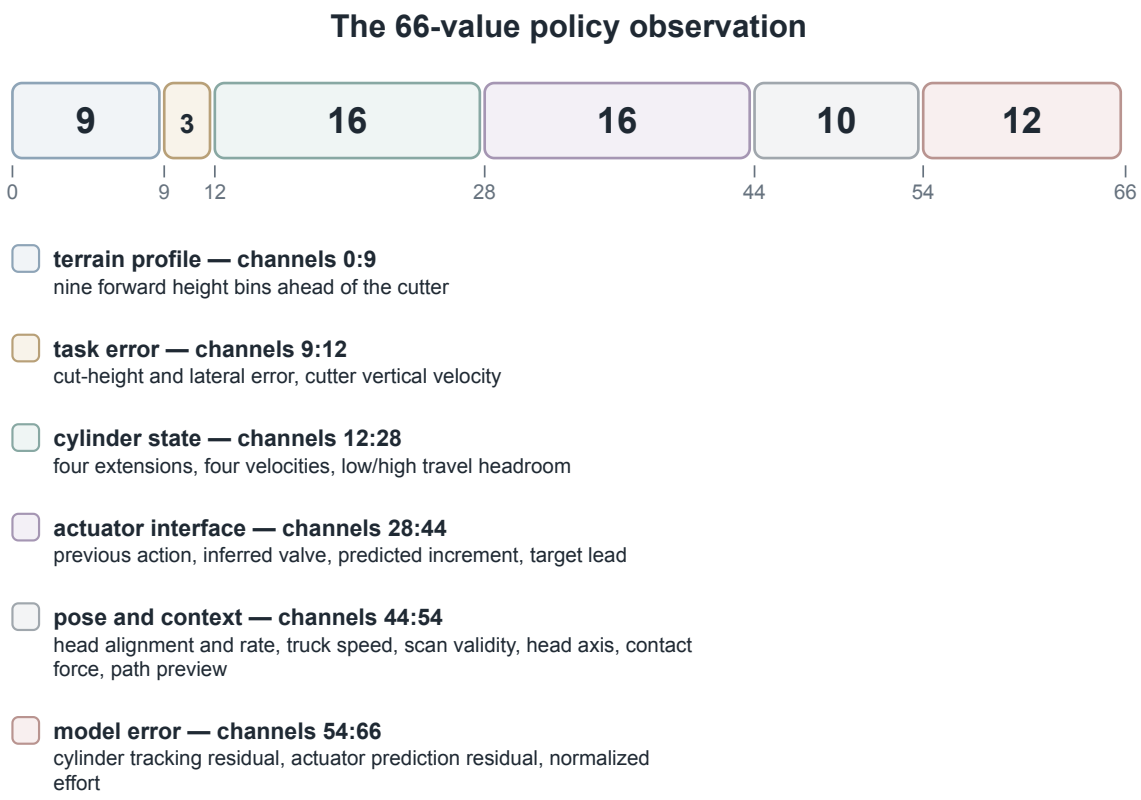

Figure 6.3: The 66-value policy observation, drawn to scale by group. The channels combine measurements, derived quantities and actuator-model estimates. No trajectory-family label, physical factor vector or latent estimate enters the actor. AF keeps this 66-channel input unchanged and carries its causal history separately.

The last group of twelve channels describes the actuator rather than the task, carrying its standing error as the cylinder tracking residual, the actuator prediction residual and the normalised cylinder effort. They exist as a direct consequence of the policy being memoryless. Forming an error requires differencing position across two steps and comparing against a prediction issued on the previous step, and a feedforward network can construct neither from a single observation, so without these channels the model error the policy is asked to compensate is absent from the input rather than merely hard to see. This was measured: across ten training blocks the p01 head alignment sat at 0.79–0.80 and did not respond to changes in the feedforward source or the levelling gain.

This group is also what makes the matched pair comparable. Under the ROM the tracking residual is near zero, because the ROM reaches what it is told, while under a learned ID–FD path it is the model's standing error. The same observation contract therefore gives each policy an honest view of its own actuator's shortcomings, which is what Section 6.9.2 requires.

That honesty cuts two ways. Because the residual channel is genuinely informative for IF and genuinely near zero for R, the same sixteen channels hand IF a closed-loop correction signal that R has structurally little use for, independent of which actuator is more accurate

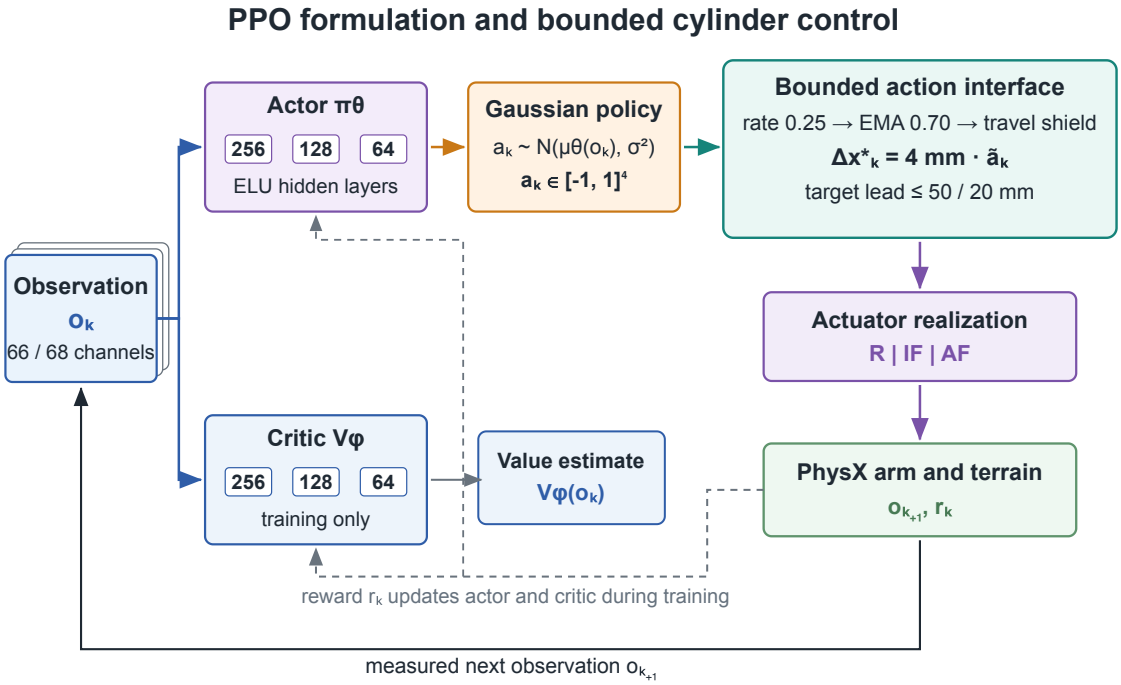

Figure 6.4: PPO actor–critic formulation and deployed control path. The actor maps the observation to a four-dimensional Gaussian action. Rate limiting, exponential filtering, the travel shield, command scaling and the target-lead cap bound that action before the selected R, IF or AF actuator path drives the PhysX arm. The critic and reward update are training-only; measured state closes the deployed loop.

open-loop. Some part of the R–IF gap in Chapter 7 may therefore reflect the value of correcting against a standing residual rather than the value of the underlying actuator representation, and the shared observation contract cannot by itself separate the two.

6.2.3 The Action and Its Bounds

The policy emits $\boldsymbol{a} \in \mathbb{R}^4$, sampled from its Gaussian at training time and taken at the mean for evaluation. Between that sample and the PhysX cylinder target sit five bounding stages. Figure 6.4 places them in the complete PPO actor–critic loop. The critic is used only during training; the actor and bounded action interface form the deployed controller. Each bounding stage can only *reduce* authority and cannot introduce motion the policy did not request.

The rate limit bounds $|\Delta\boldsymbol{a}|$ at 0.25 per control step and is held *constant* across every curriculum stage rather than used as a curriculum knob, by the reachability argument above: the teacher's own command stream has a structural per-step bound near 0.20, so a student limit below that makes part of the label set unreachable. The exponential filter ($\beta = 0.70$ at the final stage) smooths the command. The limit shield scales the command down, and only in the direction that consumes headroom, over the last 20 mm of cylinder travel, replacing the teacher's hard clamp into the learned range. The command scale converts the filtered action into 4 mm of cylinder travel per unit action, and the target integrator accumulates the increments in reference space, permitted to lead the measured extension by at most 18 mm.

This chain describes the cylinder-increment interpretation of the action, which R and IF use. Under AF the same bounded action is read as a valve command and scaled to span exactly the admissible valve range, so no further clamp binds at nominal. The bounding stages that act in cylinder space therefore do not have the same meaning on that stack, which is one reason its results are read against its own frozen base (Section 6.9.2).

That last cap is an anti-windup device and it is load-bearing. Relaxing it from 18 mm to 50 mm halved the lateral error on the two families demanding a sustained standing offset (`lane_change_hold` from 0.4833 to 0.2310, `boom_sweep` from 0.3745 to 0.1602), showing that the clamp was bounding the standing effort those paths need, while the same relaxation on cylinder 4 drove head alignment from 0.9507 to 0.4636. The final task therefore resolves the trade per axis, and this is the configuration every reported R result was produced under: 50 mm on cylinders 1 to 3 and 20 mm on cylinder 4.

The chain replaces the residual band, since exploration is bounded by how much motion a single action can command rather than by proximity to a baseline, and it fixes the plant's capability ceiling: four millimetres of cylinder travel per step is the entire authority budget, which produces the 0.240 m/s lateral rate limit of Section 3.6.2 from which the reference paths are derived.

6.2.4 The Reward Function

The reward of the final R task has ten bounded terms, listed in Table 6.2.[1] Three are positive and describe the task. Seven are penalties and describe how the machine must behave while doing it. All are bounded, and all retain a live gradient beyond the acceptance boundary, so that a policy already inside a gate still has a direction to improve in. Three rules govern the ten terms, each adopted after a measured failure described below.

No Gaussian Shaping

The earlier reward priced tracking as $w \exp[-(e/s)^2]$, which is effectively flat beyond about $2s$ and therefore supplies almost no gradient where the failing path families live. Every shaping term is instead linear or Huber in the error. Writing $x = |e|/s$, the two shapes used are

$$B(e; s, c) = \min(x,\, c), \qquad H(e; s, c) = \min\Big(c,\, \begin{cases} \frac{1}{2}x^2, & x \le 1 \\ x - \frac{1}{2}, & x > 1 \end{cases}\Big). \tag{6.2}$$

B has the same gradient at 0.5 m of error as at 0.05 m, so a policy far from the reference is still told which way to move; H buys quadratic precision near zero where cut height needs it, without a vanishing far field.

Bounded Terms

A term that can grow without limit can dominate the sum and destabilize the critic. Each contribution is clipped, which gives the reward a published budget: $6 + 12 + 4 = 22$ for the task terms against -24 for the safety terms. A non-finite value is replaced by -25, just below that worst case, so a numerical fault can never outscore legitimate behaviour.

[1] It replaces an inherited formulation carrying 53 weighted reward fields, at which size the optimum was not something a designer could reason about: three separate terms turned out to be working against the task without being noticed for thirteen training campaigns.

Table 6.2: The ten reward terms of the final R task, with weights and shaping scales. B and H are the bounded shapes of Equation (6.2). The lateral and head terms each combine a coarse and a fine scale, so far-field correction and near-field precision are priced separately.

Term	Weight	Shape and purpose
Cut-height tracking	+6	H, scale 0.04 m, cap 2.5; the primary mowing-quality term
Lateral tracking	+12	B, coarse 1.50 m and fine 0.60 m mixed 0.6/0.4; the primary line-following term
Ground contact	+4	pays for the cutter being in contact
Head alignment	−12	B on the deficit below 0.98, fine scale 0.02 and coarse scale 1.00 mixed evenly
Contact loss	−3	charges directly for loss of contact
Action rate	−1	B on the mean absolute command change, scale 0.25; penalizes chatter
Tool acceleration	−1	B, scale 300 m/s^2; mechanical-wear proxy on the cutter
Contact force	−1	B on force above 30 kN, scale 150 kN
Cylinder headroom	−2	B on encroachment inside a 0.02 m margin, scale 0.10 m
Hard impact	−4	indicator on the impact event

Fixed Sign Convention

The three task terms are non-negative and the seven safety terms non-positive. This is what makes worst-case (CVaR) weighting safe to add later: scaling a non-positive term by a factor ≥ 1 can only lower the return, whereas up-weighting a *positive* reward for the worst environments pays the policy to be bad, as measured in the earlier stack.[2]

One apparent inconsistency is deliberate. The tool-acceleration term saturates at 300 m/s^2 while the corresponding gate sits at 120 m/s^2. The reward scale is a saturation scale for a bounded penalty rather than a target, so the gate remains the binding criterion and the reward's role is only to keep a gradient against gross excursions. The action-rate term is held mild for a related reason: cylinder 4 owns head pitch, so a rate tax falls hardest on the fast corrections that keep the head down as boom attitude shifts.

Every term is logged individually and by name during training, because the whole difficulty is that total reward can rise while the system gets worse. The reward is therefore *not* the result criterion: promotion uses the eight physical gates of Section 3.9.

[2]Three terms of that earlier formulation were removed on measured evidence rather than preference. The *action-magnitude* penalty on mean(a^2) charged for precisely the sustained offset that `lane_change_hold`, `offset_return` and `boom_sweep` structurally require, so the policy learned to return to neutral, which is the one thing "hold an offset" forbids. The *head barrier* was replaced because the head group dwarfed lateral tracking, and the *clearance barrier* was disabled because measured contact, cylinder headroom and cut-height error proved more reliable signals of the same condition.

6.2.5 Optimisation

Policies are optimised with clipped-surrogate PPO over generalized advantage estimation, in the form of Section 2.7. The maintained configuration uses clip parameter $\epsilon = 0.2$, discount $\gamma = 0.99$, GAE parameter $\lambda = 0.95$, five learning epochs over four minibatches per update, an adaptive learning-rate schedule targeting a KL divergence of 0.02 from a base rate of 3×10^{-4}, gradient-norm clipping at 1.0, and a clipped value loss with coefficient 1.0. The terrain task collects 32 steps per environment per update. At 512 environments that is 16 384 transitions per update, chosen because the inherited 128-step horizon exceeded two minutes per update on the available GPU without adding useful batch diversity.

The final fine-tuning blocks depart from that configuration because they run under the reference anchor of Section 6.4 rather than an adaptive learning-rate schedule: clip $\epsilon = 0.1$, discount $\gamma = 0.97$, three learning epochs, fixed learning rates of 10^{-5} for the actor and 10^{-4} for the critic, entropy coefficient 0.001, an exploration standard deviation fixed at 0.15 and excluded from the gradient, and frozen observation normalisers.

Three of those departures are structural rather than tuning. The discount is reduced because the episode is long in simulation steps but locally controlled, so an action should be credited for correcting the terrain state it faces rather than for a return hundreds of steps later that the critic cannot resolve. The actor learning rate sits an order of magnitude below the critic's so that the actor is deliberately the slow variable. And the adaptive schedule is dropped because the anchored formulation already constrains the update through an explicit divergence bound; two competing step-size controllers would make neither attributable.

The exploration settings are also specific to this architecture. A zero action here means the arm does not move at all rather than that it tracks nominally, so the initial standard deviation was raised from 0.05 to 0.30 to let the policy discover the command directions before refining them. On the terrain task, where distillation supplies a competent starting point, it is reduced again and *held fixed* rather than initialized, for the reason taken up in Section 6.4. Exact values for any reported run are read from that run's frozen manifest.

6.3 Training Protocol and the Distillation Finding

6.3.1 Distillation by DAgger

Policies are initialized by imitation, not by reward. The privileged teacher is rolled out in the student's own environment and the student is fitted to the teacher's raw action by regression, in the DAgger scheme of Ross et al. [Ross 11]: the teacher's probability of *executing* the action, rather than merely labelling it, is annealed from one to zero over the first 70 % of iterations, so later training collects states the student itself visits. Labels are accumulated in a replay buffer of 262 144 transitions and refitted over three epochs per iteration in minibatches of 8 192.

Figure 6.7 shows a complete distillation run for the ROM stack, taken from the stored record. The behaviour-cloning loss falls by more than an order of magnitude while the teacher is still driving, reaches its minimum around the handover, and then *rises* once the student is alone. That rise is the distribution shift DAgger exists to expose rather than a failure: the student is now visiting states the teacher never chose, and the teacher's advice

on those states is new information. The closed-loop task metrics remain near the teacher's preflight values without matching them.

6.3.2 Objective Mismatch in Unconstrained Fine-Tuning

Unconstrained PPO fine-tuning from distilled checkpoints increased total reward while degrading the primary tracking objective, in the worst case by a factor of 12.9. The critic remained healthy, with explained variance in the 0.85–0.95 range. This motivates the anchored formulation of Section 6.4, which is used for the final policies.

The reward includes ground contact and cut height as well as lateral tracking. It can therefore increase while a gate-relevant tracking measure degrades.

This is a claim about two functionals over the same policy space rather than about the optimiser. Training maximizes the expected return $J(\vartheta)$, while acceptance is decided by the vector of physical gates $\boldsymbol{G}(\vartheta) \preceq \boldsymbol{g}^\star$ of Section 3.9. Writing $\mathcal{F} = \{\vartheta : \boldsymbol{G}(\vartheta) \preceq \boldsymbol{g}^\star\}$ for the feasible set, the measured content of the finding is

$$\arg\max_{\vartheta} J(\vartheta) \notin \mathcal{F}, \qquad \text{while} \qquad \vartheta_{\text{distilled}} \in \mathcal{F}. \tag{6.3}$$

Distillation lands inside $\mathcal{F}$ because the teacher was constructed to satisfy the gates, not to maximize J. Unconstrained ascent on J from that point therefore leaves $\mathcal{F}$ by construction. This is the optimiser working correctly on the wrong functional, and it is why the remedy is a constraint on the ascent rather than a repair of the reward.

Two consequences follow, both of them protocol. Unconstrained fine-tuning is inadmissible, and both the pre- and post-fine-tune checkpoints are reported in every case. And reward decomposition, critic explained variance, policy entropy and the gate metrics themselves are logged *during* training rather than only at the end, since a training curve reporting only the scalar reward would have shown a success here.

The general form applies beyond this machine: when a competent teacher exists and the evaluation criterion is not the reward, reward-based fine-tuning is not automatically an improvement step. Distillation transfers the intent, PPO then optimises the proxy, and the two point in measurably different directions. That divergence grows with distance travelled from the policy where proxy and intent were measured to agree, so the remedy is to bound the distance explicitly rather than to abandon reward optimisation.

6.4 Reference-Anchored PPO

Equation (6.3) says that reward ascent leaves the feasible set. It does not say that reward ascent is useless, only that it is unbounded. The construction of this section keeps PPO and constrains how far it may travel from a policy already known to satisfy the gates. Writing π_{ref} for a frozen reference policy and J for the expected return, each training block solves

$$\max_{\vartheta} J(\vartheta) \quad \text{s.t.} \quad \mathbb{E}_{\boldsymbol{o}\sim d_{\pi_\vartheta}}\Big[\mathrm{KL}\Big(\pi_{\text{ref}}(\cdot \mid \boldsymbol{o}) \,\|\, \pi_\vartheta(\cdot \mid \boldsymbol{o})\Big)\Big] \le \delta_{\text{ref}}, \tag{6.4}$$

with the expectation taken over the states the policy itself visits.

6.4.1 Critic Warm-Up

After distillation the actor already behaves sensibly, but the critic has never been fitted to the long terrain return, so the first PPO updates would move a useful actor using a value estimate that is not yet meaningful. The actor is frozen for 150 iterations while only the value loss is optimised, under the asserted invariant $\vartheta_{\text{after}} = \vartheta_{\text{before}}$. The observation normalisers are frozen at the same point, since otherwise the same raw observation is normalised differently from one iteration to the next and the actor's behaviour changes while its weights do not.

6.4.2 The Reference Anchor

Let π_{ref} be a frozen copy of the accepted warm actor. It is queried only on states the live policy visits and never supplies an executed action, so PPO remains the sole controller of the plant. The penalty is the *forward* divergence, which for diagonal Gaussians is available in closed form,

$$D_{\text{ref}} = \mathrm{KL}\big(\pi_{\text{ref}} \,\|\, \pi_\vartheta\big) = \sum_i \left[\log \frac{\sigma_{\vartheta,i}}{\sigma_{\text{ref},i}} + \frac{\sigma_{\text{ref},i}^2 + (\mu_{\text{ref},i} - \mu_{\vartheta,i})^2}{2\sigma_{\vartheta,i}^2} - \frac{1}{2}\right], \tag{6.5}$$

at the cost of one additional forward pass and no sampling. Because both scales are fixed at $\sigma = 0.15$, this collapses to a pure quadratic on the drift of the commanded action,

$$D_{\text{ref}} = \sum_i \frac{(\mu_{\text{ref},i} - \mu_{\vartheta,i})^2}{2(0.15)^2}, \tag{6.6}$$

which is the second reason for fixing σ: with a free standard deviation, entropy collapse shrinks D_{ref} while the commanded action moves, so the constraint would report compliance during the drift it exists to prevent.

The direction is load-bearing. Forward KL is mass-covering: it charges π_ϑ wherever π_{ref} has density and π_ϑ does not, and therefore penalizes the *abandonment* of reference behaviour. The reverse direction is mode-seeking and would permit the policy to collapse onto a subset of reference behaviour at no cost, which is the observed failure, in which one competence was dropped to buy another. It follows that the term is a behaviour-cloning anchor evaluated on the policy's own state distribution, rather than the reverse-KL regularizer of soft reinforcement learning. Its lineage is demonstration-regularized policy gradient [Rajeswaran 18], not the Boltzmann-optimum family.

The actor objective is

$$\mathcal{L}(\vartheta) = \mathcal{L}_{\text{PPO}} + \mathcal{L}_V - \beta\,\mathcal{H}(\pi_\vartheta) + \lambda_{\text{ref}}\,\mathbb{E}_k\Big[\mathrm{KL}(\pi_{\text{ref}}\|\pi_\vartheta)\Big], \qquad \lambda_{\text{ref}} = 0.10. \tag{6.7}$$

6.4.3 Trust Regions and Rollback

A penalty is not a constraint: at a fixed coefficient, a sufficiently large advantage always buys its way past. Feasibility is therefore also verified on the update taken. Two bounds are imposed,

$$D_{\text{update}} = \mathrm{KL}(\pi_{\text{old}}\|\pi_\vartheta) \le 0.01, \qquad D_{\text{ref}} \le 0.05, \tag{6.8}$$

and they do different work. D_{update} is the classical step-size control that keeps the surrogate a valid local model of J. D_{ref} is the new object: because the anchor is *fixed* within a block rather than moving with the policy, it bounds cumulative drift. A sequence of updates each individually satisfying D_{update} can still walk arbitrarily far from feasibility, and that is the trajectory the earlier campaigns took, monotonically, with the reward rising the whole way. Only a constraint referred to a fixed feasible point can observe it.

Enforcement is in two stages. During minibatch optimisation the epoch is early-stopped as soon as D_{update} exceeds its bound, and after the update both divergences are recomputed over the full rollout. If either bound fails, or any quantity is non-finite, the actor and its optimiser state are restored *atomically*. On a divergence rejection the critic fit is retained when finite, since the bounds constrain the policy rather than the value function; on a non-finite update both networks are restored. In the final block, five of one hundred logged updates were rejected, all on the reference bound and none on the update bound, so the drift caught was of the kind ordinary PPO clipping does not see.

6.4.4 The Block Ratchet

The reference is not fixed forever. Within a block it is frozen. When a checkpoint is scored and passes the gates, it is promoted to reference,

$$\vartheta_{\text{ref}}^{(m+1)} = \vartheta^\star, \qquad \vartheta^\star \in \mathcal{F}, \quad \vartheta^\star \in B_{\delta_{\text{ref}}}\left(\vartheta_{\text{ref}}^{(m)}\right), \tag{6.9}$$

and the actor's optimiser moments are reset, because momentum accumulated relative to the old anchor is not meaningful relative to the new one. The critic's are retained: it is fitting the same return, and its calibration is expensive to rebuild.

This is the structural point of the construction. The policy is at all times within a bounded divergence of a checkpoint that has been verified against $\boldsymbol{G}$, and the anchor advances only through verified checkpoints. The evaluation criterion therefore enters the training loop instead of appearing only at the end. The optimiser still ascends the proxy. It simply can never travel far from the last point at which proxy and intent were measured to agree. In this construction distillation is no longer the endpoint: it is $\vartheta_{\text{ref}}^{(0)}$.

Algorithm 6.1 summarises the procedure in five stages. The first two produce the initial verified anchor. Stages 3 and 4 repeat for each PPO block, and Stage 5 returns the last promoted checkpoint.

6.4.5 Theoretical Status

The components are standard and citable. The bound on D_{update} is conservative policy iteration and its trust-region descendants [Kakade 02, Schulman 15]; the atomic rollback is that literature's backtracking line search reduced to a single trial point; the penalty in Equation (6.7) is a Lagrangian relaxation of the constrained problem (6.4) at a fixed multiplier, with the rollback supplying the feasibility enforcement a fixed multiplier cannot.

What does not follow is a certificate. Bounding the divergence from a verified policy does bound gate degradation in form, since the chain rule of KL over trajectories, Pinsker's inequality and the boundedness of every gate metric give $|\mathbb{E}_{\pi_\vartheta}[g] - \mathbb{E}_{\pi_{\text{ref}}}[g]| \le 2G_{\max}\sqrt{H\delta_{\text{ref}}/2}$ for a metric g with $\|g\|_\infty \le G_{\max}$ over a horizon H (Appendix B.6 derives this bound step

Algorithm 6.1: Five-stage reference-anchored PPO procedure. The teacher is used only in Stage 1. All PPO rollouts are controlled by π_ϑ alone, and the frozen reference moves only when a checkpoint passes every gate.

input : teacher π_T; gate test $\boldsymbol{G}(\vartheta) \preceq \boldsymbol{g}^\star$; M PPO blocks; anchored loss and divergence bounds from Equations (6.7) and (6.8)
output : last promoted actor $\vartheta^\star$ satisfying the gate test

`// Stage 1 - Distil a feasible actor`
$\mathcal{D} \leftarrow$ DAgger states labelled by teacher mean action $\boldsymbol{y}^T$
$\vartheta \leftarrow \arg\min_\vartheta \mathbb{E}_{(\boldsymbol{o},\boldsymbol{y}^T)\sim\mathcal{D}}\left[\|\mu_\vartheta(\boldsymbol{o}) - \boldsymbol{y}^T\|_2^2\right]$
remove π_T from the controller `// all later rollouts are teacher-free`
`// Stage 2 - Warm the critic and set the first anchor`
freeze ϑ and the observation normalisers
for $j \leftarrow 1$ **to** 150 **do**
  roll out π_ϑ and set $\psi \leftarrow \psi - \eta_{\text{critic}} \nabla_\psi \mathcal{L}_V(\psi)$
end
verify $\vartheta_{\text{after}} = \vartheta_{\text{before}}$
$\vartheta_{\text{ref}} \leftarrow \vartheta$; $\vartheta^\star \leftarrow \vartheta$ `// initial verified anchor`
for *block* $m \leftarrow 0$ **to** $M - 1$ **do**
  `// Stage 3 - Train inside the frozen reference bound`
  freeze ϑ_{ref} for the whole block
  for *each PPO iteration in block* m **do**
    collect an on-policy rollout with π_ϑ as sole controller
    compute and normalise GAE; save snapshot $\mathcal{S}$; set $\vartheta_{\text{old}} \leftarrow \vartheta$
    for 3 *epochs over the rollout minibatches* **do**
      if $\mathrm{KL}(\pi_{\text{old}} \| \pi_\vartheta) > \delta_{\text{upd}}$ **then** stop the update epochs
      update actor and critic using Equation (6.7)
    end
    compute $(D_{\text{upd}}, D_{\text{ref}})$ on the full rollout `// full rollout`
    if D_{upd} *or* D_{ref} *is non-finite* **then**
      restore the complete snapshot $\mathcal{S}$
    else if $D_{\text{upd}} > \delta_{\text{upd}}$ ***or*** $D_{\text{ref}} > \delta_{\text{ref}}$ **then**
      restore the actor and its optimiser moments from $\mathcal{S}$; retain the new critic fit
  end
  `// Stage 4 - Evaluate the block and promote a passing checkpoint`
  score every saved checkpoint on at least three held-out terrain meshes
  aggregate each gate to the worst mesh and, for tracking gates, the worst path family
  $\mathcal{C} \leftarrow \{\vartheta_c : \boldsymbol{G}(\vartheta_c) \preceq \boldsymbol{g}^\star\}$
  if $\mathcal{C} \neq \varnothing$ **then**
    choose $\vartheta^\star \in \mathcal{C}$ using the fixed checkpoint selection rule
    load $\vartheta \leftarrow \vartheta^\star$ and set $\vartheta_{\text{ref}} \leftarrow \vartheta^\star$
    reset the actor optimiser moments; retain the critic state
  else
    leave ϑ_{ref} unchanged; do not promote the block
  end
end
`// Stage 5 - Return the deployable policy`
return $\vartheta^\star = \vartheta_{\text{ref}}$

by step). At the production horizon of 2808 steps and $\delta_{\text{ref}} = 0.05$ the right-hand side exceeds sixteen, which for a metric bounded in $[0, 1]$ is vacuous, and there is a further measure mismatch because the constraint is taken under the live policy's state distribution while the chain rule requires the reference's. The construction is therefore a principled restriction of reward ascent to a neighbourhood of a verified policy, not a guarantee: the gates certify each promotion.

6.5 R: Reference Actuator Baseline

R is the reference condition of this thesis. The policy emits desired cylinder motion, the frozen command ROM converts it to a cylinder position target, and PhysX closes the loop. There is no online adaptation anywhere in the stack. The truck is scripted over a 26 m strip at one constant speed drawn per episode from 2 to 7 km/h, the eleven path families of Section 3.6 are distributed across environments within a single run, and the terrain mesh is drawn per environment so that no policy can memorize one ground surface.

Its status is that of a reference condition rather than a competitor. The command ROM is a Simscape-trained model with a bounded closed-loop calibration against IsaacLab and PhysX, and ROM-plus-PhysX *is* the deployment plant, so R's actuator coincides with the plant. The consequence for how its tracking advantage may be read is stated once, in Section 6.9.2.

The policy is produced by the three-layer construction of this chapter: distillation from the privileged DLS teacher to a feasible student, critic warm-up with the actor frozen, and reference-anchored PPO in blocks under Equations (6.7) and (6.8). The final block produced checkpoints 280 to 329. The last iterate was not taken, because PPO can raise mean return while damaging a single path family. Candidates were scored on held-out meshes against the eight gates, aggregated to the worst mesh and, for tracking-critical gates, the worst path family. Chapter 7 reports the final common-protocol evaluation.

6.6 IF: Non-Adaptive Learned Baseline

IF subclasses the R task and changes exactly one thing. The policy still emits desired cylinder motion, but that command is now realised by the frozen inverse model of Section 5.2, which converts it to a valve command, followed by the frozen strict forward model of Section 5.3, which converts the valve command to a predicted extension increment. PhysX receives the same bounded cylinder target as before. Task, terrain families, meshes, observation contract, action chain, reward, gates and the PPO update are inherited unchanged, so the actuator representation is the only free variable between the two stacks.

The stack is the honest control for R, whose actuator coincides with the deployment plant, and it is the stack in which the inverse model is carried in the command path and is adequate there.

Robust is used here in the sense of Section 2.6: the stack was trained across the randomised plant family and applies one fixed response to whichever member of that family it meets, performing no online estimation and conditioning on no plant identity. It does not mean the policy is blind to plant variation, since the twelve model-error channels carry a standing

residual that is itself plant-dependent. What the stack lacks is not information about the consequences of plant variation but any representation of *which* plant produced them.

Two properties of the composed path shape its behaviour, and both were established offline. The valve command is accumulated rather than issued absolutely and is clipped to the per-axis limits $[0.95, 0.95, 0.80, 0.80]$, so the actuator path carries its own saturation boundary independently of the action chain. Only the predicted extension increment is integrated; the velocity output is retained as a diagnostic (Section 5.3).

The stored run manifest shows that the actor learning rate, entropy coefficient, seed, terrain seed, environment count, curriculum stage, frozen normaliser, and warm-up match R field for field. This makes the pair a controlled comparison. Chapter 7 reports the common-protocol results.

Figure 6.5 summarises the two non-adaptive actuator paths used as baselines. Both use the same policy interface and PhysX task; only the actuator representation changes.

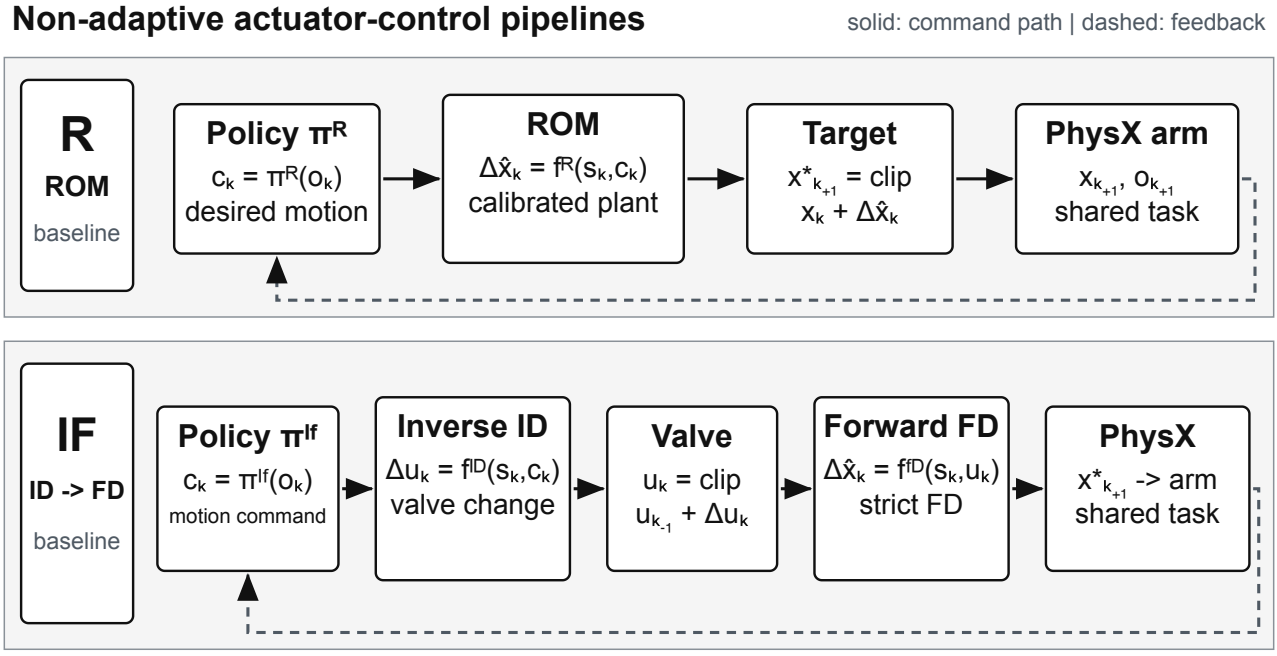

Figure 6.5: The non-adaptive R and IF control pipelines. R sends the policy's desired cylinder motion through the calibrated command ROM. IF composes the independently identified inverse and strict forward models before applying the same bounded cylinder target in PhysX. Both paths use the same observation, action limits and task.

6.7 AF: The Adaptive Stack

Table 6.3 previews what the rest of this section and Section 6.9.2 establish in full: AF shares the task, terrain, reward, observation and evaluation gates with R and IF, but not their action contract or their training protocol, and the comparisons that follow respect that split rather than papering over it.

Table 6.3: What is matched between AF and the R/IF pair, and what is not. Section 6.9.2 gives the full matching rules.

Dimension	Matched with R/IF?	Detail
Task, terrain, reward, gates	yes	identical across all three stacks
Base 66-channel observation	yes	AF carries its causal history separately
Action contract	no	AF emits the valve command directly, not a bounded c
Training seed	no	AF trains at a different seed
Actor learning rate	no	lower than the R/IF value
Frozen-actor warm-up	no	AF's base is the already-distilled policy, frozen; there i

AF is not constructed as a matched replacement for IF and is not used for a direct causal ranking against it. The adaptive stack answers a different question: whether a history-conditioned correction can improve a frozen direct-valve controller once the saturated inverse-model command path has been removed. The matched actuator-representation comparison remains R against IF; adaptation is evaluated within AF against its own frozen and capacity-matched controls.

AF places online adaptation on the forward model alone. The policy emits the valve command directly, the frozen strict forward model of Section 5.3 converts that command to a predicted extension increment, and PhysX receives the same bounded cylinder target as the other two stacks:

$$\pi_{\text{frozen}} + \text{history residual} \longrightarrow \boldsymbol{u} \longrightarrow f_{\text{FD}} \longrightarrow \text{PhysX}. \tag{6.10}$$

The inverse model is absent from the command path. The final AF stack consists of the frozen direct-valve base actor, a jointly trained causal history encoder and bounded residual head, and the frozen strict forward model. The offline AAM of Chapter 5 is not part of this runtime path. Task definition, terrain distribution, reward, evaluation gates and the downstream PhysX environment remain fixed. Because AF uses a different command interface, cross-stack claims are not based on AF-against-IF ranking. Within AF, disabling the residual exactly recovers the frozen base, and the zero-history arm preserves the same trainable capacity, providing the controlled comparisons used in Chapter 7.

That the inverse model is genuinely out of the loop is a verified property rather than a claim about intent. Under the direct-valve configuration the adaptation-site instrumentation reports zero valve elements attributable to the inverse model and zero adapter applications at that site across a complete run, where all three counters would be non-zero if the inverse model were in the path. The actuator key recorded inside the checkpoints still reads `id_aam_fd`, retained so that previously trained checkpoints remain loadable; it is a legacy identifier, not a description of the runtime path.

6.7.1 Omission of the Inverse Model

The inverse model's place in two of the three stacks follows from what each stack is for. The command ROM together with PhysX *is* the deployment plant (Section 5.4.1), so R uses the plant itself and fixes the reference condition. The inverse and forward models are identified independently of it, from the high-fidelity Simscape data of Chapter 4, and

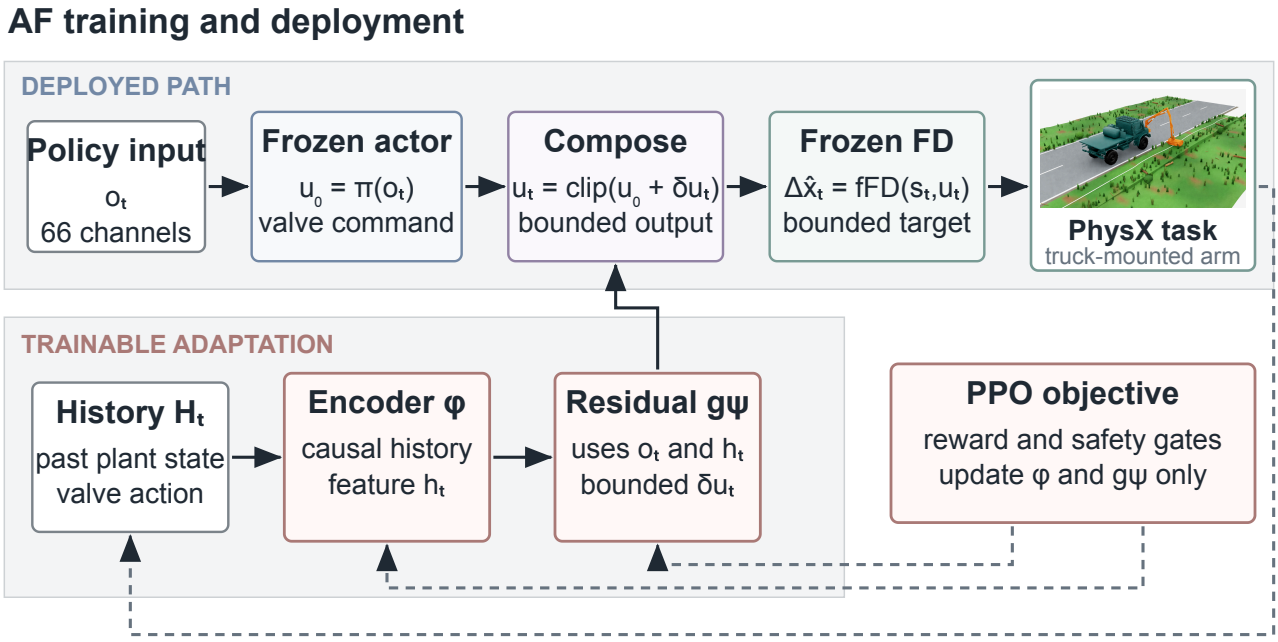

Figure 6.6: Training and deployment path of AF. The frozen base actor emits the valve command $\boldsymbol{u}_0$. A temporal encoder and residual head use causal state–action history to add the bounded correction of Equation (6.11). Only these two modules receive the PPO task gradient; the base actor and strict forward model remain frozen. The composed valve command drives the forward model and the shared PhysX verge-following task. The actor input remains 66 channels; the history window is carried separately and is not part of that count. Dashed paths denote PPO updates and rollout feedback. No privileged factor vector or inverse model enters the AF command path.

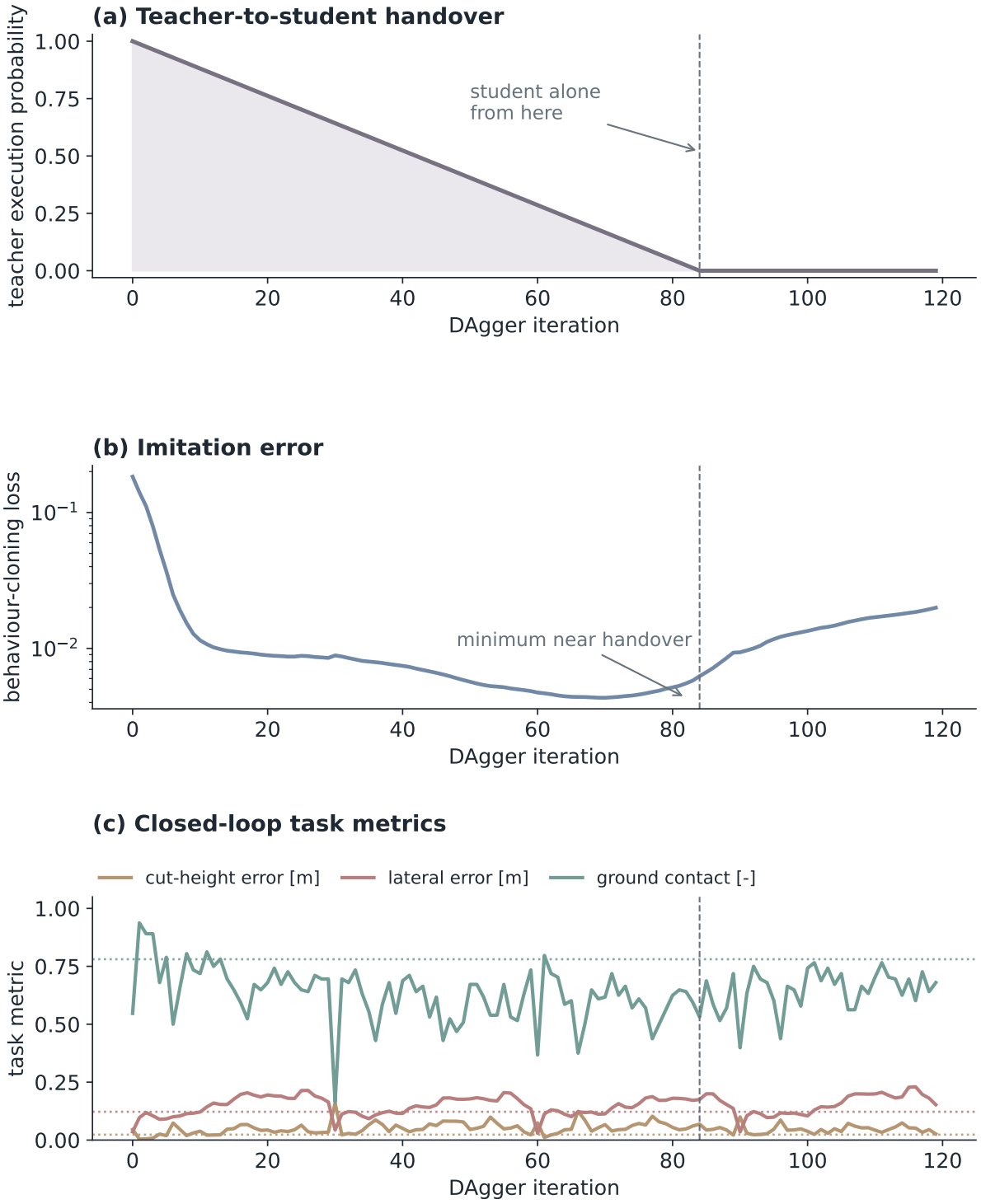

Figure 6.7: A measured distillation run for the R stack. (a) The teacher's execution probability falls to zero over the first 70 % of iterations. (b) Behaviour-cloning loss falls by over an order of magnitude under teacher execution, then rises as the student visits new states. (c) Cut-height error, lateral error and ground-contact fraction; dotted lines mark the teacher's preflight values from Table 6.1.

neither has seen the deployment plant during training, so IF prices the cost of substituting an independently identified pair for the plant itself. That question requires the inverse model: a policy commanding desired cylinder motion needs a map from that intent to a valve command, and supplying that map from identification data alone is exactly the substitution being priced. The R–IF gap is interpretable only because the composition is complete.

The adaptive stack asks a different question. The inverse model remains appropriate for IF and its one-step transfer is established in Section 5.2.2. In the adaptive command path, however, an additional correction must retain authority after the command is formed.

The same transfer study recorded the failure mode that later disqualified it for adaptation. The model predicts a valve-command *increment*, and the deployed command is the accumulated, clipped quantity $\boldsymbol{u}_k = \mathrm{clip}(\boldsymbol{u}_{k-1} + \Delta\boldsymbol{u}_k, -\boldsymbol{u}_{\max}, \boldsymbol{u}_{\max})$ with per-axis limits [0.95, 0.95, 0.80, 0.80], so under sustained demand the accumulation walks the command onto its own clip and holds it there. Fed its own predictions, the composite left the reference by more than a metre in *every one* of the 118 runs, with its pre-clip valve command at the limit on 89.1 % of elements while the ROM valve on the same runs never reached its bound. Closed-loop instrumentation over the 33-cell panel measures the same thing in the terms that matter for adaptation: the command sat at its pre-clip limit on 79.9 % of valve elements, 34.5 % of the corrections the adapter computed were clipped away after being computed, and on 37.9 % of elements the correction could not move the command at all.

The quantity being adapted and the quantity being saturated are therefore the same, and the consequence is not a small effect but an uninterpretable one: a null measured through such a path cannot distinguish a task that is not adaptation-limited from credit that was clipped before it could reach the metric. The disqualification is specific and retracts nothing claimed for IF, where the command is consumed as issued and saturation is a limit on authority rather than a defect of the map. It binds only where a second signal must be added to that command and survive the same clip.

The final stack therefore emits the valve command directly. The residual acts on the same bounded command as the policy, rather than on an increment that is subsequently accumulated and clipped.

6.7.2 Sufficiency of the Forward Model

Removing one of the two learned actuator models raises the question of whether the remaining one carries the path. Sufficiency here is a specific claim: the forward model must convert a commanded valve position into a one-step extension increment with genuine valve sensitivity, because that increment is the only quantity the controller integrates.

The evidence is the strict retraining of Section 5.3.1. Before it, zeroing the valve input at a 0.5 s horizon changed the rollout-error ratio to only 0.94, meaning the network had extrapolated smooth closed-loop kinematics rather than learned the valve's effect. After strict retraining on the open-loop plateau data the valve-zero error ratio rises to 1.81, extension rollout RMSE falls from 75.2 mm to 12.4 mm at 0.5 s and from 157.9 mm to 27.3 mm at 1.0 s, and per-axis one-step extension R^2 improves from [−5.7, 0.60] to [0.76, 0.98].

Sufficiency is established on the channel that is used and is not claimed beyond it. Three qualifications carry forward. Only the extension increment is integrated, the velocity output remaining weak on two axes and retained as a diagnostic. Zero-order hold remains competitive below 0.25 s, so the forward model earns its place at the horizons the controller actually spans rather than uniformly. And the valve authority the model represents is strongly operating-point dependent, varying by a factor of 15–75 across pressure quartiles and capable of changing sign, which is why a constant valve Jacobian was rejected.

6.7.3 The Residual Controller

Adaptation enters as an additive, bounded residual on the valve command:

$$\boldsymbol{u} = \mathrm{clip}\Big(\boldsymbol{u}_0 + \alpha \tanh\Big(g_\psi(\boldsymbol{o},\, \phi(\boldsymbol{H}_t))\Big),\, -1,\, 1\Big), \qquad \boldsymbol{u}_0 = \pi_{\mathrm{frozen}}(\boldsymbol{o}), \tag{6.11}$$

where ϕ is a temporal-convolution encoder over a window of (state, action) pairs and g_ψ is a small multilayer perceptron producing a per-cylinder correction. Only ϕ and g_ψ are trained: 53 836 parameters against a 59 588-parameter frozen base.

Three construction choices make the comparison in Chapter 7 a single-variable one, and each was verified rather than assumed. The final layer of g_ψ is zero-initialised and $\tanh(0) = 0$, so at construction the residual is exactly zero and the wrapped policy reproduces the frozen base bit for bit, at a measured maximum absolute action difference of 0.0. Every arm of an ablation therefore begins as the identical policy, and any end-state difference is attributable to the residual having been learned rather than to the task having been relearned. The correction is bounded by $\alpha \tanh(\cdot)$ regardless of what the head produces: with head weights drawn from $\mathcal{N}(0,\, 50)$ the output saturates at exactly α and remains finite, so a bad update degrades accuracy instead of emitting a non-finite action, which matters because a non-finite action written into the history buffer would poison the entire window. And the residual scale can be set to zero at evaluation time without touching any other part of the stack, in which case the stack reduces to the frozen base exactly.

This last property improves on the earlier latent-conditioned design, whose actor was itself a 68-input network, so that disabling the residual restored the baseline actuator path but not the baseline actor and the ablation was never quite single-variable. Under AF the base actor is the distilled policy unmodified, the residual carries the whole of the context dependence, and disabling it recovers the baseline in full.

The history window is carried in the observation vector rather than in a buffer held inside the module, because PPO calls the actor many times per update epoch and a module-held buffer would be advanced by the optimisation loop and desynchronise from the simulation. The causal convention is fixed throughout: the estimate at step t uses $\boldsymbol{x}_{t-k:t-1}$ and $\boldsymbol{a}_{t-k:t-1}$ and does not include step t.

6.7.4 Freezing the Base Policy

Freezing the base is structural rather than a tuning preference, and the alternative was closed by measurement before the residual route was adopted. Under a trajectory-regularised fine-tune with a reference-KL anchor the policy is stable but cannot acquire a new input dependence, because the anchor's reference is context-blind by construction,

so any context-dependent behaviour increases KL to that reference and is penalised. A gradient multiplier of 1000 applied to the context columns alone moved action sensitivity only from 0.001 % to 0.004 % of action scale over 400 iterations, and masking the anchor at $\boldsymbol{z} = \mathbf{0}$ reached 0.005 %. The anchor is not the blocker, however: without it PPO walks away from the distilled behaviour, trading line tracking for ground contact because that raises its objective.

A residual head has no anchor. It is new capacity whose only function is to depend on the latent, so nothing opposes that dependence. Measured on the same stack, the trained residual's action responds to its history at 3.35 % of action scale, roughly 840 times the anchored route.

6.7.5 Training of the Adaptation Encoder

The encoder ϕ and the head g_ψ are trained together under the reinforcement objective rather than against an auxiliary identification target. This is the concurrent scheme of Ji et al. [Ji 22]. The resulting representation is therefore optimised for control utility under the training objective, rather than for reconstruction of privileged plant parameters.

An offline dynamics objective does not by itself guarantee a control-relevant history representation. The adaptation module is therefore assessed behaviourally in Chapter 7, through action sensitivity to history and the closed-loop comparison with the frozen base.

6.7.6 Consistency With the Plant Dynamics

Two consistency requirements follow from the system model of Chapter 3, and both are met by construction.

The first is causality. Each history row carries observable quantities only, namely joint positions, joint velocities, cylinder pressure differences and the previous valve command. The privileged factor vector, the true latent, the reference, future rows and any hidden simulator state are excluded, and the estimate at step t is formed without step t, so nothing enters the deployed path that would be unavailable on the physical machine.

The second is a qualification rather than a guarantee. Under AF the pressure differences supplied to the encoder are recursive states of the frozen forward model, not measurements of the plant, so the encoder observes the plant through the forward model's representation of it and its estimate is consistent with the true dynamics only to the extent that the forward model is. Section 6.7.2 bounds that extent. This limit on the interpretation of the latent is carried into Chapter 7 rather than resolved here.

The stack does not claim a physically named estimate. The residual is trained to improve the task, and the quantity ϕ extracts is whatever serves that objective; it is not constrained to correspond to seal condition, compliance, supply pressure or any other named factor. Recovering named physical quantities would require an explicit plant model and a different estimator, noted as future work in Chapter 8.

6.7.7 Scope of the Claims

The stack is complete and its interfaces are verified. The residual authority α is a genuine tuned hyperparameter and is reported as one rather than presented as a natural constant.

The deployed value is $\alpha = 0.5$. At $\alpha = 0.15$ the residual is not merely less effective but actively worse than the base it corrects, by 28.3 % on cut-height error paired across 528 environments, with the interval on the paired difference excluding zero. A bounded correction needs enough authority to matter, and below that threshold it contributes noise rather than adaptation; no value of α should therefore be read as incidental. AF cannot be ranked against R and IF because the action contract is different. That restriction is a comparison rule, not a judgement of the adaptation result. The adaptation claim is made inside AF, on the identified forward model: frozen base, capacity-matched residual without history, and history-conditioned residual. Chapter 7 separates architecture-level evidence that the correction can reach the plant from method-level evidence about whether the improvement replicates across training seeds.

6.8 IsaacLab Integration and Timing

Four rates run concurrently and Table B.1 lists them: PhysX integrates at 120 Hz, the policy acts at 60 Hz under a decimation of two, and the learned actuator models advance on a 20 Hz grid of 0.05 s model ticks. The AF history encoder and residual are evaluated with the policy at 60 Hz. No single one of these is "the control frequency", and reporting a scalar rate for this system would misdescribe every one of the three stacks.

The relationship between the 60 Hz and 20 Hz layers is where integration errors hide, so it is fixed by construction rather than by interpolation. A learned actuator model is evaluated on every third control step and its predicted increment is held across the two intervening steps. The action presented to the model is the action of the tick on which it fires, not an average, and the target integrator accumulates in reference space, because integrating against the measured extension turns a static hold into a slow ratchet (Chapter 3).

Order within a tick is fixed and asserted at run time. The history encoder consumes the window ending at the previous tick and produces its estimate before the actuator path runs, so no quantity computed at the current tick can influence its own estimate. History buffers are per-environment and cleared on episode boundaries, since the vectorised environments reset asynchronously and a shared buffer would leak one episode into the next. Both properties are checked rather than assumed: a runtime check replays a trace and asserts bit-exact agreement between the online estimate and an offline causal recomputation. Checkpoint loading then compares the stored normalisation statistics, feature contract and history length of every learned component against what the environment expects, and aborts on a mismatch.

6.9 Constraints and Fair-Comparison Rules

6.9.1 Limits and Safeguards

Authority is bounded at four independent places, and the redundancy is deliberate because each bound protects against a different failure. The action chain of Section 6.2 limits the rate of change of the command, fades authority as cylinder travel headroom is consumed, and caps how far the integrated target may lead the measured extension. The actuator path clips the valve command to its per-axis limits. The environment terminates an episode on the safety conditions of Section 3.9 rather than letting it accumulate reward

in a state an operator would stop. And every learned component falls back to a defined behaviour on a non-finite output.

Preflight runs before any training or evaluation and refuses to start on a violated precondition. The check that does the most work asserts that the commanded reference stays inside the measured actuator authority ceiling of Section 3.6.2: the cylinders move the cutter laterally at most 0.240 m/s, so a path family demanding more is infeasible by construction and no reward, critic or optimiser change could recover it. Terrain is derived from that ceiling, and the preflight assertion keeps the two from drifting apart.

6.9.2 Matching Rules and Comparison Controls

This thesis does not present a three-way ranking of the stacks. Two distinct comparisons are made, each with its own matching rule and each carrying one claim; conflating them would produce a table whose rows are not commensurable.

The Matched Pair: R Against IF

The representation claim rests on R and IF, which agree on every recorded training field and differ only in the actuator path. Held identical across the pair are the verge task and its eleven path families, the terrain generator and the held-out mesh set, the reward function of Table 6.2, the action chain and its authority budget, the eight evaluation gates and their aggregation to the worst mesh and worst path family, episode length, actor and critic capacity, and the three-layer training construction of distillation, critic warm-up and reference-anchored PPO. That pair is a controlled single-variable comparison and is the only place an across-stack number is read as such.

One departure is declared. Because the command ROM plus PhysX *is* the deployment plant (Section 5.4.1), R's tracking advantage over IF is not evidence that a reduced-order model approximates the actuator better than an identified pair: one of the two defines the target.

Non-Comparability of AF and IF

AF does not share the action contract of the other two, since its policy emits the valve command itself where the other two emit a bounded desired cylinder increment that a downstream path realises. The quantity the policy chooses is not the same quantity in the two cases, so the action chain that makes the other comparison fair does not extend to this one. This is a declared property of the implementation, and it is why no row of any table ranks AF against IF on task error. The adaptive claim was never a claim about beating the non-adaptive stack.

The Paired Comparison: AF Against Its Own Frozen Base

The adaptation claim is made within the adaptive stack, against the frozen base policy it was built from, and the matching is exact. The arms share the frozen base policy, the residual architecture, the trainable capacity of 53 836 parameters, the training budget and the reward. Because the head of g_ψ is zero-initialised and $\tanh(0) = 0$, every arm begins as bit-for-bit the same policy, verified at a maximum absolute action difference of 0.0. The single manipulated variable is whether the residual may see the (state, action) history it

is conditioned on. A capacity-matched control is trained at every seed, so the comparison is made within a seed rather than across one at all five (Section 7.5.5).

The arms are paired environment by environment, so scenario variation is differenced out rather than averaged over. The adaptive and zero-history arms share the same architecture, trainable capacity, training budget, reward and evaluation conditions; the manipulated variable in that comparison is access to live history. The primary evaluation scores every arm at the saved checkpoint nearest its own training-reward peak, chosen by one rule for every arm and without consulting held-out performance. The fixed final checkpoint is reported beside it as a robustness analysis. Both are given in Section 7.5.

Fine-Tuning Budgets Must Be Matched or Reported

Comparing a distilled baseline against a separately optimised adaptive policy and attributing the difference to adaptation is not permitted, for the reason established in Section 6.3.2: reward ascent moves a policy off the evaluation criterion, so an unmatched budget is itself a treatment. The paired construction above satisfies this rule by design. The rule is stated because it constrains any future stack added to this comparison, not because the present one is at risk of violating it.

7. Experimental Evaluation: Representation, Robustness, and the Value of Adaptation

This chapter evaluates the closed-loop stacks defined in Chapter 6 as an evidence chain. Sections 7.2 to 7.4 first ask whether the actuator representation matters in closed loop and how much plant variation the trained controllers already absorb. Section 7.5 then asks what remains for a residual to recover inside AF, whether live history explains that recovery, and whether any component varies with cutter mass. Section 7.6 states the conditions under which the evidence supports online adaptation.

7.1 Experimental Protocol

7.1.1 Test Conditions and Comparisons

All closed-loop results use held-out terrain meshes, eleven trajectory families, a truck speed of 4 km/h, a 26 m scoring window, and sixteen reserved plant configurations. The matched comparison is between R and IF. Adaptation is assessed only inside AF, by comparing a frozen base, a capacity-matched residual without history, and a history-conditioned residual.

The results are paired by environment or by mesh and trajectory family. Error metrics are reported separately for lateral tracking and cut height because the two errors describe different aspects of the task. Confidence intervals are obtained by paired bootstrap. Checkpoints are selected using the validation or training criterion specified for each experiment and are then evaluated once on the held-out set.

7.1.2 Comparative Scope and Scoring

R and IF form the cross-stack comparison. AF has a different action contract and is therefore not ranked against that pair. Its results are interpreted only through within-stack comparisons.

Table 7.1: Closed-loop performance over 33 paired held-out conditions. The matched stacks differ only in the actuator representation.

	R	IF
lateral error, mean (m)	0.1026	**0.0703**
lateral error, median (m)	0.0392	**0.0281**
lateral error, p95 (m)	0.4664	**0.2665**
cut-height error, mean (m)	0.0084	**0.0063**
cut-height error, median (m)	**0.0013**	0.0031
ground contact	**0.9623**	0.9529
hard impacts	0.0947	**0.0905**
tool acceleration p95 (m/s^2)	39.90	**39.02**

One training run exists per stack, so the cross-stack numbers describe two deployed controllers and carry no estimate of policy-training variance. The matched pair differs only in the actuator path, which is what supports the representation claim; the seed-level statements in Section 7.5 come from five runs of a single stack and are paired within seed.

7.1.3 Scoring Horizon

All reported closed-loop results use the full 26 m trajectory rather than a shorter selection window. The earlier 8.9 m selection window covered only about one third of an episode and is therefore treated as an evaluation-protocol limitation, not as evidence about adaptation.

7.2 Closed-Loop Task Performance

The first comparison reports closed-loop tracking of R and IF on the held-out meshes and trajectory families.

The task is scored on two independent error channels, drawn side by side in Figure 7.1: lateral error is deviation from the planned cutting line in the horizontal plane, and cut-height error is deviation from the intended vertical cut. They are kept separate throughout because a controller can improve one while worsening the other. Ground contact, impacts and tool acceleration are reported alongside them as checks on motion quality.

Table 7.1 compares the matched pair over 33 held-out (mesh, family) conditions. Median errors are low for both stacks, and the p95 values move in the same direction as the means, so the improvement also reduces the larger tracking deviations rather than only the average.

Figure 7.2 resolves the aggregate by trajectory family. IF reduces mean lateral error from 0.103 m to 0.070 m and mean cut-height error from 8.4 mm to 6.3 mm over the 33 paired conditions. The advantage is not uniform: IF is better on six of the eleven families and R on five, with the largest difference on the double S-curve. The difference is concentrated on the more demanding families. Both stacks follow the simpler paths closely.

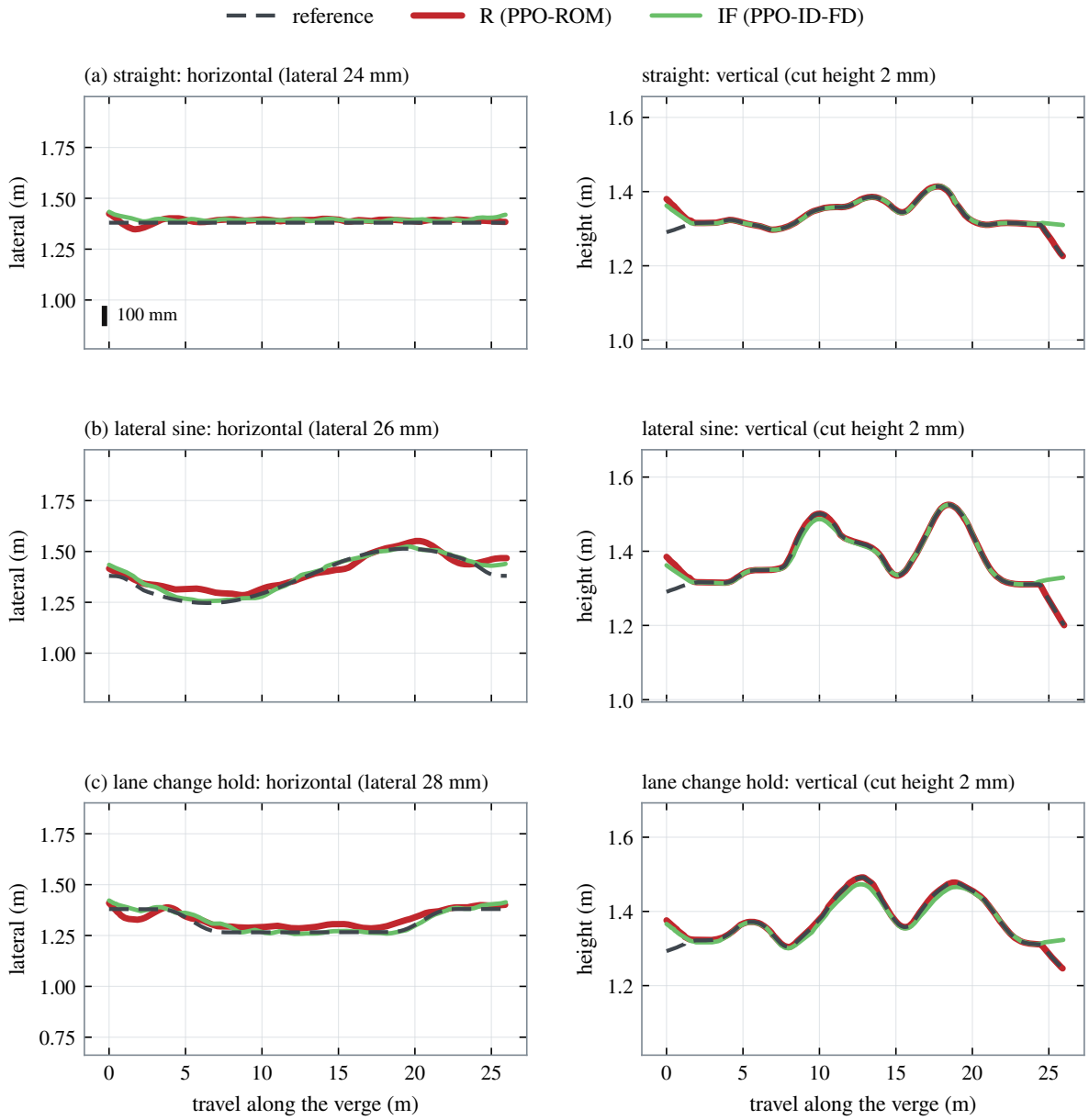

Figure 7.1: Cutter path against the reference on held-out mesh 1337, for three structurally different trajectory families: a flat line, a smooth periodic sweep and a step-and-hold. The left column is the horizontal plane and the right column the vertical plane over the same drive. The vertical reference is not recorded directly and is recovered as $z_{\text{ref}} = z_{\text{cutter}} - e_{\text{cut}}$, so both columns compare reference against actual. In the vertical panels the reference is largely hidden beneath both traces, which is what a millimetre-scale cut-height error looks like drawn at path scale.

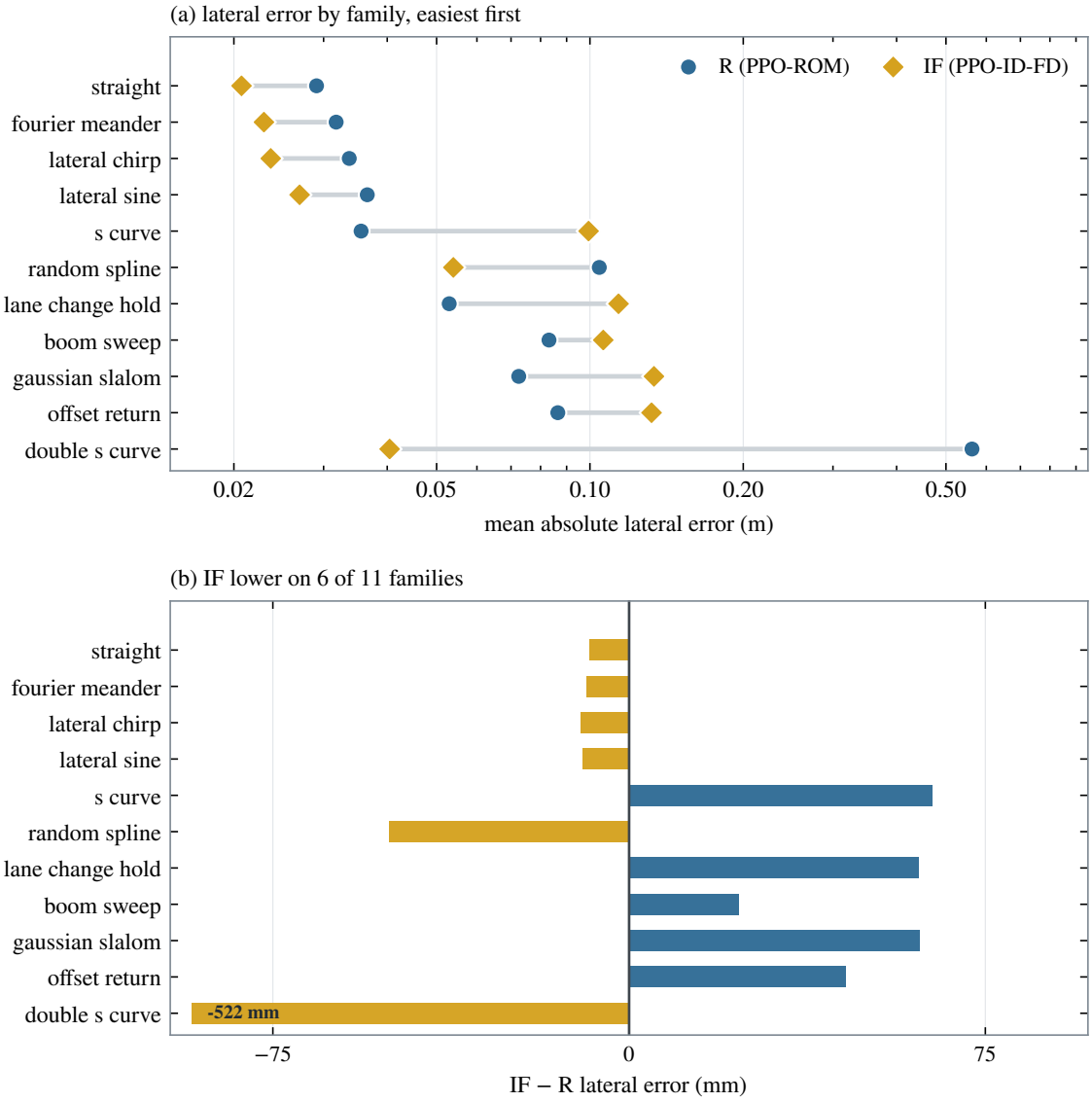

Figure 7.2: Lateral error by trajectory family for the matched pair. The aggregate difference is concentrated in the more difficult families.

Table 7.2: Transport latency along the actuation chain, in milliseconds. Values are medians over open-loop excitation runs at 60 Hz.

	excitation amplitude			
Link	0.05	0.15	0.35	0.70
command → differential pressure	117	117	117	150
command → cylinder velocity	183	167	133	117
command → cylinder extension	117	117	100	100
command → joint position	117	117	100	100

Table 7.3: Response to a 100 mm commanded step. Values are median times over sixteen environments.

Stack	detected response	rise	settled	overshoot
R	333	617	2633	57 %
IF	350	717	2183	73 %

Times are in milliseconds.

7.3 Actuator Representation and Dynamic Response

The same matched pair is now examined in the actuation channel: transport delay, closed-loop step response, and command effort.

7.3.1 Plant Delay and Step Response

Open-loop excitation measured a command-to-pressure delay of approximately 117 ms. Cylinder extension and joint motion began at the same time, which confirms that the dominant delay is hydraulic rather than kinematic. The similar values across excitation amplitudes indicate that this delay is a stable property of the simulated actuation chain.

The commanded-step experiment measures the closed-loop response to a 100 mm change in the reference line. Rise time is measured from 10 to 90 % of the step. Settling time is the first point at which the error remains within 20 mm for 0.5 s.

IF settles about 0.45 s earlier, whereas R has lower overshoot. The command rate limit alone requires at least 417 ms to complete this step, and both stacks require more than 2 s to settle. Reducing hydraulic delay alone would therefore not remove the observed settling time; the command rate, the policy response and the reference handling would have to change with it.

Closed-loop settling is much slower than the hydraulic delay. Command-rate limits and policy dynamics dominate the response time. The actuator path changes the shape of the step, not the main settling time.

7.3.2 Control Effort

The reversal count identifies rapid changes in command direction that the root-mean-square magnitude does not reveal, and Figure 7.3 pairs each stack's effort with the tracking error

Table 7.4: Control effort on a held-out mesh. The command and its rate are in normalised command units, so only their ratio between stacks is meaningful. Bold marks the better value of the pair.

Stack	RMS command (norm.)	command-rate RMS (norm. s^{-1})	reversals (s^{-1})	lateral error (mm)
R	0.586	7.55	40.0	120.1
IF	0.827	10.68	**10.6**	**26.8**

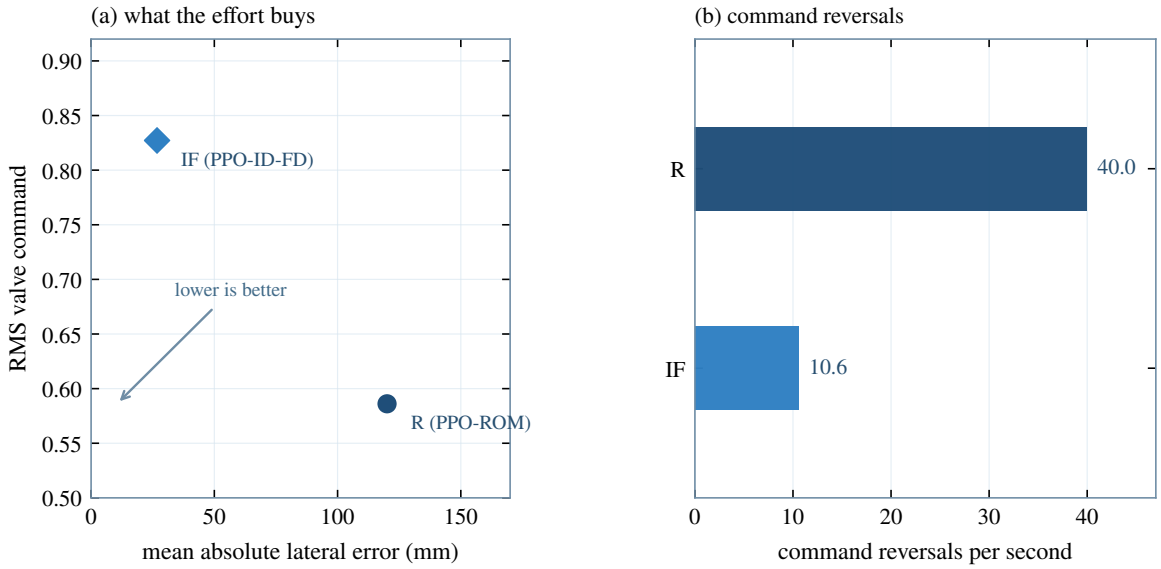

Figure 7.3: (a) Control effort against the tracking error it buys for the matched pair; the lower left is better. (b) Command reversals per second, which the root-mean-square command does not reveal.

it buys. A lower RMS command is not preferable when it comes with frequent reversals and larger path error.

R uses a smaller command magnitude but reverses almost four times as often. IF uses more actuation, reverses less, and tracks more closely. Effort is therefore reported together with the tracking it buys.

7.4 Generalisation and Robustness

The preceding sections use the held-out evaluation set. This section establishes the robustness already present before adaptation is considered, and then applies a common authority reduction to all stacks.

The matched stacks generalise to the held-out meshes, trajectory families, and reserved plant configurations used in this chapter. The randomisation sweep showed little change in tracking over the tested mechanical variation. Its hydraulic interpretation is limited because the inverse and forward plant does not apply the hydraulic factors, while the reduced-order model constrains them to its identification range.

The held-out evaluation therefore establishes transfer within the tested simulation population. It does not establish performance on arbitrary hydraulic conditions or physical hardware.

7.4.1 Delivered Stress

The sweep therefore supports robustness to the tested changes in tool mass and centre-of-mass height. It does not establish robustness to unbounded hydraulic variation. A common authority perturbation was used to compare all stacks more directly. This separates a reduction in realised actuator motion from the specific randomisation factors used during deployment.

7.4.2 Actuator Authority

Actuator authority was reduced by scaling the realised actuator displacement:

$$\Delta x_k^{\text{applied}} = \kappa \, \Delta x_k^{\text{actuator}}, \qquad \kappa \in (0, 1]. \tag{7.1}$$

Here, $\kappa = 1$ represents nominal authority and smaller values reduce the motion delivered to the plant. The same perturbation is applied after each stack's actuator representation, which makes the comparison independent of the individual factor parameterisations.

Table 7.5: Authority at which median lateral error first reaches 1.5 times its full-authority value.

Stack	nominal error (mm)	threshold (mm)	onset κ
R	40.69	61.04	0.10
IF	23.66	35.49	0.20
AF frozen	45.18	67.78	0.10
AF adaptive	27.03	40.55	0.30

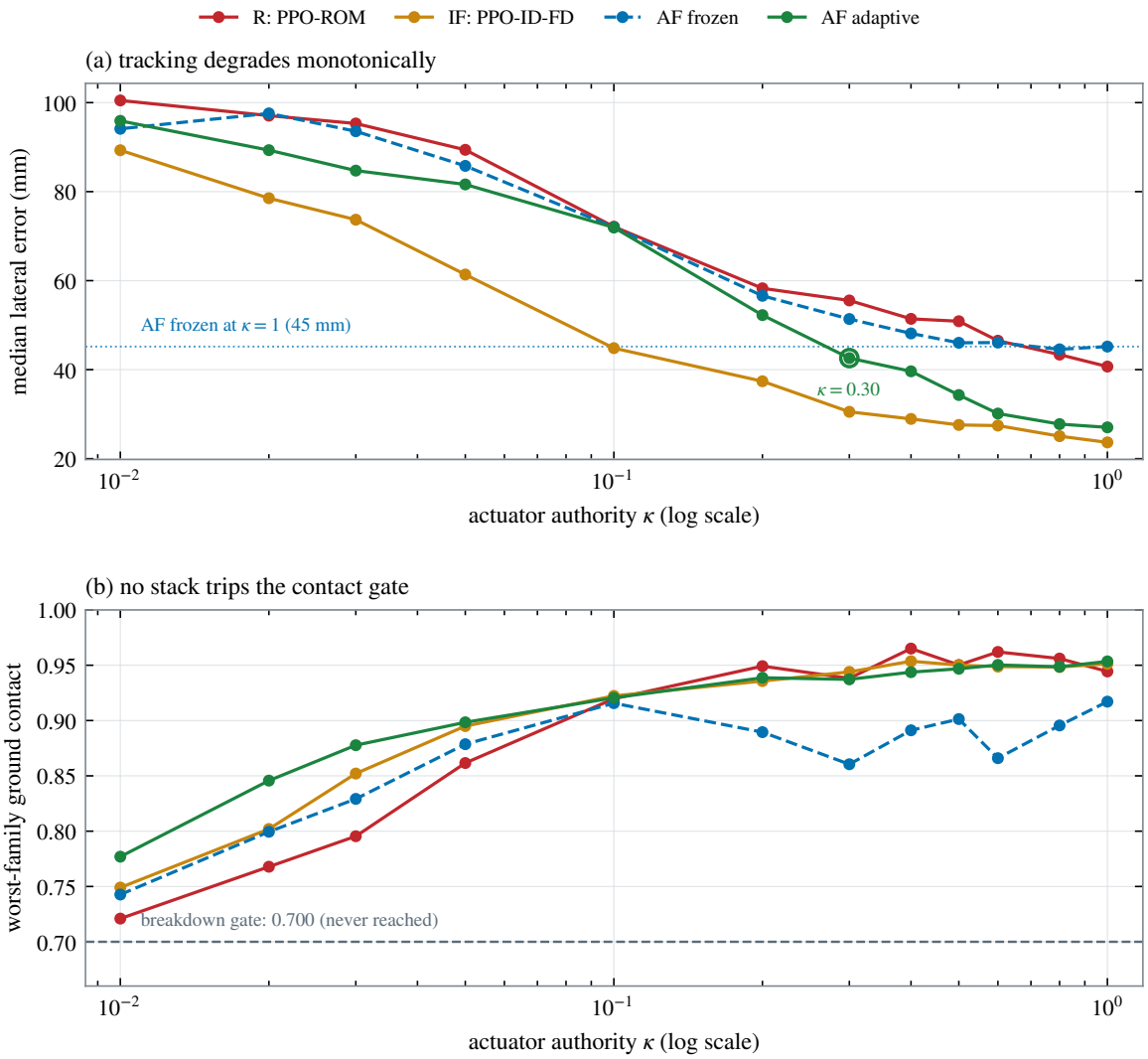

Figure 7.4: Lateral tracking and ground contact as actuator authority is reduced. The frozen and adaptive arms are compared within AF.

The learned controllers retain a broad margin under authority loss. A hard failure boundary was not reached within the tested range, so the experiment establishes an authority margin rather than a breakdown threshold. IF has the lowest lateral error at every rung of the cross-stack comparison.

No stack reached the contact or safety gate. Within AF, the adaptive residual holds the frozen residual's nominal lateral error down to $\kappa = 0.30$. This is an authority margin inside AF. It is not a ranking of AF against R or IF.

7.4.3 What Robustness Leaves for Adaptation

The robustness result sets the scale for the following experiment. The frozen controllers already absorb substantial variation, and the authority sweep does not reach a contact or safety failure. The adaptation study therefore measures residual performance differences beyond a robust learned controller. Modest, channel-specific, or seed-dependent gains are compatible with this setting because the baseline leaves only a limited deficit to recover. This is also why the capacity-matched control matters: with a small deficit to recover, extra residual capacity and live plant information can produce similar gains.

7.5 Evaluation of Online Adaptation Mechanisms

Two online adaptation mechanisms were built for this machine and they act at different points in the control chain. The explicit mechanism corrects the inverse command with an estimated plant latent and is measured in Chapter 5; its outcome is recalled in Section 7.5.1 because it determines the design tested here. The implicit mechanism conditions a bounded residual on a causal history window and applies it to the valve command. The experiments below remain inside AF and use the identified forward model from Chapter 5. They measure three distinct effects: improvement over the frozen base, improvement due to live history beyond matched residual capacity, and variation of the residual advantage with cutter mass. Keeping these effects separate prevents additional model capacity from being interpreted as online identification.

7.5.1 The Explicit Mechanism and Its Delivery Limit

The explicit mechanism is not re-scored in this chapter. Two results from Chapter 5 carry it. Its estimated latent improves the adapter objective over a zero-context control on all five seeds (Section 5.6.1), and it changes no closed-loop tracking metric (Section 5.6.2). The reason is measured rather than inferred: the cylinders it corrects are saturated for most of the scored window, so a bounded additive correction is clipped before it reaches the plant. The two results are compatible, and together they place the constraint at the interface rather than at the estimator. The sections below therefore test the same adaptation idea through a path that retains authority, and every comparison in them is between arms of that second mechanism.

7.5.2 Experiment and Measurement

The online-adaptation experiment changes cutter mass during each episode. The change time and direction vary across environments. The three compared arms share the frozen base policy, residual architecture, training budget, and reward. The base has no residual, the control has a residual with a zeroed history window, and the adaptive arm receives recent state-action history.

Table 7.6: Primary comparison of the history-conditioned residual with the frozen base, five training seeds at the training-reward-peak checkpoint and 880 paired environments. An asterisk indicates that the paired bootstrap interval excludes zero. Outcome labels follow the pre-registered rule: joint improvement means both channels resolved, Pareto improvement means one resolved and the other inside its non-inferiority bound.

Seed	cut-height gain	lateral gain	outcome
17	+39.1 %*	+84.9 %*	joint improvement
23	+35.5 %*	+4.0 %*	joint improvement
101	+74.7 %*	+5.9 %*	joint improvement
42	+43.4 %*	−16.1 %*	trade-off, fails
7	+4.5 %	+55.7 %*	Pareto improvement

The primary plant configuration assigns nominal and heavy cutter masses across the environment batch and changes the assignment at a jittered time. The varying event time prevents the residual from relying on one fixed schedule. Since the history-conditioned residual is active throughout the episode, this experiment tests continuous conditioning on the operating point as well as the response to the event.

Results use 880 paired environments across five held-out meshes. Lateral and cut-height effects are evaluated separately. An A/A series on this instrument places the noise floor at approximately 10 % on cut height and 1.5 % on lateral error, and those two figures fix both the resolution limit and the non-inferiority bounds used below. The paired design ensures that each arm is assessed under the same terrain, trajectory, and mass condition.

Two protocol decisions are pre-registered, and both are stated before the numbers because both of them move seeds.

Checkpoint. Every arm is scored at the saved checkpoint nearest the peak of its own training-reward curve, selected by one parser for every arm, control and adaptive alike. Training reward is measured on the training plants and the held-out meshes are never consulted in the selection, so this is model selection on training signal rather than on the test set. The rule exists because a fixed final checkpoint catches late-training collapse: seed 7 peaks at 14729 mean reward and ends at 6957. The final checkpoint is retained and reported below as the robustness analysis.

Success. An arm succeeds if one tracking metric improves with its paired interval excluding zero while the other either improves or stays inside its non-inferiority bound. The stricter bar, both channels improving with intervals excluding zero, is reported alongside it because it is the criterion the earlier method versions used. The relaxed rule was fixed because the stricter one records a change smaller than the instrument can resolve as a failure.

7.5.3 History-Conditioned Residual Across Training Seeds

The percentages compare paired environments. A positive value means that the adaptive arm has lower error than the frozen base for the median paired environment. The bootstrap marker indicates that the paired difference is resolved by the stated statistical test.

Four of five seeds succeed, so the median seed passes. Three of them improve both channels with resolved intervals. Seed 7 improves lateral tracking by 55.7 % while its cut-height

Table 7.7: The same comparison at the fixed final checkpoint, no selection step, 880 paired environments.

Seed	cut-height gain	lateral gain	outcome
17	+62.6 %*	+78.6 %*	joint improvement
23	+40.9 %*	+21.3 %*	joint improvement
101	−1.1 %	+28.6 %*	Pareto improvement
42	−23.3 %*	−18.2 %*	both worse
7	−84.8 %*	−72.1 %*	both worse

change of +4.5 % stays inside the 10 % bound, which is an improvement at no measurable cost rather than a trade. Seed 42 is the failure: it gains 43.4 % on cut height and gives up 16.1 % laterally, past the 1.5 % bound. Under the stricter both-channels bar the same table returns three seeds. The residual therefore improves the median training run, and Section 7.5.5 asks what in the residual produces that improvement.

7.5.4 Robustness to the Checkpoint Rule

The pre-registration commits to reporting the fixed final checkpoint alongside the selected one, and to treating a disagreement between the two as a result rather than as a detail.

Three of five seeds succeed at the final checkpoint under the same rule, and two under the stricter both-channels bar. The median passes on either reading, so the direction of the result does not depend on the checkpoint rule. Its margin does. Seeds 17 and 23 improve both channels under both rules and seed 101 improves at no measurable cost under both. The rules disagree on seed 7, which succeeds at its reward peak and degrades both channels by 84.8 % and 72.1 % at the final checkpoint, after the reward collapse that motivated the selection rule. Seed 42 fails under both, trading channels at the peak and losing both at the end.

7.5.5 Capacity Versus Live History

The capacity-matched control separates the effect of the residual from the effect of the history it receives. This comparison is necessary because a larger residual can improve control even when it receives no information about the current plant, and Table 7.6 cannot distinguish the two.

One control is trained per seed with the history window zeroed and the input width unchanged, so the control and the adaptive arm share the frozen base, architecture, trainable parameter count, training budget, reward, optimiser and plant schedule. The single manipulated variable is whether the numbers the encoder reads are the real history or zeros. Both arms are scored at the same selected checkpoint as Table 7.6.

Read the two halves in order. Added capacity alone reproduces most of Table 7.6: the zero-history control beats the frozen base on four of five seeds under the same success rule, with cut-height gains of 63.9, 77.2, 90.9 and 95.0 %. The increment from live history is not present on the median seed. Only seed 42 improves both channels against its own control, and it is the one seed where added capacity achieved nothing. Seeds 17 and 7 buy

Table 7.8: Decomposition of the residual advantage at the selected checkpoints. The left pair isolates added capacity, the right pair isolates live history at equal capacity. Same 880 paired environments and the same success rule as Table 7.6.

	capacity alone (base → control)		live history (control → adaptive)	
Seed	cut-height	lateral	cut-height	lateral
17	+63.9 %*	−1.2 %	−21.1 %*	+80.3 %*
23	+77.2 %*	+42.2 %*	−21.2 %*	−22.8 %*
101	+90.9 %*	+56.2 %*	−5.2 %	−29.9 %*
42	+1.3 %	−62.2 %*	+38.4 %*	+124.9 %*
7	+95.0 %*	+39.7 %*	−46.0 %*	+12.2 %*

large lateral gains with cut-height losses past the 10 % bound, seed 23 is worse on both channels, and seed 101 shows no resolved improvement. One of five seeds succeeds on this axis, so the median seed fails it.

The final-checkpoint analysis does not overturn this and is not uniform. On the full mesh set only seed 17 carries a scored control arm there, where live history improves both channels by 51.3 and 81.6 %. On the three-mesh subset all five are available and again one of five succeeds: seed 17 improves both channels, seeds 23 and 42 trade cut height for lateral tracking, and seeds 101 and 7 are worse on both. Live history therefore produces a resolved closed-loop benefit on particular combinations of seed and checkpoint.

One mechanism is a candidate and is not settled by these data. Every arm in this table conditions on the full observation over the window, 70 features per step and 1400 numbers in total, of which almost all describe terrain and task state rather than plant identity, while the head that consumes them holds 53k parameters. Rapid motor adaptation avoids exactly this by never training its encoder with the policy gradient, fitting it instead by regression onto a low-dimensional privileged vector [Kumar 21]. Restricting the encoder input to plant-relevant channels is the direct test and is left to future work (Section 8.5).

The two statements the section supports are therefore separate. Added residual capacity reproduces the median gain against the frozen base, so live history is not required for that median result. Live history is required for seed 42 at the selected checkpoint, for seed 17 at the final checkpoint, and for the additional lateral term in the dose study of Section 7.5.6. The method-level statement is that a bounded residual on the valve path improves tracking. The adaptation-specific statement is that live history is not the source of the median gain under this encoder, and is the source of the gain on particular seeds and on the load-dependent lateral channel.

7.5.6 Dependence on Cutter-Mass Dose

The cutter-mass dose was varied and compared with the 1.0 condition, in which the event schedule runs but the mass does not change. In the higher-dose runs, half of the environments start at the heavier mass and the later flip exchanges the nominal and heavy assignments. The dose therefore changes both the starting mass distribution and the event magnitude. This experiment measures dependence on cutter mass; it does not isolate a transient response to the mid-episode change.

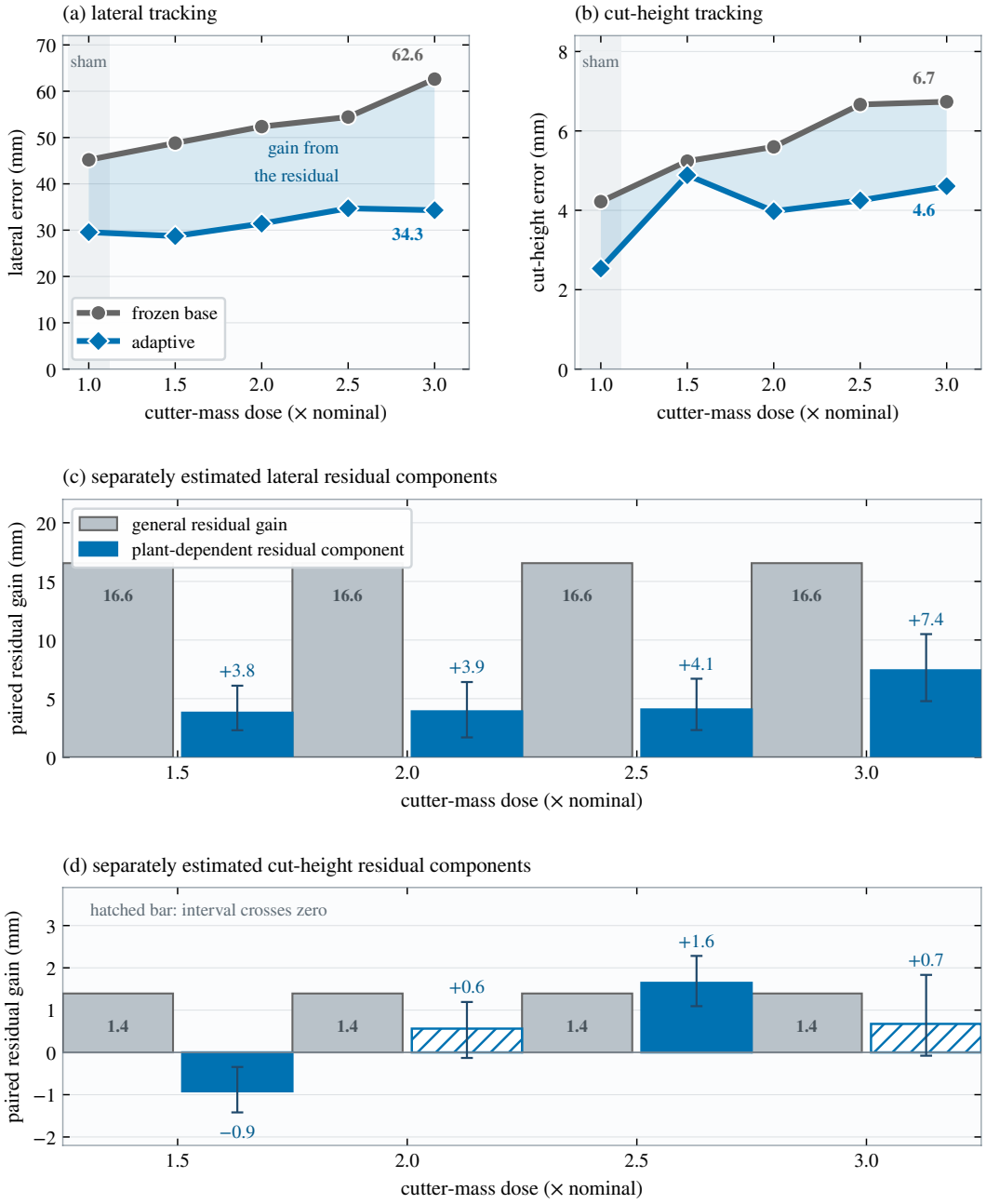

Figure 7.5: Task performance across cutter-mass dose on both metrics. The 1.0 condition measures the residual advantage at nominal mass. The additional component at higher doses includes static mass dependence and any response to the later mass exchange, which this design varies together. The lateral gap in panel (a) widens with every step of the dose; the cut-height gap in panel (b) does not, and the two panels carry their own scales. Panels (c) and (d) place the separately estimated components beside each other on each metric rather than implying an exact additive decomposition. Bars are drawn open where the bootstrap interval crosses zero, which happens twice on cut height and never on the lateral axis.

Table 7.9: Lateral residual advantage across cutter-mass dose. The last two columns compare each dose with the 1.0 condition. Every interval excludes zero.

Cutter-mass dose	total gain (mm)	gain above 1.0 (mm)	interval (mm)
1.0 (sham)	16.557	n/a	n/a
1.5	20.045	3.796	[2.303, 6.105]
2.0	20.487	3.912	[1.693, 6.415]
2.5	20.490	4.073	[2.318, 6.697]
3.0	23.705	7.410	[4.785, 10.503]

Table 7.10: Cut-height residual advantage across cutter-mass dose. The total gain excludes zero at every dose. The excess above the 1.0 condition does not: it is negative at 1.5 and unresolved at 2.0 and 3.0.

Cutter-mass dose	total gain (mm)	gain above 1.0 (mm)	interval (mm)
1.0 (sham)	1.393	n/a	n/a
1.5	0.474	−0.919	[−1.419, −0.345]
2.0	1.958	0.562	[−0.131, 1.192]
2.5	3.016	1.644	[1.093, 2.282]
3.0	2.005	0.674	[−0.078, 1.833]

The lateral advantage above the 1.0 condition rises from 3.8 mm at a 1.5 dose to 7.4 mm at a 3.0 dose. At the largest dose this component is about 31 % of the total lateral gain. The result establishes a plant-dependent component of the residual advantage. It does not identify that component uniquely with live history or with the mid-episode event.

Cut height behaves differently, and Table 7.10 reports it on the same design. The residual improves cut-height tracking at every dose and all five paired gains exclude zero, so the benefit itself is not in question. The part above the 1.0 condition is: it is negative at a 1.5 dose, unresolved at 2.0 and 3.0, and resolved only at 2.5. Panel (b) of Figure 7.5 shows the same behaviour in the levels, where the adaptive arm gives up most of its nominal margin at the 1.5 dose and the gap does not then widen with the dose, and panel (d) shows it in the component the lateral result is stated on: the same decomposition applied to cut height changes sign at the lowest dose and returns an interval containing zero at two of the remaining three. The cut-height gain is therefore treated as a general residual effect rather than as a mass-dependent one. The two axes also carry different amounts of error to remove: the frozen base runs at 45 mm to 63 mm laterally against 4.2 mm to 6.7 mm in cut height, so a dose term of the lateral size could not appear on this channel.

The internal pressure state of the three stacks through the same load change is reported in Appendix B.5. It is an actuator-model state rather than a calibrated pressure, so it supports the shape of the response and not its level, and it is not used to quantify any result in this chapter.

7.6 What Determines Effective Online Adaptation

The experiments identify four conditions that govern whether online adaptation adds value beyond a robust learned controller.

A control-relevant deficit must remain. The transferred controllers already absorb substantial plant variation and retain a broad authority margin. Adaptation has little to recover when that robustness keeps the tracking error nearly unchanged.

The relevant variation must be observable. The identifiability results of Chapters 4 and 5 show that plant information is present only under suitable excitation. A history encoder cannot recover a factor that the measured trajectory does not reveal.

The correction path must retain authority. The explicit mechanism produced no closed-loop effect because clipping removed the correction before it reached the plant, although its estimate was the more informative of the two. The residual on the valve path has bounded authority and can change the realised command.

History must add value beyond residual capacity. The residual improves tracking on four of five seeds, and an equal-capacity residual reading zeros improves it on four of five as well, so the median gain is capacity rather than history. The increment favours live history on seed 42 here, on seed 17 at the final checkpoint, and in the lateral dose response. It is conditional, and estimator quality alone does not establish it.

The answer the chapter supports has two levels and both have to be reported together. Online plant context is usable, and the level at which it is usable is set by the interface it acts through. The explicit mechanism improves the model that consumes its estimate and changes nothing in the loop. The bounded residual on the valve path improves task tracking on the median training seed under the pre-registered rule and under the fixed final checkpoint, which is a method-level result about the residual controller, and most of that median gain is residual capacity: live history separates from a capacity-matched control on one seed of five, and a load-dependent lateral component appears in the dose study. Effective online adaptation here therefore depends jointly on a residual deficit surviving robust training, on that variation being observable, on the correction path retaining authority, and on live history adding something beyond capacity. Stated without those four, the residual result will be misread as plant identification.

8. Conclusion and Future Work

This thesis asked how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation adds a further increment to the robustness those learned models already provide. The six research questions of Section 1.4 are answered in the order they were asked, which is also the order in which the answers depend on one another.

8.1 Summary of Findings

The findings are stated in the order of the research questions. The first four concern how hydraulic actuation is learned, transferred and controlled, and the last two concern robustness and online adaptation. Figure 8.1 makes the dependency between those experiments explicit.

The inverse and forward representation identified in Simscape transfers to the deployment simulation without retraining (RQ1, RQ2). The inverse model was trained on Simscape data alone and evaluated on all 118 IsaacLab deployment runs, drawn entirely outside that training distribution, where its teacher-forced one-step extension error is millimetre-scale and nearly flat across domains: 2.36 mm on the combined runs against 3.36 mm on nominal. This claim is limited to the one-step regime used by the policy. The inverse model remains frozen downstream. The reduced-order model received a small bounded calibration against the deployment plant.

Predictive accuracy must be accompanied by a control-authority test (RQ1). Input ablation and open-loop excitation showed that a learned actuator must be tested for dependence on the command, not only for prediction error. With the accepted excitation design, removing the valve input increases rollout error by a factor of 1.81. This is a general validation requirement for learned models used in control.

Two actuator representations both support closed-loop control, and they do not produce the same controller (RQ3, RQ5). The command-conditioned reduced-order model reaches $R^2 = 0.998$ on the internal valve state and 0.97 on extension, with a diagonally dominant extension–command Jacobian of condition number 2.14, and it

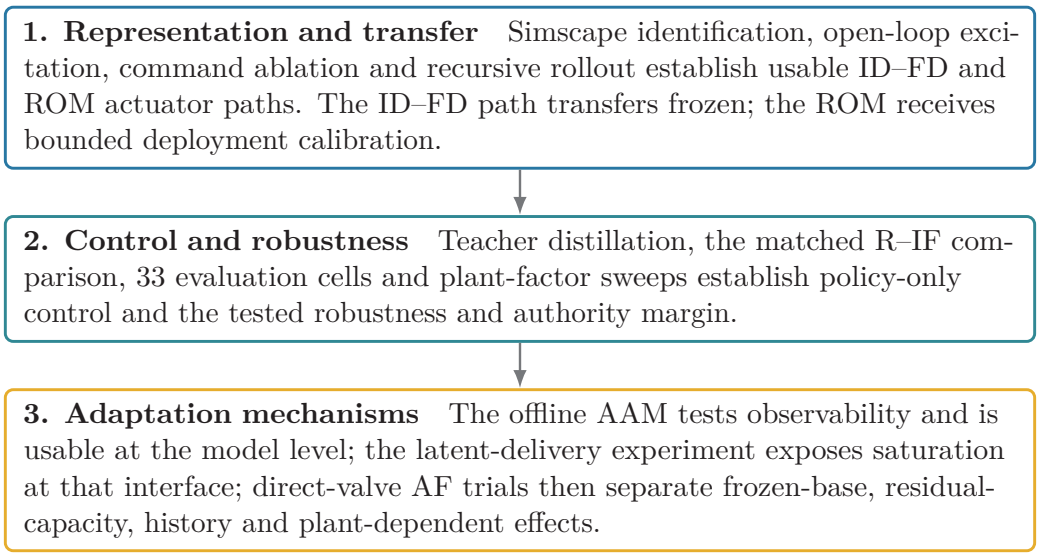

Figure 8.1: Experimental evidence chain used to answer the research questions. Each stage constrains the claim made at the next stage; the final adaptation claim is interpreted only after transfer and robustness have been established.

runs inside GPU-parallel training. The inverse–forward chain composes two separately learned models in series, which could have compounded their errors and did not. The two representations differ in the control strategy they induce from the same policy interface. R spends the least command and reverses about 40 times per second, against 10.6 for IF, which converts greater actuator effort into lower lateral error. The choice of representation therefore affects the resulting controller and is not an implementation detail.

R and IF agree on every recorded training field and differ only in the actuator path, so the difference above *is* attributable to the representation. AF is a separately trained run at a different seed, a lower actor learning rate and no frozen-actor warm-up, reported as a deployed controller rather than a ranked competitor. Only one training run exists per stack, so seed variance is not estimated even for the matched pair, and claims about the adaptation path are carried instead by the paired within-stack comparison of Section 7.5.5.

A learned policy can replace the classical controller at runtime (RQ4). The damped-least-squares controller supplies privileged demonstrations during training and is then removed; the distilled policy emits desired cylinder motion directly into the learned actuator and is the entire deployed controller. Over 33 evaluation cells IF holds a mean lateral error of 70.3 mm, a median of 28.1 mm and 95.29 % ground contact (Table 7.1). The deployed policies remain anchored to the demonstrated behaviour because the reward is an imperfect proxy for the evaluation gates.

The robust stacks retain a broad margin without online adaptation (RQ2, RQ5). Every closed-loop result is measured on held-out meshes, eleven trajectory families scored individually, and plants drawn from sixteen reserved factor configurations disjoint from training by construction. Under those conditions IF holds the lateral criterion in 32 of 33 cells at a median of 0.0281 m, and the frozen inverse model beneath it, identified on Simscape alone, transfers to all 118 deployment runs with millimetre-scale one-step error, flat across domains. This is the broadest claim the work supports and it requires no plant estimation.

The factor sweep supports the tested mechanical variation, but not unbounded hydraulic variation. On the inverse and forward path, hydraulic factors are not applied; on R they remain inside the identification range (Section 7.4.1). Reducing realised actuator motion to one per cent of nominal did not trigger the safety or ground-contact gates. The result therefore establishes an authority margin, not a hard robustness boundary (Section 7.4.2).

Online plant context is usable, and whether it reaches the closed loop is set by the correction interface (RQ6). The explicit mechanism estimates a plant latent from recent interaction history. Its latent-conditioned adapters beat a zero-context control on the adapter objective on all five seeds, and the same correction changes no closed-loop metric, because it is added to an inverse command already resting against its limit for most of the scored window. Moving the correction onto the valve path restores the authority the residual needed. At the pre-registered endpoint the history-conditioned residual there succeeds on four of five training seeds, so the median seed passes; three of those seeds improve both channels. The fixed final checkpoint returns three of five under the same rule and preserves the direction, so the result does not depend on the checkpoint rule. Capacity-matched controls then locate the source of that gain: an equal-capacity residual reading zeroed history also beats the frozen base on four of five seeds, so added residual capacity rather than live history accounts for the median improvement. Live history separates from that control on one seed at the selected checkpoint and on a different seed at the final one, and the cutter-mass dose experiment resolves a plant-dependent lateral component that grows with load, while coupling the starting mass with the later event. The supported finding is therefore a method-level gain from the bounded residual, with online adaptation as a conditional increment whose evidence is seed- and channel-specific.

Table 8.1 separates the experiments that support this conclusion. This distinction prevents the closed-loop gain of an enlarged residual network from being reported as evidence that live history caused it.

8.2 Design Implications

The results carry design consequences for each of the three parts of this work: how a hydraulic actuator is identified and transferred, how a policy is trained to act through it, and how plant adaptation is fitted around both. They are stated below in that order, because each stage constrains the next.

Identification and Transfer

Design the excitation for identifiability, and test the model by input ablation. Task-control data can make state and command nearly redundant. Dedicated excitation makes command effects identifiable. A learned actuator should therefore be accepted on both prediction accuracy and loss of performance when the command input is removed.

Embed known physics and bound learned corrections. The reduced-order model propagates pressure, force, velocity, and extension through explicit updates. Its learned terms are bounded, preventing unphysical state changes inside the training loop. This structure also represents the pressure dependence of valve authority directly.

Prefer a frozen model with a bounded, gated correction to retraining. The inverse model transferred across the deployment plant family at millimetre scale in every

Table 8.1: Experimental decomposition of the adaptation claim. The rows answer different questions at different levels of the control chain and are not interchangeable evidence.

Experiment	Controlled comparison	Supported conclusion
Latent adapter objective	estimated, privileged and zero latent on the adapter's own objective	Online plant context is usable at the model level: both latent arms beat the zero-context control on five of five seeds.
Latent-delivery test	estimated, privileged and zero latent through the inverse-model correction path	Better plant context does not recover extension or velocity; saturation leaves insufficient correction authority at that interface.
Primary seed table	history residual against its frozen base, paired within seed	The residual improves tracking on four of five seeds and the median seed passes; three of five at the final checkpoint.
Capacity control	history-conditioned residual against an equal-capacity zero-history residual	Capacity alone reproduces most of that gain; live history separates from zeroed history on one seed of five at each checkpoint rule.
Cutter-mass dose and sham	plant perturbation against the corresponding sham condition	A plant-dependent lateral component is resolved, but starting mass and the later exchange remain coupled.

domain, from 2.36 mm to 3.36 mm one-step, and with an ordering that does not follow plant severity: the nominal plant scores worst and the combined draws best (Table 5.2). It was therefore deployed frozen everywhere downstream. Where a deployment offset did have to be absorbed, it was absorbed by a correction capped at 3 mm and gated to vanish at zero command, with every hydraulic parameter frozen and the original data retained as replay. That construction preserves the identified physics, keeps a commanded hold a hold, and confines the deployment-specific part of the model to a component whose authority is bounded by construction.

Control Methodology

Constrain policy updates when the reward is not the evaluation criterion. The physical evaluation gates and the reward are not identical. A forward-KL anchor to a verified reference policy bounds policy drift, and the anchor advances only through checkpoints that satisfy the gates. Reward terms should represent quantities that are also treated as evaluation constraints.

Where no baseline is inherited, the action contract is the safety argument. The deployed policy is not a residual: a zero action leaves the arm stationary, and nothing is summed with the policy output at runtime. Authority is therefore bounded by the chain the action passes through rather than by proximity to a nominal controller, with a rate limit, a filter, a shield that fades authority as travel headroom is consumed, a command scale, and an anti-windup cap on how far the integrated target may lead the measured extension. That last cap is load-bearing and axis-dependent: relaxing it from 18 mm to 50 mm halved lateral error on the families needing a sustained standing offset, while the same relaxation on the head-pivot cylinder drove alignment from 0.9507 to 0.4636.

Plant Adaptation

Measure identifiability and correction authority before training an estimator. The identifiability envelope determines whether plant identity is present in the data. The delivered-stress check determines whether a perturbation reaches the plant and whether a correction can influence the command. Both tests should precede estimator training.

Where the estimate is used also matters. A dynamics model can integrate a changing estimate over time, whereas a memoryless policy receives that change as input variation. Filtering on the policy path may therefore be useful, but its appropriate time constant remains to be determined on the deployed stack.

8.3 Contributions

Section 1.4 stated six contributions. Their standing at the end of the work is as follows, with the supporting evidence as summarised above rather than repeated.

Transferable hydraulic actuator learning is delivered: both representations were identified from Simscape and realised in IsaacLab, and the inverse model transfers without deployment-specific retraining. The **reduced-order actuator model** is delivered and carries everything downstream, being accurate enough to serve as the deployment plant and cheap enough to evaluate inside GPU-parallel training, which the stiff implicit solver of the model it replaces cannot support; it was calibrated against IsaacLab and is therefore the reference condition rather than an independent competitor.

The **control-relevant identification and validation** procedure is delivered and is the contribution most likely to transfer independently of this machine, covering both the input-ablation test that exposed a forward model with no command authority and the identifiability audit that established the excitation envelope.

Policy-only control through learned hydraulics is demonstrated. After the classical teacher is removed, IF holds a mean lateral error of 70 mm and a median of 28 mm over 33 held-out cells at 95 % ground contact (Section 7.2). Worst-cell aggregation still misses two gates; that bound is stated in Section 8.4.

The **decision procedure for online adaptation** is delivered, and its main product is the discipline that produced it rather than any single number. It separates residual plant sensitivity, observability, correction authority and residual capacity before a gain is attributed to adaptation. The same evaluation discipline also exposes a broader control-design lesson: a quantity which is gated but not priced is traded away by the optimiser.

Where online adaptation is effective is answered at two levels rather than with a single verdict. The explicit mechanism is informative at the model level and has no closed-loop effect through the saturated inverse command, and the implicit mechanism that replaced it acts on the valve path, where authority remains. The primary evaluation yields four successful seeds and one trade-off, and the final-checkpoint analysis returns three successes, so the median training run improves under both rules. The capacity-matched comparison then locates the source of that gain: a residual of the same capacity reading zeroed history reproduces most of it, while live history separates from that control on one seed of five under either checkpoint rule. The cutter-mass dose study resolves a plant-dependent lateral component, but does not separate static mass dependence from the later mass

exchange. The supported method-level claim is a bounded residual on the valve path. The supported adaptation-specific claim is narrower: live history contributes where residual capacity alone does not, and where load variation stays visible in the scored channel.

Two enablers are recorded rather than claimed. The causal plant estimator and the bounded adaptation path behave as specified, with no privileged access, no future information, bounded output and reversion to the robust actuator path on invalid input. Of the eight-gate promotion criterion, one property is worth reuse: five gates separate the matched pair and carry the comparative claims, while lateral, hard impacts and tool acceleration do not (Section 3.9.1).

8.4 Limitations

Scope of plant validation. The evidence is simulation-to-simulation, and the deployment plant is itself a learned model. No claim here is a claim about hardware.

Scope of hydraulic observability. The differential pressure the estimator consumes is produced by the actuator model rather than measured, so transferring the model-level result to hardware requires instrumentation the FME 500 does not carry. An observation contract remains unresolved alongside this: every result uses the pressure-inclusive row, while the stricter deployable alternative excludes measured differential pressure. Since the estimator's advantage is concentrated in that channel, the deployable variant should be expected to perform worse, and by how much is not established.

Scope of controller comparison. On worst-mesh, worst-family statistics both stacks fail the lateral gate and IF does not satisfy the head-alignment gate under worst-case aggregation (Section 7.2), so neither stack satisfies the whole eight-gate evaluation contract. The closed-loop comparisons resolve differences to a few per cent over 33 paired conditions and do not establish equivalence. The available plant variation bounds the disturbance rather than the architecture, and more tightly than the sweep's labels suggest: a stack shown to absorb a 40% reduction in valve effectiveness and a nominal doubling of every deployment factor has been shown to tolerate what actually reached it, which on the inverse/forward plant is a mechanical perturbation only. The three stacks are matched on evaluation but not on training budget, so the comparison describes three deployed controllers rather than three equally tuned training procedures.

Scope of training variation. Policy-training variation is not quantified for the matched R and IF pair, and it bounds two claims differently. R and IF are each represented by one training run, so the 33 paired cells measure terrain and trajectory variation and say nothing about how far a second run of either would move; the actuator-representation result rests on that assumption. The adaptation result includes five training seeds and meets the primary criterion on four, so it establishes a median improvement and not a uniform one. The defensible claim is an architecture-level correction on the identified valve path, together with a plant-dependent lateral term under the tested cutter-mass doses.

Scope of the adaptation evidence. The commanded step cannot finish faster than 417 ms at the command-rate limit, while the plant answers in about 117 ms. That loop is a hard case for adaptation understood as a reaction to a single event and favours continuous conditioning on the operating point. Appendix B.5 is consistent with the second reading: the adaptive advantage is already present before the load change. Section 7.5.6 shows a

cutter-mass-dependent lateral term relative to the 1.0 condition, which the coupled dose design does not resolve into a transient adaptation response.

8.5 Future Work

The evidence in this thesis is simulation-to-simulation. The next step is to evaluate the learned actuator models and distilled policy on the target machine.

That step begins with instrumentation, because the FME 500 as delivered carries no cylinder-chamber pressure transducers, and the differential pressure the models propagate exists here only as an actuator-model state. Fitting pressure transducers, extension sensing and a rod load pin converts three modelled quantities into measured ones. The collection instrument is already designed and validated: the open-loop excitation schedule of Chapter 4 was built to make the valve causally identifiable, and the identifiability audit of Section 4.7.5 applies to machine data unchanged.

One simulation experiment should precede that step, because it targets the weakest link in the present result. Every history encoder trained here reads the full observation, 1400 numbers over the window, of which almost all describe terrain and task state rather than plant identity, and a head of this capacity can fit that noise. Restricting the encoder input to plant-relevant channels, or fitting it by regression onto a privileged plant vector instead of through the policy gradient as rapid motor adaptation does [Kumar 21], is the direct test of whether the missing history effect is a property of the plant or of the encoder.

Hardware data would support a bounded correction to the frozen inverse model, with hydraulic parameters retained and the original dataset used for replay. Input ablation should verify that the correction remains dependent on the valve command. The reference-anchored policy provides a suitable starting point because it limits policy drift from a verified checkpoint.

The central design lesson is conditional. Robust training absorbs a large part of the plant variation before online adaptation is introduced. A remaining correction is useful only when the variation is observable, the fixed controller leaves a deficit to recover, and the command path retains authority. Apparent adaptation gains must also be separated from gains due to residual capacity. Evaluating these conditions together provides a defensible basis for deciding when online adaptation is justified on a hydraulic manipulator.

A. Dataset Ranges and Audit

Chapter 4 describes how the Simscape and IsaacLab collections were generated. This appendix retains only the numerical plant ranges, the open-loop audit and the frozen deployment split needed to interpret or reproduce those collections.

A.1 Randomised Plant Ranges

Table A.1: Randomised hydraulic parameters. The seal-wear coordinate is drawn independently for each cylinder and drives three physically linked quantities.

Parameter	Nominal	Randomised range	Acts on
Entrained air fraction ε_{air}	0.005	0.001 – 0.080	fluid compliance
Dead-volume fraction ε_{vol}	0.020	0.005 – 0.080	fluid compliance
Discharge coefficient c_d	0.70	0.56 – 0.84	valve flow gain
Seal wear $w_{\text{seal},i}$	0	0 – 0.25 per cylinder	clearance, leakage, spool lag
Relief setting p_{set}	27.6 MPa	24.8 MPa – 30.4 MPa	pressure ceiling

Table A.2: Randomised mechanical parameters. Rigid-body perturbations are applied independently to each of the fourteen bodies.

Parameter	Nominal	Randomised range
Body mass	CAD value	±20 %
Centre-of-mass offset	CAD value	±5 %
Inertia tensor and products	CAD value	consistent with mass draw
Joint stiffness, J1 and J2	60 kN/m	40 – 120 kN/m
Joint stiffness, J3 and J4	60 kN/m	40 – 120 kN/m
Joint stiffness, J5	25 kN/m	15 – 60 kN/m
Joint damping	250	100 – 600

Table A.3: Six physical factors in the IsaacLab deployment collections. One factor vector is drawn per run and held fixed throughout that run.

Coordinate	Range	Domain	Acts on
e_{flow}	$[-0.25, 0.25]$	hydraulic	valve flow gain
$e_{\text{compliance}}$	$[-0.25, 0.10]$	hydraulic	effective chamber compliance
e_{seal}	$[0, 1]$	hydraulic	seal wear, leakage, spool lag
e_{pressure}	$[-0.15, 0.15]$	hydraulic	supply/relief pressure level
e_{mass}	$[0.8, 1.5]$	mechanical	cutter mass scale
e_{COM_z}	$[-0.10, 0.05]$	mechanical	cutter centre-of-mass height

A.2 Open-Loop Export Audit

Table A.4: Audit of the strict open-loop excitation export. The extraction stage computes and stores every check.

Check	Result	Interpretation
Rows, runs, columns	25 600 / 4 / 49	four complete runs
Median sample time	0.01 s	as commanded
Maximum sample-time error	9.1×10^{-15} s	numerically exact grid
Maximum Δextension error	0	targets match state differences
Maximum Δvelocity error	0	targets match state differences
Maximum $\Delta(\Delta p)$ error	0 Pa	targets match state differences
Non-finite values	0	no solver escapes
Valid fraction	0.937	startup, gaps and shutdown excluded
Command rank	4	all four channels independent
Lagged excitation rank	80	history matrix full rank
Valve saturation	8.5 % per channel	limits reached but not dominant
Stroke coverage	12 % – 33 %	above floor, below target

A.3 Frozen Deployment Collections

The task-controller and open-loop context-identification collections share the same factor design, domain composition and run-level split. Each contains 118 runs with one fixed factor vector per run. Model-level identification results use the open-loop collection for the identifiability reason given in Section 4.7.5.

Table A.5: Composition of each frozen deployment collection. The 102 training runs are divided into 82 fitting and 20 validation runs; the remaining 16 runs form the held-out test split.

Domain	Runs	Factors varied
`combined`	64	all six
`hydraulic`	40	flow, compliance, seal, pressure
`mechanical`	12	mass scale, centre-of-mass height
`nominal`	2	none
Total	118	

Splitting is performed at run level. Since the factor vector is constant within a run, a window-level split would place near-duplicates of the same plant in both fitting and evaluation sets. The frozen manifests store the run identifier, factor key and split label, making the separation directly auditable.

B. Implementation and Reproducibility Record

The methods chapters define the architectures and training procedures. This appendix retains only the timing contract, exact input layouts, frozen artifact identifiers and reproduction entry points that are cumbersome to include in the main text.

B.1 Timing Contract

Table B.1: Rates of the simulation, control and learned components. The offline AAM and deployed AF encoder are listed separately because they use different history grids.

Layer	Rate	Period
PhysX physics substep	120 Hz	8.33 ms
Policy and control step	60 Hz	16.67 ms
AF history encoder and residual	60 Hz	16.67 ms
Inverse, forward and ROM models	20 Hz	50 ms
Offline AAM inference	20 Hz	50 ms
Simscape logging grid	20 Hz	50 ms

The offline AAM reads 20 rows on the 20 Hz grid, giving a 1.0 s window, and its five-estimate smoother adds 250 ms of delay. Deployed AF instead reads 20 control-step rows, giving a 333 ms window, and applies no output smoother.

B.2 Frozen Artifacts

Table B.2: Frozen artifacts and their SHA-256 digests, truncated to the first 32 hexadecimal digits. Residual arms are evaluated on the frozen AF base.

Artifact	Role	SHA-256 (truncated)
`aam.pt`	offline AAM	`cd6a7f0b118b96c7ce66b889b9b3e422`
`zhat_adapter.pt`	ID adapter	`30629a4f83db5c2bed3c777bcf82dc57`
`best_id_cylinder_control.pt`	inverse model	`8e26c52e82d89efe193e001d0a8b07a6`
`best_fd_cylinder_strict.pt`	forward model	`1f41e44309a8f320430a3382ca88ce5f`
`rom_s6_aligned/model_598.pt`	R policy	`ce7e3ab9649b1193f123f9f91107bbf8`
`idfd_s6_aligned/model_598.pt`	IF policy	`06533b8e8614b0ad982dd8c238a17772`
`distilled_v1/phase1_init_70.pt`	AF frozen base	`641e476cfcd765a8d68a17980d6cbbf2`
`rebal17_history/model_399.pt`	AF residual, seed 17	`ea39150fd2919ce96aa1acdbbdc9d65d`
`rebal23_history/model_399.pt`	AF residual, seed 23	`63da26e9b9710b2f54715d7d70e82d59`
`rebal101_history/model_399.pt`	AF residual, seed 101	`f5a400f3b7e66b46a0cca11142a67d70`
`rebal42_history/model_399.pt`	AF residual, seed 42	`5cafe571d429ea8181989ad87c2ad558`
`rebal7_history/model_399.pt`	AF residual, seed 7	`4e5c3041987c20685e928ac0d5fac192`
`rebal17_nohistory/model_399.pt`	zero-history control	`c8b001b27ded55f2f32d5fe47d022a0a`

B.3 Learned-Actuator Interfaces

Both actuator networks use a linear projection to 512 units, four pre-norm residual blocks, final normalisation and a linear output projection. Their exact interfaces are given in Table B.3.

Table B.3: Input and output layouts stored in the frozen inverse and forward model checkpoints.

	Inverse model	Forward model (strict)
Input width	28	20
Input layout	extension(4), velocity(4), $\Delta p(4)$, force(4), target Δextension(4), target Δvelocity(4), $\Delta t(4)$	valve(4), extension(4), velocity(4), $\Delta p(4)$, force(4)
Output width	4	12
Output layout	valve command(4)	Δextension(4), Δvelocity(4), $\Delta(\Delta p)(4)$
Target mode	normalised Δ valve command	normalised Δ cylinder state
Parameters	2 124 292	2 124 300

Only the predicted extension increment of the forward model is integrated in the control loop. Its velocity output is retained as a diagnostic.

B.4 Policy Observation Layout

The deployed actor receives 66 values in every stack. AF carries its history window separately and does not widen the actor input. The final twelve values make actuator prediction and cylinder tracking errors visible to the otherwise memoryless actor.

Table B.4: Policy observation layout. Indices are zero-based and half-open.

Index	Width	Contents
0–9	9	verge height profile ahead of the cutter
9	1	cut-height error
10	1	lateral error
11	1	cutter vertical velocity
12–16	4	measured cylinder extensions
16–20	4	measured cylinder velocities
20–24	4	cylinder headroom, low side
24–28	4	cylinder headroom, high side
28–32	4	previous applied action
32–36	4	current valve commands
36–40	4	actuator-predicted Δextension
40–44	4	commanded target lead
44	1	head-down alignment
45	1	head-down alignment rate
46	1	truck speed
47	1	scan valid fraction
48–50	2	cutter local $+Z$ axis in world XY
50	1	normalised cutter-terrain interaction force
51–54	3	path preview at 0.5, 1.0 and 2.0 m
54–58	4	measured minus commanded cylinder state
58–62	4	actuator prediction residual
62–66	4	normalised cylinder effort

B.5 Internal Model Pressure State, Not a Physical Pressure

The pressure-state analysis is mechanistic support only. It verifies that the internal hydraulic state changes across the load event; it is not used to quantify adaptation performance or hydraulic accuracy. Figure B.1 shows the internal state per cylinder for all three stacks on the same plant and aligned on the same event.

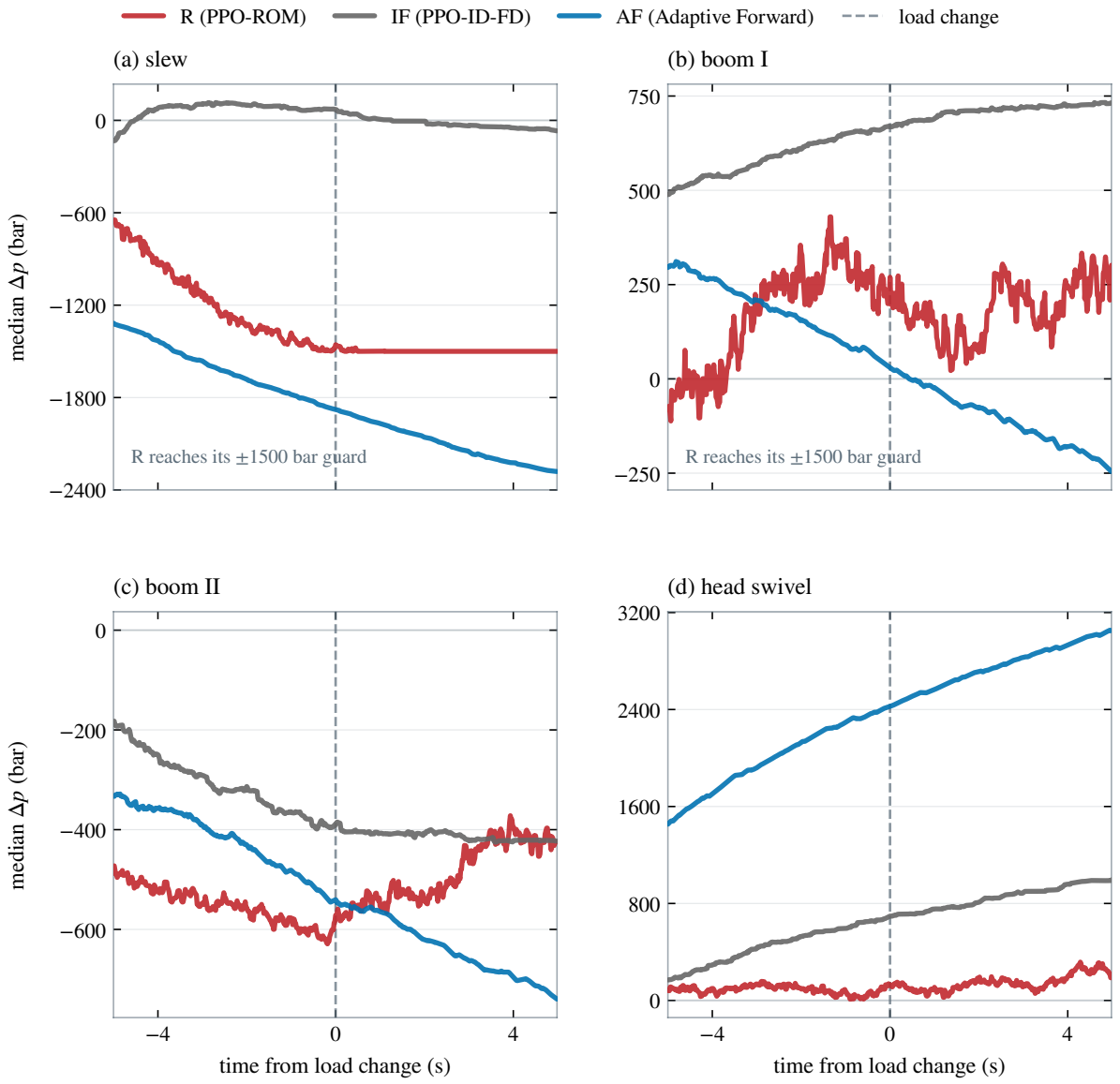

Figure B.1: Median differential pressure per cylinder through the change of load, 151 environments aligned on their own flip instant. The quantity is each stack's own actuator-model state, internal to the command ROM for R and the forward model's recursive state for the other two, so the panels show what each controller believes about the hydraulics rather than a common measurement, and each panel carries its own scale. The slew and boom I panels mark where R sits on its 1500 bar numerical guard.

Two properties of the signal govern how the figure is read. The pressure state is not a calibrated pressure on any stack: in this window the medians run to 2295 bar and the peaks to 3784 bar, which is past what a cylinder of this size could develop. The rod-force channel carries no supervised weight and pressure-increment accuracy is only $R^2 = 0.06$ to 0.09 (Section 5.4). On R the excursions are additionally clipped by the 1500 bar numerical guard, which binds on the slew and the first boom cylinder and not on the other two. There is therefore no common physical level the panels can be read against, and each

cylinder is drawn on its own scale. What the figure supports is the shape of the response either side of the change, not the level.

Read that way, the traces show that the model-derived pressure state responds through the event. The adaptive advantage is already present before it, which is consistent with continuous conditioning on the operating point. The uncalibrated state is not evidence for the magnitude or cause of the tracking improvement.

B.6 Derivation of the Reference-Divergence Gate Bound

Section 6.4.5 states, without showing the steps, that a bound on the per-step reference divergence D_{ref} implies a bound on gate degradation of the form $|\mathbb{E}_{\pi_\vartheta}[g] - \mathbb{E}_{\pi_{\text{ref}}}[g]| \leq 2G_{\max}\sqrt{H\delta_{\text{ref}}/2}$. This appendix derives that bound and recomputes it at the production horizon, since the number quoted in the main text does not match evaluating the formula it cites.

Setup

Let P_π denote the distribution over length-H trajectories $\tau = (s_0, a_0, \ldots, s_{H-1}, a_{H-1})$ induced by rolling out policy π under the fixed environment transition kernel. Because the transition kernel is shared by π_{ref} and π_ϑ, it cancels from the log-ratio between the two trajectory distributions, and the chain rule of KL divergence reduces to a sum of per-step, per-state action divergences:

$$\mathrm{KL}(P_{\pi_{\text{ref}}} \,\|\, P_{\pi_\vartheta}) = \mathbb{E}_{\tau\sim P_{\pi_{\text{ref}}}}\left[\sum_{t=0}^{H-1} \mathrm{KL}\Big(\pi_{\text{ref}}(\cdot \mid s_t) \,\|\, \pi_\vartheta(\cdot \mid s_t)\Big)\right]. \tag{B.1}$$

This step is exact only if the per-step divergences on the right are evaluated on states drawn from π_{ref}'s own rollout distribution. In the actual training loop D_{ref} is evaluated on states visited by the *live* policy π_ϑ (Section 6.7.4), not by π_{ref}, so what follows is the idealised argument the main text already flags as carrying a measure mismatch, not a derivation that closes that gap.

Chain rule bound

Bounding every one of the H per-step terms by the training-time average bound δ_{ref} of Equation (6.8) gives $\mathrm{KL}(P_{\pi_{\text{ref}}} \| P_{\pi_\vartheta}) \leq H\,\delta_{\text{ref}}$.

Pinsker's inequality

For any two distributions P, Q on a common space, total variation distance $\delta_{\text{TV}}(P,Q) := \sup_A |P(A) - Q(A)|$ satisfies $\delta_{\text{TV}}(P,Q) \leq \sqrt{\frac{1}{2}\mathrm{KL}(P\|Q)}$. Applied to the trajectory distributions,

$$\delta_{\text{TV}}(P_{\pi_{\text{ref}}}, P_{\pi_\vartheta}) \leq \sqrt{\tfrac{1}{2}H\delta_{\text{ref}}}. \tag{B.2}$$

Bounded-functional bound

For a trajectory functional $g : \mathcal{T} \to [0, G_{\max}]$, such as any of the eight gate metrics, the standard identity $\sum_\tau |P(\tau) - Q(\tau)| = 2\,\delta_{\mathrm{TV}}(P, Q)$ gives

$$\left|\mathbb{E}_{P_{\pi_{\mathrm{ref}}}}[g] - \mathbb{E}_{P_{\pi_\vartheta}}[g]\right| \le G_{\max} \sum_\tau |P_{\pi_{\mathrm{ref}}}(\tau) - P_{\pi_\vartheta}(\tau)| = 2\,G_{\max}\,\delta_{\mathrm{TV}}(P_{\pi_{\mathrm{ref}}}, P_{\pi_\vartheta}). \tag{B.3}$$

Combining Equations (B.2) and (B.3) gives the quoted bound,

$$\left|\mathbb{E}_{\pi_\vartheta}[g] - \mathbb{E}_{\pi_{\mathrm{ref}}}[g]\right| \le 2\,G_{\max}\sqrt{\tfrac{H\,\delta_{\mathrm{ref}}}{2}}. \tag{B.4}$$

Numeric value at the production horizon

At $H = 2808$ steps and $\delta_{\mathrm{ref}} = 0.05$, $\sqrt{H\delta_{\mathrm{ref}}/2} = \sqrt{70.2} \approx 8.38$. For a gate metric normalised to $[0, 1]$, $G_{\max} = 1$, so Equation (B.4) evaluates to $2 \times 8.38 \approx 16.76$. This is the value the formula in Section 6.4.5 produces; the main text's statement that the bound "exceeds eight" understates it by close to a factor of two, and the correct reading is that the bound exceeds sixteen for a metric bounded in $[0, 1]$. The substantive conclusion is unaffected and, if anything, strengthened: a bound of 16.76 on a $[0, 1]$-valued metric is vacuous more emphatically than a bound of 8.38 would be, and neither closes the measure mismatch flagged above. The construction remains a principled restriction of reward ascent to a neighbourhood of a verified policy, not a certificate.

B.7 Reproduction Entry Points

The closed-loop results of Chapter 7 are generated by the scripts under `experiments/E1` through `experiments/E6`. Each experiment directory records the evaluated model, mesh set, trajectory families, carrier speed and scored horizon in its manifest. AF arms are trained and scored under `PPO_AAM_FD/experiment_scripts`; the paired seed statistics are recomputed from `outputs/rif_delay` by `experiments/E3/code/seed_table.py`. Model-level results use the frozen collections described in Appendix A.3. Together with the artifact digests in Table B.2, these records identify the data, models and evaluation code behind the reported results.

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